



POWERING INDONESIA'S GREEN GROWTH:

International Experience and Lesson Learned in Stimulating
Renewable Energy Investment through Central Bank
and Government Incentives



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Indonesia Clean Energy Development (ICED) Program

POWERING INDONESIA GREEN GROWTH:
International Experience and lesson learned in
stimulating Renewable Energy Investment through
Central Bank and Government Incentives – Case
study from 6 countries

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Executive Summary

The purpose of this report is to:

- Summarize the lessons learned from the USA, China, Brazil, India, Bangladesh, and Thailand to assist Indonesia in stimulating commercially proven small- and medium-scale, utility-interactive renewable energy generating projects, with an emphasis on five target technologies: hydroelectric, biomass, solar, wind and biogas.
- Focus on measures taken by central banks as well as administrative, fiscal and other financial incentives adopted by governments in the five case study countries and Indonesia in order to stimulate investment in the target technologies.
- Draw conclusions about the actions, measures, and incentives that have proven successful in the six case study countries and other countries that Bank Indonesia and the Government of Indonesia (GOI) may want to replicate.

Case studies were developed to pinpoint each country's approach to renewable energy development, focusing on the target technologies, particularly utility-interactive projects of 10 MW or less. The study did not incorporate findings on how new and mature technologies in the United States, Europe and other countries are being employed to convert biomass, or biowaste, into ethanol and biodiesel which are, by law, then blended into conventional transportation fuels such as gasoline and diesel.

Four of the case study countries are ranked in the top ten countries in renewable energy development by Bloomberg: the USA, China, India, and Brazil. Although neither Thailand nor Bangladesh is ranked as highly, each provides additional valuable perspectives in certain areas.

Given the limitations in the information available (some countries post much more information than others on their renewable energy development activities), each country's case study was organized along similar lines. The main areas investigated were:

- Socioeconomic context
- Legislative initiatives affecting renewable energy development (outside the country)
- Legislative initiatives affecting renewable energy development (inside the country)
- Renewable energy research and development (R&D)
- Use of fiscal incentives
- Alternative generation standards
- Cap and trade frameworks
- Specific alternative generation technologies.

The case studies and other research provided a foundation from which the authors could draw comparisons among the case study countries and Indonesia to determine where the GOI might want to consider placing more focus in meeting its own renewable energy

objectives, particularly in the smaller utility-interactive projects.

A number of these comparisons may prove beneficial to GOI decision-makers in comparing the approach of the case study countries and Indonesia, in supporting renewable energy development through:

- Reducing domestic fossil fuel subsidies
- Setting renewable portfolio standards
- Establishing a domestic cap and trade market
- Incorporating feed-in tariffs
- Extending fiscal incentives
- Providing an enabling infrastructure for project finance
- Conducting research and development
- Disseminating information.

Key Findings

The report's conclusions focus on administrative, fiscal and financial initiatives that could be taken by the GOI to stimulate small-scale, utility-interactive, renewable generation. This is followed by measures the central bank could take to improve prudent lending to this target market.

Government of Indonesia Initiatives

According to *REN21 Renewables 2012: Global Status Report*, the GOI has set an aggressive renewable portfolio standard (RPS) requiring the Government and PLN to produce or procure renewable energy for up to 25% and 15%, respectively, of Indonesia's primary energy and electricity production by 2025. Within that target, there appear to be certain set-asides for each of the target technologies examined here, in addition to that set by GOI for geothermal technology (a 6.3% share of primary energy by 2025). The term "set-aside" or "carve-out" refers to a provision within Indonesia's RPS that requires PLN to use specific renewable resources (geothermal, wind, solar, hydro, biofuel and biomass energy), each of which would account for a defined percentage of primary energy by 2025. It is not clear whether this is a fixed set-aside or merely an estimate of what is possible. Nonetheless, the overall objective of Indonesia's RPS is relatively aggressive than those of the case study countries. Aggressive objectives require bold policies and measures in order to enable the objectives. However, it is currently unclear whether the GOI is able to move quickly enough to institute the measures necessary.

For example,

- Subsidies on fossil-fuels and electricity will need to be reduced fairly rapidly, as is already widely recognized by GOI policy makers, because it does not make much sense to be paying subsidies for fossil fuels while at the same time paying preferential tariffs and/or extending corporate incentives to a large percentage of

the utility-interactive stations that generate electricity from renewables. The experience of Brazil, which completely eliminated subsidies to the fuel and electricity sectors in a relatively short period of time, is a case study that should be read by policy makers.

- Since most small utility-interactive projects will be implemented in outlying areas, policy coordination at the national, provincial, district and municipal levels needs to be much strengthened. As noted in this report, developers face a variety of obstacles in obtaining the permits needed to advance their projects. At the same time, local officials lack the capacity to properly assess the unsolicited proposals that are being forwarded. One way of attempting to resolve this problem is for PLN, and the provincial, district and municipal governments to at times rely on “cluster tendering”: preparing, approving and tendering several projects at once, with tenderers given the option of bidding on all, some or just one of the projects being considered. This is likely to be more efficient than encouraging unsolicited proposals from any entity that feels like submitting one. The recently proposed 150 MW solar PV tender, in coordination with a new feed-in tariff (FIT) for solar photovoltaic (PV), seems to be a cluster tender for the wide dissemination of solar PV in Indonesia.

While unsolicited proposals tend to exacerbate issues related to policy coordination or capacity at the local level, cluster tendering ameliorates these issues. It enables local governments to coordinate their efforts with each other and PLN during the planning and project preparation stages. For example, preliminary environmental clearance and/or other social safeguard issues can be dealt with early on with assistance from the Ministry of Environment; management performance specifications and standards can be drafted and incorporated in the project agreement that is annexed to the tendering documents, including specifics on reporting and corporate responsibility standards; the timetables and conditions related to the issuance of permits and licenses can be divulged in the same documents.

The bidding criteria could be based on a parameter such as the tariff (with the proviso that bids cannot exceed the FIT) and/or the quality, experience and financial wherewithal of the bidders. By using competitive tension for the rights to the projects, the participation of the IIGF (through its guarantee mechanism), IIF and PIP (through its debt instruments) may be allowed or encouraged, if needed. Cluster tendering will lead to better quality developers, will spread the risks of site selection as well as implementation, and may be key to “professionalizing” this segment of the market. In any case, it is a good way for local officials to “learn by doing.” A test case of cluster tendering is emerging now that ICED and IIGF have promised to collaborate in developing one.

- A sound review of the FITs and, particularly, corporate incentives, is needed to ensure adequacy. While the FITs adopted by PLN for geothermal, small hydro and

biomass appear adequate, a comparison of Indonesia's corporate incentives with those of the case study countries would suggest a need for reassessment.

- Indonesia does not offer production tax credits (PTC), investment tax credits (ITC) or grants to small projects when the project may need them for their financial sustainability. Some countries offer a variety of ITC and/or grant programs to encourage the use and development of renewables and energy efficiency. Most programs offer support for a broad range of technologies, while a few focus on promoting a single technology, such as PV systems. Most PTC, ITC and/or grant programs are designed to pay down the cost of eligible systems or equipment, making unviable projects reasonably viable at the posted FIT. This is especially important in Indonesia, where developers are supposed to cover their own transmission costs to the grid. Projects that otherwise could be located in ideal sites have to be moved elsewhere because they are too far from the grid, making the project too expensive to implement.

To deal with this problem, government could move to create a public benefit fund (PFB). Most PFBs were developed by US states during the electric utility restructuring era of the late 1990s to ensure continued support for renewable energy, energy efficiency and low-income energy programs. These funds are commonly supported through a very small surcharge on electricity consumption (e.g., in the case of the US, \$0.002/kWh). This charge is sometimes referred to as a "system benefits charge." PFBs commonly support rebate programs, grant and loan programs for renewable generation, research and development, and energy education programs. Such a fund should only benefit small utility-interactive projects.

Alternatively, if a PFB is not possible, such a fund could be created from the national budgetary savings that result from the gradual withdrawal of subsidies to fossil fuels or some other mechanism such as a carbon tax, as India and the US city of Boulder, Colorado have recently implemented.

- Reportedly, Indonesia has requested assistance in developing an emission trading scheme (ETS) from the World Bank's Partnership for Readiness; this assistance may have begun on 12 November 2010 with an expression of interest that the GOI delivered to the Bank. An ETS, sometimes referred to as a "cap and trade," is a market-based approach used to control pollution by providing economic incentives for achieving reductions in the emissions of pollutants. A central authority such as the Ministry of Environment sets a limit or cap on the amount of a pollutant that may be emitted by a firm. The cap is allocated or sold to firms in the form of emissions permits that represent the right to emit or discharge a specific volume of the specified pollutant. Firms are required to meet the cap or pay penalties for failing to do so. Alternatively, firms can purchase carbon emission reduction (CER) certificates equivalent to their emissions above their assigned cap. Firms that need to increase their volume of emissions must buy CERs from those with lower emission levels than those set by cap. The transfer of CERs is referred to as a trade.

In effect, the buyer is paying a charge for polluting, while the seller is being rewarded for having reduced emissions. Thus, in theory, those who can reduce emissions most cheaply will do so, achieving the pollution reduction at the lowest cost to society. Renewable generation plants emit small levels or, in some cases, no emissions, and hence are entitled to CER certificates, for up to an amount that equals the cap allowed for the typical “reference” power station. The developer can sell the CERs in the open market to polluters who cannot meet cap and hence use the proceeds as “equity” or as a “special dividend” resulting from the start-up operations of the renewable power station. This important incentive is not being funded by the GOI, but the government is enabling it through the newly created cap and trade market.

- The GOI, as far as the authors of this report are aware, has not yet considered creating a renewable energy laboratory, such as the US National Energy Renewable Energy Laboratory (NREL), which is solely dedicated to the research, development, commercialization and deployment of renewable energy. Most of the case study countries have an agency that is responsible for this kind of research and development. In Bangladesh, for example, the University of Dhaka has assumed this role. Alternatively, or simultaneously, a university or government ministry could create the Indonesian equivalent of the database in the USA which describes each state’s practices, incentives and lessons learned in implementing renewable energy and energy efficiency. The latter is housed at North Carolina State University and paid for by the Department of Energy. Since Indonesia’s provinces and districts have considerable autonomy, the measures and incentives provided by each such local government would be of great interest to other local governments with less experience.

Bank Indonesia Initiatives

In addition to scrutinizing the GOI’s administrative measures and corporate incentives for small utility-interactive projects, this report examines Indonesia’s evolving legal and regulatory framework as it applies to green banking and renewable energy and, where possible, has compared it to that of case study countries, all of which are reasonably successful in attracting renewable energy investment. As mentioned above, four of the sample countries are rated in the top ten in recruiting private investment into renewable energy.

On the basis of the analysis conducted, it appears that Indonesia is implementing what is regarded as a generally favorable framework for attracting investment into renewables. In particular, the emerging IRR of Law 32 [2009], *Environmental Protection and Management*, if issued in the manner described in Section 2.2 of this report, will be of great benefit in creating awareness among financial institutions of the importance of assessing environmental issues and social concerns in prudent lending. That, in turn, is likely to be a key driver within the banking sector in understanding current and evolving technologies in the use of renewable energy for generating electricity, processing biofuels for transportation, and incorporating energy-efficient practices in corporate operations. All

evidence points to the fact that China and Bangladesh have enjoyed a tremendous interest in renewables since their green banking acts were passed in 2010 and 2011, respectively.

In addition to the issuance of IRR of Law 32, to further ensure prudent lending practices in Indonesia, Bank Indonesia should consider the following additional measures:

- Creation of a debt fund, as was done in Bangladesh, to enable commercial banks to borrow long-term, fixed-rate funds, in order to offer renewable energy developers fixed-rate financing on the basis of locked-in margins and matched maturities. This is particularly important to developers implementing small-scale (less than 10 MW) utility-interactive power plants that rely on renewable feedstocks. A fund of this nature would have the advantage to the lender (as well as the developer) of reducing liquidity and interest volatility risks.
- Increased frequency of workshops, through ICED, that improve the banking community's skills in the extension of credit based on cash flow techniques as well as the identification, assessment and mitigation of risk with an emphasis on "project structuring," i.e., improving the risk profile of projects through using the international and domestic mechanisms and instruments described in Section 3.2.6. Based on the authors' limited contact with senior bankers, there does not appear to be a widespread inclination to use available instruments to mitigate risk and improve credit quality.
- Improved access to information regarding renewable energy by creating a first-class library and online database that is readily accessible to banks, developers and other parties in Bank Indonesia offices throughout the country, or in suitable alternative locations.
- Assurances that a dedicated team, such as exists currently within USAID's ICED Project, continues to assist lenders and developers once the project ends in September 2013.

As a closing note, according to the UNFCCC, Indonesia's land use change and forestry emissions (largely from deforestation) as a percentage of GHG emissions in 2005 was 50.8%, which implies deforestation is a primary source of GHG emissions. In 2010, Norway embarked on a \$1-billion program to save Indonesia's forests, which was renewed in May 2013. However, forests are still being cleared by fire on Sumatra, as recently evidenced by the haze in Malaysia and Singapore, which the international press recently reported. The GOI needs to embark on a ban, or strict enforcement, of deforestation, and possibly jail time for infractions of the ban. New renewable energy additions bode well for Indonesia's targets of alternative generating technology energy as percentage of the RPS of primary energy and electricity produced, but barely puts a dent in GHG emissions if deforestation is allowed to continue.

Acronyms

AMDAL	Analisis Mengenei Dampak Pemantauen Lingkungan
BI	Bank Indonesia
CAGR	Compounded annual growth rate
CDM	Clean Development Mechanism
CER	Carbon emission reduction
CERCLA	Comprehensive Environmental Responsibility, Compensation and Liability Act
CO ₂	Carbon Dioxide
CIT	Corporate income tax
EU	European Union
ETS	Emissions trading scheme
FIT	Feed-in tariff
GDP	Gross domestic product
GEF	Global Environment Fund
GHG	Greenhouse gasses
GOC	Government of China
GOI	Government of Indonesia
ICED	Indonesia Clean Energy Development
IEA	International Energy Agency
IGA	International Geothermal Agency
IIF	Indonesia Infrastructure Fund
IIGF	Indonesia Infrastructure Guarantee Fund
IMF	International Monetary Fund
IRR	Implementing rules and regulations
ISO	International Organization for Standardization
ITC	Investment tax credit
JI	Joint implementation
MCC	Millennium Challenge Corporation
MEMR	Ministry of Energy and Mineral Resources
MOE	Ministry of Environment
MOF	Ministry of Finance
MTOE	Metric tons of oil equivalent
MW	Megawatts
NREL	National Laboratory for Renewable Energy
OECD	Organization of Economic Co-operation and Development
PBF	Public benefit fund

PIP	Pusat Investasi Pemerintah
PTC	Production tax credit
PURPA	Public Utility Regulatory Policies Act
PV	Photovoltaic
R&D	Research and development
RPS	Renewable portfolio standard
SREC	Solar Renewable Energy Certificates
TOE	Tons of Oil Equivalent
TPES	Total primary energy supply
UKL-UPL	Environmental management efforts-environmental monitoring efforts report
UNEP	United Nations Environment Program
UNFCCC	United Nations Framework Convention on Climate Change
USA	United States of America
VAT	Value-added tax
WAHLI	Indonesian Forum for the Investment
WCI	Western Climate Initiative

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1 Overview

1.1 Objectives

This report has three objectives:

- Summarize the lessons learned from six countries – Bangladesh, Brazil, China, India, Thailand, and the United States – to assist Indonesia in stimulating commercially proven small- and medium-scale utility-interactive renewable energy generating projects, with specific emphasis on five technologies: hydroelectric, biomass, solar, wind and biogas (together, the “target technologies”).
- Focus on the 1) measures taken by central banks and 2) administrative, fiscal and other financial incentives adopted by the governments of the case study countries to stimulate investment in the target technologies.
- Draw conclusions from a survey of the six countries on actions/measures/incentives that have proven successful in other countries and which Bank Indonesia (BI) and the Government of Indonesia (GOI) may want to replicate.

1.2 The World Renewable Energy Situation

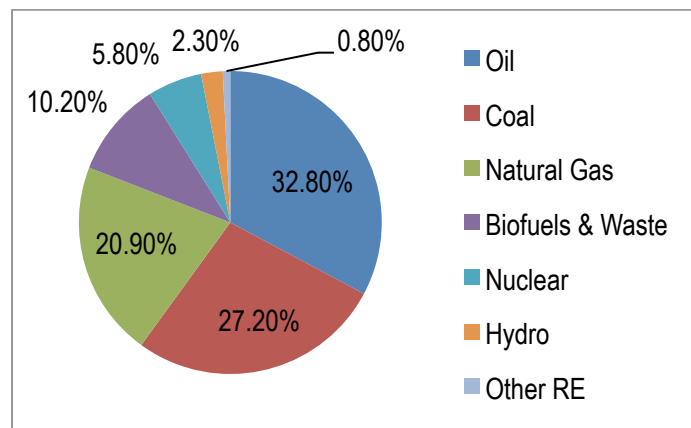
According to the International Energy Agency (IEA), in 2009, the primary sources for the world’s 12,150 million metric tons of oil equivalent (MTOE) were: fossil fuels 80.9%, biofuels and waste 10.2%, nuclear 5.8%, hydro 2.3%, and other renewable energy 0.8% (see Figure 1).

Growth and investment in alternative technologies have increased greatly in recent years and are expected to continue, according to the Bloomberg New Energy Finance Report *Global Trends in Renewable Energy Investment 2013*.

Highlights of this trend include:

- New investments in renewable energy have grown 26% annually from 2004-2012 on a compounded annual growth rate (CAGR) basis, but declined by 12% between 2011 and 2012, from \$279 billion to \$244.4 billion.
- Solar power grew 36% annually from 2004-2012 on a CAGR basis, but declined by 11% between 2011 and 2012, from \$158.1 billion to \$140.4 billion. Worldwide, solar photovoltaic (PV) installed capacity stood at more than 100 GW at the end of 2012, with major installations in Europe, China and United States.

Figure 1
World Total Primary Energy Supply, 2009



Source: IEA Key World Energy Statistics, 2009

- Between 2004 and 2012, wind power grew by 24% annually on a CAGR basis, but declined 10% between 2011 and 2012, from \$89.3 billion to \$80.3 billion. At the end of 2012, the worldwide installed wind capacity was 282,482 MW, with major installations in Europe, Asia, and the United States.
- Biomass waste and biofuels grew by 4% annually from 2004-2012 on a CAGR basis. However, between 2011 and 2012, they declined 34% and 40%, respectively, from \$12.9 billion and \$8.3 billion to \$8.6 billion and \$5.0 billion, in 2012.
- The International Geothermal Association (IGA), has reported that 10,716.7 MW of geothermal have been installed in 25 countries with major installations in the USA, Philippines and Indonesia, which together hold slightly more than 50% of total installed capacity. This represents a 20% increase in geothermal power online capacity since 2005. The IGA projects that installed geothermal capacity will grow to 18,500 MW by 2015.

Five factors stand out as the leading drivers of growth in the renewable energy industry:

- High fossil-based energy costs related to increased demand and uncertainties in supply
- Economies of scale and falling capital equipment costs for alternative technologies, improving both yield and efficiency, and thereby improving competitiveness
- Legislative action by governments worldwide (a key focus of this paper)
- Greater public awareness of environmental concerns, particularly regarding the greenhouse gas (GHG) emissions associated with fossil fuels

- The Kyoto Protocol, through which certain countries have committed to reduce GHG emissions, and which in turn motivates those governments to implement policies that decrease the use of fossil-sourced energy and increase the use of alternative technologies.

Alternative energy is also increasingly the subject of policy discussion in light of structural changes in global macro conditions, including:

- **Continued global economic growth:** The world is witnessing a realignment of economic activity, with Brazil, Russia, India, and China leading the way. Continued increases in energy production will be necessary to support this growth.
- **Continued population growth:** The world's population continues to grow and, in 2008 – for the first time in history – more than 50% of the global population, or 3.3 billion people, will be living in urban areas (towns and cities). By 2030, this urban population is expected to grow by 60% to a projected 8 billion people, creating unprecedented energy and land use concerns.
- **Energy security:** Political instability in petroleum-rich nations is leading to a broader re-evaluation of energy policy, particularly regarding reliance on imported oil.
- **Climate change:** Scientific evidence regarding the threats posed by climate change and the extent to which this change is caused by human activity (e.g., GHG emissions associated with energy production and use) is raising important questions about how to achieve sustainable growth and the degree to which alternative sources of energy can substitute for fossil fuels.

Investment in new global renewable energy capacity was \$279 billion in 2011 alone, a year-over-year increase of nearly 22%. In 2012 it declined 12% to \$244.4 billion. In its *New Policies Scenario*, the IEA's 2012 World Energy Outlook projects a cumulative outlay of \$37 trillion over 2012-2035 for global investments in energy supply infrastructure, which is equivalent to 1.5% of global GDP on average during that period. Sixty-one percent of the total investment is expected to be made in countries outside of the Organization for Economic Co-operation and Development (OECD). Oil and gas supply is projected to account for \$19 trillion, or a little over 50%, of total investment.

Although the boom in the investment in, and development of, alternative technologies is global, a few regions are at the forefront of this growth. The Frankfurt School UN Environment Program Collaborating Center for Climate and Sustainable Energy Finance's *Global Trends in Renewable Energy Investment 2013*, prepared by Bloomberg New Energy Finance, concludes that:

- China remains at the top for total renewable energy investment, at \$64.7 billion in 2012, up 22% from 2011. The USA is second, with \$34.2 billion, down 36% from 2011. China has continued to benefit from a \$46 billion green stimulus package,

which it announced at the height of the financial crisis in 2008. By the end of 2010, 70% of the funds had been spent. The Chinese government's support for renewables, as discussed further here, includes reduced corporate income taxes, significant reductions in value added taxes, other tax incentives, feed-in tariffs, research and development (R&D) incentives, and subsidies for energy conservation technologies improvement.

- The top-ten countries for renewable electricity capacity, excluding corporate and government R&D and hydroelectric power greater than 50 MW, are China, USA, Germany, Japan, Italy, UK, India, South Africa, Brazil and France.
- However, if one ranks these countries on a per capita basis, Germany would be at the forefront followed by Italy, UK, Japan, South Africa, USA, France, China, Brazil and India.
- By region, the European Union (EU) was home to nearly 32.7% of new global renewable investments at the end of 2012, which amounted to \$244 billion. China and the EU together accounted for 60% of the total investment in renewable energy.
- For large hydroelectric power (>50 MW), the top six countries were China, Russia, Vietnam, Brazil, Iran and India. Of the top ten countries in new alternative energy investments, excluding R&D, four are case study countries (Table 1).

Each of these renewable energy technologies (hydroelectric, biomass and biogas, wind and solar) has its own “environmental footprint,” with frequently contested positive and negative features. Each faces unique economic challenges that make it difficult for these technologies to compete with energy generated by long-established, subsidized, fossil fuels.

In 2012, new investments in wind and solar alone represented 90% of new global renewable energy investments. These two power sources grew 24% and 36%, respectively, from 2004-2012 on a CAGR basis. On the other hand, geothermal, biofuels and biomass, and waste grew 5%, 4%, and 4% annually from 2004-2012, respectively.

Table 1
New Investment in Clean Renewable Energy, 2012

Country/	New Investment (\$billions)	% Change Y-O-Y
China	\$64.7	22%
USA	34.2	-36
Germany	19.8	-35
Japan	16.0	73
Italy	14.1	-53
UK	8.8	-12
India	6.4	-50
South Africa	5.7	NM
Brazil	5.3	-38
France	4.6	-31

Source: Bloomberg New Energy Finance 2013.

This paper concentrates on the economic challenges associated with growth in the use of energy from renewable sources and, in particular, the way policymakers in the sample countries are dealing with them. Decision-makers in Indonesia have concluded that alternative technologies should be more widely deployed. Thus, this paper explores the options for enhancing the competitiveness of and increasing the demand for such technologies. The discussion of policy options adopts the view that current market forces, by themselves, may not be sufficient to generate the desired expansion in alternative energy production. Hence, this paper considers the effectiveness of such policy options as:

- Central bank practices that may have been adopted in the case study countries to influence the flow of credit into renewable energy
- Funds created by central banks or governments to facilitate the commercial financing of these facilities, either by extending credit on preferential terms or grants
- Government measures: administrative as well as incentivizing, such as:
 - Reducing fossil fuel use or electricity-based subsidies with the intent of making renewable generation or the processing of biofuels for transportation more competitive
 - Establishing renewable portfolio standards (RPS) in order to increase the proportion of electricity generated by utilities from renewable energy
 - Using feed-in tariffs (FITs), which require utilities to purchase bulk electricity from an alternative, renewable energy source, generally at a higher than market rate

- Implementing fiscal incentives to encourage investment in the renewable energy generation of electricity, such as production tax credits (PTC), investment tax credits (ITC), accelerated depreciation, elimination of sales taxes or import duties on purchased equipment, and elimination of the value-added tax (VAT) on bulk sales of electricity to the utility
- Imposing carbon-based taxes to discourage the use of heavy fossil fuels and/or create tradable carbon emission reduction (CERs) credits and other related market-based incentives, such as carbon-based “cap and trade” systems that aim that increase competitiveness of low-carbon alternatives using market mechanisms
- Providing other indirect support for renewable energy technologies such as government support for private sector R&D efforts to commercialize alternative energy technologies.

Many countries are already using these and other policy tools to encourage the supply of and demand for energy generated by alternative technologies. This paper outlines the policies used globally to overcome market barriers for these technologies and concludes with a survey of such practices in the USA, Brazil, China, India, Bangladesh, and Thailand to provide a useful reference point for future decisions.

1.3 Renewable Energy in Indonesia

In the last decade, Indonesia has moved from a robust energy exporter to an energy importing nation, creating policy concerns on several levels. For example, rising production costs, energy subsidies, and climate change impacts have stimulated new interest in renewable energy technologies. Few countries offer the renewable energy potential of Indonesia. The country is home to 40% of the world’s known geothermal resources and offers significant opportunities in wind, solar, biomass, as well as biofuels.

Historically, the GOI has maintained high energy subsidies for the country’s energy consumers. When supply was high and demand low, the subsidies stimulated economic growth with low, stable energy prices. But as demand began to outpace supply with the corresponding rise in prices, the subsidies became a burden on the nation’s fiscal budget and a deterrent to investments in cleaner, renewable resources.

Indonesia’s fuel and electricity or energy subsidies have the negative consequence of making growth itself an unaffordable expense, as the government must pay more money for every new unit of energy developed and sold – money that cannot be channeled into economic growth needs or social programs.

The next four tables compare Indonesia to the case study countries in four key areas to provide some context for the discussions that follow. Unfortunately, 2009 is the only year for which these statistics are available.

Table 2 compares Indonesia to the case study countries in terms of the share of primary energy sources used to meet domestic requirements, excluding electricity. Oil, for example, accounts for 33% of total primary energy, which places Indonesia in the middle of the sample countries. Bangladesh, China, and India rely less on oil than Indonesia, while Thailand, the USA and Brazil are more reliant. As might be expected, Indonesia also relies significantly on waste and biofuels as a primary source of energy, in an amount that is not dissimilar to the other developing countries in the sample, excluding China.

The US and China dwarf the other countries in their use of primary energy sources, excluding electricity, with 2,160 and 2,257 million tons of oil equivalent energy, respectively. Each uses around 10 times the amount of primary energy employed in Indonesia.

Table 2
Percentage of Total Energy Sources, Excluding Electricity, 2009
(millions of metric tons of oil equivalent energy)

Source	Indonesia	Bangladesh	Brazil	China	India	Thailand	USA
Oil	33.1	15.9	40.2	16.8	23.6	39.2	37.1
Biofuels & Waste	26.0	29.8	32.1	9.0	24.5	20.0	3.9
Natural Gas	17.4	51.7	7.2	3.3	7.2	25.8	24.7
Coal Peat	15.1	2.1	4.6	67.3	42.3	14.5	22.5
Hydro	0.5	0.5	14.2	2.3	1.4	0.5	1.1
Geothermal	7.9	-	0.3	0.5	0.3	-	0.7
Solar & Wind							
Nuclear	-	-	1.4	0.8	0.7	-	10.0
Percent	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Energy (TOE) in millions	222	26	241	2.257	675	103	2.160

Source: *International Energy Agency Statistics 2011*. TOE is tons of oil equivalent =11.63 MWh or 41.868GJ.

In 2009, 86.7% of Indonesia's electricity came from fossil fuels (coal, oil and natural gas) (Table 3). This is less than that of Bangladesh (95.9%) and Thailand (91.1%), but above China (80.6%), India (83.8%), USA (69.1%) and Brazil (8.1%). The USA makes significant use of coal, natural gas and nuclear in generating electricity (87.7%), while renewable energy – hydro (7.1%) and other sources (4.0%) – made up 11.1% of its electricity production in 2009. Only 8.1% of Brazil's electricity production comes from fossil fuels; it has a high dependence on hydro at 83.8%. Along with Indonesia, Bangladesh, China, India and Thailand are highly dependent on fossil fuels (>80%) for electricity production.

Table 3
Comparative Electricity Production, 2009 (in GWh)

Source	Indonesia	Bangladesh	Brazil	China	India	Thailand	USA
Coal/Peat	64,976	638	9,782	2,913,122	616,584	29,596	1,892,661
Oil	35,467	1,832	14,639	16,494	26,099	710	50,445
Natural Gas	34,351	33,840	13,332	50,813	111,206	104,943	949,776
Fossil Fuels	134,794	36,310	37,753	2,980,429	753,889	135,249	2,892,882
Hydro	11,381	1,552	390,988	615,640	106,909	7,148	298,410
Geothermal	9,295	-	-	153	-	1	17,046
Nuclear	-	-	12,957	70,134	18,636	-	830,210
Biofuels	-	-	23,095	2,351	1,995	5,981	49,850
Wind	-	-	1,238	26,900	17,933	1	74,226
Waste	-	-	-	-	-	-	22,441
Solar PV	-	-	-	321	27	9	1,698
Solar Thermal	-	-	-	-	-	-	816
Other	-	-	437	-	-	-	635
Total	155,470	37,862	466,468	3,695,928	899,389	148,389	4,188,214

Source: International Energy Agency Statistics 2011

Table 4 shows that as of 2009, the USA had the most diversified sources of renewables and waste for its generation of electricity and energy production. It is particularly strong in hydro, which provides 64.2% of its renewable energy supply for electricity production, and its biofuels production (ethanol from corn) is the largest in the world. Brazil is second in this case; both countries are required by law to blend ethanol into gasoline consumption.

Other salient facts in these countries' electricity generation include:

- Hydroelectric power in Indonesia, Bangladesh, Brazil, China, India and Thailand accounts for more than 50% of electric production
- China and the USA are the largest users of solar PV and wind energy, by far
- India is third behind China and the USA in wind energy
- The USA and Brazil are the largest producers of liquid biofuels for transport, while China, India, Brazil and Indonesia topped the USA in the use of solid biofuels in 2009
- The USA and Indonesia were first and second in geothermal resources, while Indonesia topped the USA in energy production in 2009.

Table 4
Comparable Renewables and Waste Electricity (GWh) and Energy (Tj) Production, 2009

Source	Indonesia	Bangladesh	Brazil	China	India	Thailand	USA
Electricity							
Hydro	11,381	1,552	390,988	615,640	106,909	7,148	298,410
Geothermal	9,295	-	-	153	-	1	17,046
Solid Biofuels	-	-	23,057	2,351	1,995	5,849	40,478
Biogases	-	-	-	-	-	132	9,281
Wind	-	-	1,238	26,900	17,933	1	74,226
Solar PV	-	-	-	321	27	9	1,698
Solar Thermal	-	-	-	-	-	-	816
Municipal Waste	-	-	-	-	-	-	16,909
Industrial Waste	-	-	-	-	-	-	5,532
Liquid Biofuels	-	-	-	-	-	-	91
Total	20,676	1,552	415,283	645,365	126,864	13,140	464,487
Energy (Tj)							
Solid Biofuels	2,205,601	369,049	2,627,335	8,152,659	6,920,355	837,075	1,948,306
Liquid Biofuels*	355	-	20,987	1,868	256	669	34,657
Geothermal	669,240	-	-	75,191	-	36	341,983
Solar Thermal	-	-	16,768	300,633	11,204	-	65,129
Biogases	-	-	-	322,384	-	1,521	213,091
Municipal Waste	-	-	-	-	-	-	299,398
Industrial Waste	-	-	-	-	-	-	120,217

*Thousands of metric tons

Source: *International Energy Agency 2011.*

The percentage of renewable energy in total primary energy supply (TPES) and TPES per capita (using IMF population statistics) are shown in Table 5. According to the International Energy Agency, 2009:

- All of the countries have more renewable energy as a percentage of TPES than the USA, but the latter's TPES per capita was the highest by a wide margin. Indonesia ranks second in the percentage of renewable energy to TPES, but fifth in TPES/capita.
- China was second from the bottom in the percentage of renewable energy to TPES, but second-highest in TPES/capita.
- Brazil was first in the percentage of renewable energy to TPES, but fourth in terms of TPES per capita in 2009.

Table 5

Renewable Energy as a% of Total Primary Energy Supply and TPES per capita, 2009		
Country/	% Renewable Energy	TPES per Capita
USA	5.8	7.04
China	11.8	1.69
India	26.2	0.58
Brazil	46.6	1.24
Thailand	20.6	1.63
Bangladesh	30.3	0.20
Indonesia	34.4	0.86

Of the G7 countries, Japan and the UK have smaller renewable energy percentages of TPES than the USA, at 3.6% and 3.3%, respectively. Canada has a higher renewable energy percentage of TPES than China at 16.8%, which is a result of its heavy use of hydropower. In addition, Canada's TPES per capita was 7.53, which is better than that of the USA.

As shown in Table 6, the top carbon dioxide (CO₂) emitters were China, USA, EU, the Russian Federation, Japan, and Canada. The case study countries and Indonesia together contributed 51.4% of the total CO₂ emissions worldwide. These data include CO₂ emissions from fossil fuel combustion, as well as cement manufacturing and gas flaring. Together, these sources represent a large proportion of total global CO₂ emissions.¹

Table 6
2008 CO₂ Emissions (thousands of metric tons, except per capita data)

Country	CO2 Emissions (Rank)	% World Total	Per Capita (Rank)
China	7,031,916 (1)	23.50%	4.9 (55)
USA	5,461,014 (2)	18.27%	18.5 (12)
India	1,742,698 (3)	5.83%	1.3 (133)
Indonesia	406,029 (15)	1.36%	1.5 (130)
Brazil	393,220 (17)	1.32%	1.9 (124)
Thailand	285,733 (22)	0.96%	4.2 (97)
Bangladesh	46,527 (63)	0.16%	0.3 (178)
Total	15,367,137	51.4%	4.5
World	29,888,121	100%	4.5

Source: *United States Department of Energy's Carbon Dioxide Information Analysis Center*. In 2008, the world population was 6,688,000 and 3,452,192 for the case study countries, according to the *IMF World Economic Outlook* of April 2013.

¹ Emissions and sinks related to changes in land use are not included in these estimates. However, changes in land use can be important – global estimates indicate that deforestation can account for 5 billion metric tons of CO₂ emissions, or about 16% of emissions from fossil fuel sources. Tropical deforestation in Africa, Asia, and South America are thought to be the world's largest contributors to emissions from land use change.

2 Policy, Legal, and Regulatory Framework

2.1 Key Policy, Legislative and Regulatory Initiatives

Renewable energy accounts for a small but growing portion of Indonesia's electricity portfolio. Most of its renewable energy comes from the hydropower and geothermal industries, but growth in other sectors is likely. Presidential Regulation No. 5 [2006] on National Energy Policy originally mandated an increase in renewable energy production (electricity plus transportation biofuels) from 7% to 17% by 2020. The GOI intended at the time to alter its energy mix: oil: 20%, natural gas: 30%, coal: 33%, biofuels: 5%, and geothermal: 5%.

However, in 2010, the GOI issued Energy Vision 25/25, which increased the development target for new and renewable energy from 17% to 25% and altered the mix: renewable energy: 25%, oil: 30%, gas: 23%, and coal: 22%. This also included provisions to reduce energy demand by 33.85% by 2025. According to the policy, new energy sources are those produced by new technologies such as nuclear, hydrogen, coal bed methane, liquefaction of coal and gasified coal, as promulgated in Law 30/2007 on Energy.

In 2010, the Ministry of Energy and Mineral Resources (MEMR) revised the country's geothermal potential to from 27 GW to 28.1 GW, which amounts to 12 billion barrels of oil equivalent and is almost twice the country's current oil reserves of 6.4 billion barrels. At that time, Indonesia's installed geothermal capacity was just 1.2 GW, leaving it 96% of its potential undeveloped. This undeveloped potential is similar for hydropower at 75 GW (93%), biomass at 50 GW (97%) and wind (99%). Overall, Indonesia has an undeveloped renewable potential of 96% with an on-grid installed capacity of 2.9 GW and an off-grid capacity of 3.2 GW, against a total resource potential estimated at 163.3 GW.

To address investment requirements in renewable energy, incentives for renewable energy development in Indonesia can be classified into four groups:

- Financial incentives, which provide loans and loan guarantees at below-market rates to competitively bid projects
- Fiscal incentives such as tax exemptions and other tax-related subsidies
- Access to services at below-market rates
- Market price support and regulation, which provide above-market prices or demand when renewable energy is produced and sold.

Most of these incentives are briefly mentioned in Presidential Regulation No. 5 [2006]. The implementation of specific schemes is described in detail under various ministerial decrees and regulations.

Table 7
Incentive Schemes and Related Regulation

Incentive Scheme	Related Regulation	Remark
Financial		
Indonesia Infrastructure Guarantee Fund	MOF Reg. 260/2010	SOE provides guarantee to attract private capital investment in infrastructure
Geothermal Fund	MOF Reg. 2/2012	Provides finance for exploration, payable if sites prove productive
Development Credits for Biofuels and Plantation Revitalization	MOF Reg. 117/2006 MOF Reg. 79/2007	Low-cost loans for farmers and farmer groups that plant energy crops
Government financial guarantee	MOF Reg. 139/2011	Government guarantee for geothermal and hydropower projects
Fiscal		
Import duty and Value Added Tax (VAT) exemption	MOF Reg. 21/2010	Import duty and VAT exemption on machinery and capital for power plants, and importation of taxable goods
Income tax reduction	MOF Reg. 21/2010	Reduction and other incentives for income tax on energy development projects, including net income reduction, accelerated depreciation, dividends reduced for foreign investors, and compensation for losses
Provision of goods and services below market value		
Geothermal Fund	MOF Reg. 3/2012	Survey and exploration services, payable if sites prove productive
Guarantee of business viability of PLN	MOF Reg. 139/2011	Guarantee that PLN would perform and respect contracts with independent power producers
Public competitive bidding	MEMR Reg. 1/2006	Agrees favorable tariffs with most competitive company bidding for tender
Feed-In Tariffs	MEMR Reg. 4/2012 MEMR Reg. 22/2012	FIT for geothermal, biomass and mini-hydro
Mandatory utilization	MEMR Reg. 32/2008	Obligatory use of biofuels for fuel mix and mandatory use of biofuels in mining Industry

Source: *TKN Report Investment Incentives for Renewable Energy: Case Study of Indonesia*, December 2012

Since 2008, the government has offered tax incentives for foreign investment, including investors in renewable energy. A renewable energy investor is eligible for a net income reduction of 5% of the investment value each year, over a six-year period (Ministry of Finance Regulation No. 21/2010). Accelerated depreciation and amortization allow investments to be depreciated within 2–10 years, depending on the type of asset. Moreover, an income tax reduction for foreign investors allows them to pay a rate of only 10% on dividends they receive. Compensation for operating losses for foreign investors is

available for more than five years and follows certain criteria. VAT on imported goods for producing renewable energy can also be exempted. However, it is not clear if these incentives match those extended by other countries, as discussed further below.

2.2 The Potential Evolving Role of Bank Indonesia in Environmental Management

A central bank is a public institution that manages a state's currency, money supply, and interest rates. Such banks have generally not been responsible for stimulating renewable energy projects, or even advocating environmental and social policy. This is likely because of the key role central banks play in the supervision of the banking sector where, as regulators, they ensure that the financial institutions under their supervision are involved in prudent lending. New international initiatives in recent years such as Basel I, II and III, issued by the Basel Committee on Bank Organization² have, if anything, reinforced this overseer mandate.

However, the new awareness on the limits to economic development brought on by increased reliance on conventional patterns of industrialization has begun to shape new policies in support of the custodianship of the environment in many countries. These changes are accompanied by a new vision of what the central bank's role should be in the execution of such policies.

²The Basel Committee on Banking Supervision is a committee of banking supervisory authorities that was established by the central bank governors of the Group of Ten countries in 1974. It provides a forum for regular cooperation on banking supervisory matters. Its objective is to enhance understanding of key supervisory issues and improve the quality of banking supervision worldwide. The Committee also frames guidelines and standards in different areas. Some of the better known among them are the International Standards on Capital Adequacy, the Core Principles for Effective Banking Supervision, and the Concordat on Cross-border Banking Supervision.

2.2.1 Importance of Commercial Bank Regulation in Environment and Social Management

The central bank, as regulator of the banking sector, has great influence on the quality of a country's economic growth and long-term sustainability. In addition, the banking sector in any country is a major source of finance for commercial projects. Hence, it can play a crucial role in indirectly controlling unwanted emissions and pollutants at the national level. The internal environmental footprint of a central bank, or any commercial bank, is very low. Hence, the environmental impact of banks is not physically related to their banking activities, but rather to the activities of their borrowing customers. Given the importance of credit intermediation in any economy, the environmental impact of the banking system's external activity is potentially huge.

Encouraging environmentally and socially responsible investments is important to the operating performance of any bank. Environmental and social management in the banking business is a fundamental part of risk management and a critical element in prudent lending. Properly done, it increases the market value of the bank by lowering its loss ratio, as a higher-quality loan portfolio results in higher earnings.

The role of banks in environmental and social management has been recognized by international initiatives, such as the Equator Principles³ and the UNEP.⁴ In support of these initiatives, multilateral agencies have been advocating for some years now for higher environmental standards and strategies to apply to the evaluation of any new investment project. The International Organization for Standardization (ISO)⁵ has issued a comprehensive series of guidelines for incorporating environmental protection and pollution prevention objectives into industrial activity; its standards are codified in ISO 14000. The latter standard (similar to ISO 9000 Quality Management) is a family of guidelines related to environmental management that were developed to help organizations 1) minimize how their operations (or processes, etc.) negatively affect the environment (i.e., cause adverse changes to air, water, or land), 2) comply with applicable laws, regulations, and other environmentally oriented requirements, and/or 3) can be

³ The Equator Principles were developed by private-sector banks – led by Citigroup, ABN AMRO, Barclays, and WestLB – and were launched in June 2003. The banks chose to model the Principles on the environmental standards of the World Bank and the social policies of the International Finance Corporation. As of October 2009, 67 financial institutions had adopted the Equator Principles, which have become the de facto standard for banks and investors on how to assess major development projects around the world. The Equator Principles were revised in July 2006.

⁴ The UNEP Financial Initiative is a global partnership between UNEP and the financial sector. Over 200 institutions, including banks, insurers and fund managers, work with UNEP to understand the impacts of environmental and social considerations on financial performance.

⁵ The ISO is an international standard-setting body composed of representatives from various national standards organizations. Founded on 23 February 1947, the organization promotes worldwide proprietary, industrial, and commercial standards. It has its headquarters in Geneva, Switzerland.

improved. These and other initiatives provide the impetus for the banking sector to expand the use of environmental analysis into its credit policy and investment decisions.

Until recently, commercial bank supervision and/or management did not consider environmental management relevant to their business operations. Bankers, generally, were unconcerned with how their clients handled environmental risks. However, in recent years, changing perceptions have led to the belief that being passive in dealing with these issues can result in the absorption of unquantifiable risk. For example, the enactment of Comprehensive Environmental Response, Compensation and Liability Act [1980] (CERCLA) in the United States resulted in huge losses to the banks in the late 1980s, as the latter were held directly responsible for the environmental pollution of their clients and made to pay the remediation costs. This adoption of CERCLA is one reason why US banks are ahead of other countries in integrating environmental and social concerns into their business operations. In recent years, several other countries (in particular in Europe, China, and Bangladesh) have adopted policies that have made banks responsible for the misdeeds of their clients. Financial institutions in these countries are required to engage proactively with stakeholders on environmental and social policy issues, and to evaluate the impacts of their clients' commercial operations. That, in turn, has forced bank customers to take care of the environmental and social policy issues relating to their business operations.

Bloomberg New Energy Finance publishes an annual top-20 survey of Green Banks globally, which included 9 European, 4 US, 3 Japanese, 1 Australian, 1 Brazilian, 1 Canadian, and 1 South African bank in 2012. In the survey, Citigroup claimed the top spot after making investments that include California's Solano Wind Project, which in May began delivering enough new power for 44,000 homes.

The legal and regulatory integration of environmental and social compliance into the operations of commercial banks is commonly referred to as "green banking."

2.2.2 Evolving Legal and Regulatory Framework in Indonesia

Bank Indonesia and the Ministry of Environment (MOE) entered into an agreement on 17 December 2010 under which they would work together to enhance the role of green banking under a framework of environmental protection and management. The agreement, which was motivated by increasing global awareness on the application of sustainable development principles in various industries – including the banking industry – will remain in effect for three years. The scope of coordination involves, among others, the preparation of legal tools, the provision of information, education and socialization activities, as well as joint research. The two institutions will also collaborate in drafting the implementing rules and regulations (IRR) of Law 32 of 2009, *Environmental Protection and Management*, which is still a work in progress.

BI acts as the banking system's regulator. By 30 December 2013, however, its commercial bank prudential supervision functions will be transferred to the Financial Services Authority

(Indonesia), which is currently being formed. While the IRR has yet to be released, the law itself, as described further below, contains important green banking elements. Hence, once the IRR is drafted, it is expected to provide significant underlying detail with regard to the extent green banking will be applied in Indonesia.

The MOE acts as a coordinating and policy formulating body on environmental matters, and is responsible for preparing drafts of environmental legislation to be adopted by the executive branch. It also issues the guidelines to determine ambient air and effluent quality standards for industry, as well as other technical guidance, which are adopted by sectoral technical agencies to implement pollution control. Sectoral departments, such as the Ministry of Industry and the Ministry of Manpower and Transmigration, are responsible for environmental management within their technical jurisdictions, while the local governments are mandated to manage environmental concerns within their jurisdiction, as part of an integrated national environmental management program.

In recent years, the major development in the environment sector has been the passage of Law No. 32 on 3 October 2009 (New Environmental Law). Law No. 32 revoked Law No. 23 of 1997 *Regarding Environmental Management*. The latter was perceived to be ineffective in reducing environmental pollution and damage, which has been increasing since the law was enacted. Law No. 32 requires companies to issue an environmental management statement (*Surat Pernyataan Pengelolaan Lingkungan*), environmental management efforts-environmental monitoring efforts reports (*Upaya Pengelolaan Lingkungan-Upaya Pemantauan Lingkungan* or “UKL-UPL”) or an environmental impact assessment (*Analisis mengenai Dampak Lingkungan* or “AMDAL”) depending on their industry and exposure to the environment. In addition to this obligation, the Law provides a new requirement for companies to carry out an environmental risk analysis (*Analisis Risiko Lingkungan*) or obtain an environmental permit (*Izin Lingkungan*). According to the Law, the environmental permit is a precondition to obtain other business permits. The MOE produced a draft Government Regulation on Environmental Permits, which indicated that there would be a requirement for an environmental feasibility permit as well as an environmental protection and management permit. The feasibility permit covers activities that may have significant environmental impacts, but are not yet covered in AMDAL or UKL-UPL.

According to Law No. 32, each province functions as a responsible and real autonomous region and administrative territory. Therefore, the provincial and local environmental impact management agencies (*Badan Pengendalian Dampak Lingkungan Daerah* or BAPEDALDA) are expected to take up more of the enforcement tasks in their specific areas, such as adherence to the monitoring requirements of industrial operating licenses, and responding to public reports of environmental damage for businesses and/or activities that are obligated to implement environmental impact assessments. However, implementing such actions at this level of government is likely to be a challenging task, given that technical and enforcement capacity is lacking.

It is quite significant that the Law No. 32 of 2009 specifically allows an individual or a group of legal standing to sue for environmental damages. In 1989, the Indonesian Forum for the

Environment (WALHI), as a representative of the public interest, was granted standing to sue five national government agencies and the pulp-and-paper industry in the Jakarta District Court for environmental pollution, damages, and non-compliance with environmental laws. Although the court rejected the substantive issues of the suit, the case has set a strong precedent for similar legal suits against polluters and raised public awareness of legal protections against environmental pollution. Another case occurred in 2001, when the Counsel of Judges at the South Jakarta District Court ruled against PT Freeport Indonesia, based on WALHI's law suit, and decided in its verdict that the company was guilty of disseminating misleading information regarding an accident involving toxic overflows at their Wanagon Lake mining location in Irian Jaya. This accident allegedly claimed four lives on 4 May 2000.

Whether the legal and regulatory framework continues to evolve in a direction that resembles the US or European model will depend a great deal on the ultimate scope and content of the IRR. However, based on conversations with the MOE on 21 May 2013, it appears that some, or all, of the following green banking measures are likely to become mandatory:

- Banks will be expected to promote green credit as a strategy, support the economy to grow in a green, low-carbon way, manage environmental and social risks, and contribute to the transformation of economic growth.
- Senior management of each bank will be expected to develop strategic plans related to green credit, not just for renewable energy projects that will be booked, but also those that involve energy efficiency. It is not yet clear whether the annual strategic plan will include actions and measures to take with respect of clients that are polluting or in other ways damaging the environment.
- Senior bank management will be encouraged or, perhaps, directed, to appoint a senior officer to organize, lead and manage a department that specializes in long-term credit extensions involving renewable energy or energy efficiency.
- With regard to general portfolio management, banks will work within the environment and social risk categorization framework designed by the MOE, with all business firms rated (by national, provincial or municipal regulatory bodies) on their adherence to environmental and social risk guidelines. Four colors will be used to designate their degree of compliance: gold, blue, red or black. The worst color ratings (red and black) will go to those firms demonstrating the least compliance. The latter firms, depending upon circumstances, may become ineligible to receive further credit and may be required by the bank to draw up plans to reduce their negative impact on the environment.
- Where appropriate, new term loan agreements will contain covenants that borrowers will be required to agree on with respect to improving their environmental and social risk profiles.

- Extended non-compliance with the framework could mean the loss of a business license, which increases banks' risks of lending to clients with a serious deleterious environmental impacts such as those in the business of manufacturing, processing chemicals, refining oil or derivative products, producing plastics, or processing paper and pulp. In effect, this means that those firms are likely to be monitored most closely by various regulatory bodies in the finance and environment sector.

2.2.3 Approaches in the Case Study Countries

If the IRR indeed contains the above provisions, it will resemble the framework of three of the six case study countries: the USA, China, and Bangladesh. The US framework is probably the most robust because in addition to the above provisions, it has the most comprehensive legal system and capable regulators, combined with very harsh enforcement options. China and the Bangladesh have similar green banking frameworks, which became operational in 2010 and 2011.

China's approach toward green banking resembles that of the United States, but perhaps without the civil actions that can be taken by third parties. However, it also has some aspects that are very different and probably account for its potential effectiveness in execution. China relies on five-year guidelines⁶ that contain detailed plans for each region, thereby eliminating or at least reducing the coordination issues that arise in Western-style democracies where there are different levels of government and jurisdictional authorities, each executing (or disregarding) national objectives in the way the authority believes is best for its regency.

Moreover, the Government of China (GOC) devotes significant resources to communicating and socializing targets at both the provincial and district levels, as well as monitoring progress.⁷ Measures frequently used include the designation of low-carbon pilot provinces and cities, and fiscal measures to incentivize energy costs. The result is significant cohesion in the execution of national policy, an issue that does not appear to have been solved in Indonesia as yet.

The green banking initiatives that are part of China's 12th Five Year Plan include similar programs to the initiatives that appear to be contemplated in Indonesia, but go further in some areas than may be the case with the Law 32 IRR, to wit:

Bank of China requires:

⁶ Planning is a key characteristic of centralized, communist economies, and one plan established for the entire country normally contains detailed economic development guidelines for all its regions.

⁷ KPMG, China, "12th Year Plan for Sustainability, 2011-14," a four page document.

- Mandatory development of green banking strategic plans at the Board of Directors level of all financial institutions, which most likely include measures to take regarding existing clients with environmental and social concerns
- Implementation of Equator Principles, or another acceptable lending framework, in all banks and financial institutions
- Strengthening the green credit approval process based on borrowing clients' environment track record in light of:
 - Mandatory implementation of low-carbon regulations that apply to all business firms, both government and private
 - Borrowing clients' track records with regard to environment and social concerns
 - Quality of clients' strategic plans to reduce their carbon footprints
 - Required internal bank classification of clients who on the worst polluter list
 - Rejection of further credit to clients on the worst polluter list.
- Mandatory development of green banking strategic plans at the Board of Directors level, most likely to include plans and actions related to borrowing clients
- Encouraging and/or facilitating the disposal of clients' high carbon emission assets.⁸

None of the other case study countries goes this far in implementing green banking, except possibly Bangladesh, whose framework appears to be fairly similar to and perhaps influenced by China's. Nonetheless, from the standpoint of creating awareness and meaningful policies for the protection of the environment, the most effective actions have been implemented by Bank of China and Bangladesh Bank.

Each quarter, Bangladesh Bank requires the banks under its supervision to report on their green banking activities. The disclosures required include policy formulation for green banking, budget set-asides for green banking activities, amounts spent on marketing and capacity building, loans made to renewable energy or energy efficiency projects, use of the Central Bank's fixed-rate funding facility, in-house environment management, design and introduction of innovative green financial products, etc. These actions are reported by the commercial banks and evaluated by the Central Bank. Bangladesh Bank periodically identifies the top-ten banks in overall green banking activities.

Bank of China and Bangladesh Bank, incidentally, also do a lot of "leading by example." Both institutions have set themselves the goal of reducing their carbon footprints, through the following types of internal investments:

- Reducing reliance on internal carbon emitting processes, substituting all electronic processes in internal and external administration

⁸ Ibid.

- Using video-conferencing for regional meetings, rather than relying on travel
- Capturing rain water for internal use inside the bank
- Installing renewable energy electricity for internal bank needs, usually solar panels on the roof, or other complimentary means of generating electricity
- Installing waterless toilets, or developing plans to maximize their use
- Channeling state-owned banks to support clean energy policy and, also, adopt similar operating principles
- Issuing annual awards to the banks that have developed the best green banking lending and administrative practices.

3 Measures/Incentives that Stimulate Alternative Generating Technologies

3.1 Central Bank Measures

Aside from the efforts BI is making in the area of green banking, this report has uncovered other examples among the case study countries of direct measures taken by central banks to stimulate renewable energy.

- Bangladesh Bank administers two infrastructure funds through its Green Banking and Corporate Social Responsibility Department. The first is a debt fund empowered to co-finance long-term debt with commercial banks. Since the source of funding for this facility is provided through donor grants and concessional funding, it is likely that this facility can extend fixed-rate, intermediate-term debt finance, something which is risk mitigating in small-scale, utility-interactive renewable energy projects.

The second fund serves as a rediscount mechanism, enabling banks to access fixed-rate cost of funds for their portion of the long-term debt extensions to qualifying renewable energy projects. The two funds working together deliver a full package of fixed-rate debt finance to the borrowing client. Although this report did not uncover any evidence that other central banks are also employing the equivalent of a modified rediscount window to enable domestic banks to provide long-term fixed-rate finance toward the development of desired industries (agriculture is a case in point), this has been a time-honoured practice among central banks in other countries, particularly Latin America.

The commercial banks, using the “rediscount” facility, would retain the usual supervisory credit responsibilities and all of the risks associated with their participation in long-term loans.

- Currently, there may be insufficient awareness within the commercial banks of the financial tools that can be used to structure projects and improve prudential lending. Table 8 provides a sample of international and national agencies that provide technical or facilitative assistance in developing projects in emerging markets, along with their key instruments.⁹ It is unclear whether any of the agencies and banks interviewed are currently working with provincial agencies to leverage capital,

⁹ Further reading on these agencies can be easily accessed on their websites. A good primer on the World Bank and funds listed below can be found in the following publication: S. Venugopal, A. Srivastava, C. Polycarp, and E. Taylor, 2012, “Public Financing Instruments to Leverage Private Capital for Climate-Relevant Investment: Focus on Multilateral Agencies.” Working Paper developed through World Resources Institute, Washington, DC. Available online.

particularly grants and guarantees, to improve the bankability of provincial renewable energy projects. China and India seem to make a point of maximizing the input of the international mechanisms and instruments to improve the bankability of renewable energy projects. In addition, China extends special incentives to encourage this to occur. It would be very helpful to expose commercial banks to the way in which these instruments can be used to improve the risk profile of smaller-scale renewable energy projects.

Table 8
International and Domestic Initiatives Available in Indonesia and Abroad
to Structure Projects and Improve Bankability

	Grants	Loans	Guarantees	Equity
International Institutions				
World Bank Carbon Finance Unit-ERPAs				■
World Bank Partnership for Market Readiness	■			
International Finance Corporation (IFC)		■	Political and credit risk	■
Multilateral Investment Guarantee Agency (MIGA)			Political risk	
Clean Technology Fund (CTF)	■	■	Technology risk	■
Strategic Climate Fund (SCF)	■	■		
GEF	■			
USAID Development Credit Authority (DCA)			Political and credit risk	
Export Credit Agencies (G-10)		Eximbank	Political and credit risk	
Kyoto Protocol Clean Development Mechanism	■			
Indonesian Institutions				
Indonesia Infrastructure Guarantee fund (IIGF)			Political and credit risk	
Indonesia Infrastructure Fund (IIF)		■		
Pusat Investasi Pemerintah (PIP)	For geothermal exploration	■		■
MCC Green Prosperity Fund	■	■		

Although Indonesia hosts a number of international and domestic public and private financial institutions with a wide variety of financial instruments, most appear to be geared to support large, competitively bid, infrastructure facilities. A notable exception is the \$320 million Millennium Challenge Corporation (MCC) Green Prosperity Fund, which provides much needed support in small-scale utility-interactive projects, but which has a limited mandate to operate in just 13-provinces outside of Java.

Few countries have taken on the responsibility for outsourcing to a reputable firm the creation of a comprehensive data base on renewable energy technologies and making it available in selected target destinations, perhaps central bank offices around the country. Information on renewable technologies does not appear to be widely available in Indonesia, other than through USAID's ICED Project, nor is there an obvious channel through which such information can be disseminated. This may create bottlenecks in project development, potentially increasing the cost of the entire program – those of developer, lender and regulator.

However, Bangladesh has established a resource center in Dhaka, at the University of Dhaka's research center. The University of Chittagong in Bangladesh offers what appears to be an engineering degree in renewable energy engineering.

3.2 Central Government Measures

3.2.1 Effective Public Policy

Renewable energy policies and programs have a wide range of technologies and markets that they can support. These options may be constrained due to limited funding and resources as well as regulatory requirements. Well-established programs often consider the following elements when determining how to allocate resources:

- **Market Needs:** Program design to support renewable energy reflects the market potential and barriers of different technologies, and allocates appropriate incentives toward achieving this potential.
- **Diverse Technology and Market Portfolio:** Programs include a range of targeted technologies, measures, and markets including emerging and higher-cost technologies, which could have a longer financial payback.
- **Accessibility:** Program funds are allocated in a reasonably equitable manner across customer classes. This distribution helps build public support for programs, although it could also result in lower savings per dollar spent (e.g., program transactions costs could be larger for a residential program than an industrial program, or for a rural program than a national one). Programs also attempt to increase awareness and distribute available funds across their geographic service territories and demographic groupings.
- **Adaptability:** Individual programs and incentives can be adjusted to changing demand and market conditions. At the same time, programs can avoid abrupt stops or frequent changes in incentive levels, which leads to market confusion and lack of vendor/partner support.

Sections 3.2.2 to 3.2.7 summarize the case study programs taken holistically. There are considerable differences among countries but, generally, with the exception of the USA and India at the federal level, successful countries tend to have consistent central government policy, appropriate regulations, generous incentives, available finance mechanisms, and other such measures.

3.2.2 Consistency and Coordination of Policy

In Indonesia, there appears to be a lack of relationship between renewable energy development policy and targets, and continued reliance on oil and electricity subsidies. Indonesia is not alone in this matter. Most of the case study countries, with the exception of Brazil, have the same problem.

The continuation of fuel and electricity subsidies limits the attractiveness of renewable energy projects, creates a need for higher feed-in tariffs (FITs) than otherwise would be the case, and/or creates a higher level of corporate incentives. It should be noted that hydroelectric and biomass technologies are competitive, or nearing competitiveness, with conventional generation in a number of countries even in the absence of subsidies. Solar and wind will require time before they are competitive and, therefore, require relatively robust preferential tariffs. However, continued subsidies for fuels and electricity in Indonesia – while also providing FITs for geothermal, biomass, solar and wind – may create fiscal stress and, in any case, take direct resources away from other pressing social needs.

A recent report from the International Monetary Fund (IMF) concludes that energy subsidies are pervasive and impose substantial fiscal and economic costs in most regions. On a pre-tax basis, subsidies for petroleum products, electricity, natural gas, and coal reached \$480 billion in 2011 (0.7% of global GDP).¹⁰ The cost of subsidies is especially acute in oil exporting countries, which account for about two-thirds of the total.¹¹

Together, China and the USA account for about 3.9% of the world pre-tax total; the six case study countries, including Indonesia, account for 19.0%, and oil exporting countries account for about two-thirds of the total. Removing these subsidies could lead to a 13% decline in CO₂ emissions and generate positive spillover effects by reducing global energy demand.

Table 9 reflects the IMF's recent study of subsidies for petroleum products, electricity, natural gas and coal.

¹⁰ There are differences of opinion on these magnitudes. See, for example, International Energy Agency, "Redrawing the Energy-Climate Map: World Energy Outlook Special Report," 10 June 2013. This report has estimates that vary somewhat from that of the IMF, estimating the pre-tax subsidy at \$523 billion and the subsidy for renewables at \$88 billion.

¹¹ On a post-tax basis – which also factors in the negative externalities from energy consumption – subsidies are much higher, at \$1.91 trillion (2.72% of world GDP).

Table 9
IMF Energy Subsidy Reform, 2011:
Pre-tax Subsidies for Petroleum Products, Electricity, Natural Gas, and Coal

Country	% of GDP	\$ GDP, billions	\$ Amount, billions
Indonesia	3.24	846	27.4
United States	0.05	15,075	7.5
China	0.15	7,321	11.0
India	1.74	1,838	32.0
Brazil	0.00	2,492	0.0
Thailand	2.18	345	7.5
Bangladesh	5.13	113	5.4

Source: *IMF Energy Subsidy Reform: Lessons and Implications*, January 2013; and IMF population data for 2011

Aside from the issue of fossil fuel subsidies, to the outside observer at least, Indonesia appears to lack coordination and socialization among the national, provincial and district governments with regard to the implementation of national goals for renewable energy. Reportedly, there are significant differences in competencies among these levels of government.

Utility-interactive, small-scale, renewable energy generating projects are implemented at the local level, precisely where coordination, socialization and competency appear to be most lacking. There are reports of projects that are held up at the district level by delays in acquiring the land-use license, negotiating the corporate social responsibility payments to the community, or for other reasons. Delays add unnecessarily to project development expense. Moreover, competency in evaluating renewable energy projects for technical, environmental, or financial merits is also missing.

3.2.3 Renewable Portfolio Standards

A key instrument that has been successful in furthering renewable energy is the renewable portfolio standard (RPS), also referred to as the renewable energy standard – a policy that seems to be popular among the case study countries. This type of policy indirectly creates a market for renewable energy by obliging power retailers to source a certain percentage of their electricity from renewable energy facilities. An RPS usually targets a share of renewable energy as a percentage of annual energy sales in a jurisdiction; it increases over time. There are various differences in RPS legislation with respect to the percentages of renewable energy required and whether carve-outs are used to establish preference for certain technologies (see, for example, the discussion of the use of solar renewable energy certificates by the United States in Annex A).

The Public Utility Regulatory Policies Act (PURPA), enacted in 1978 by the US Congress, was the first RPS legislation in the case study countries, although the Act fell short of establishing a mandatory share of renewables relative to utilities' total generation. Instead, PURPA required the utilities to buy green power if it could be obtained more cheaply than the

utility's "avoided cost", i.e., the cost on the margin of adding conventional generating capacity. Today, the USA, China, India, and Thailand have RPSs specifying the targeted level of renewable vs. total generation over time. However, in the USA and India, this is done on a state-by-state basis, with some states free to demur from setting such standards. California, in the USA, was the first state to adopt RPSs in 1984.

Table 10 compares the RPS for each of the case study countries relative to Indonesia. In some cases, the renewable energy target is expressed as a percentage of total primary energy and in others, the proportion that renewable generation will have relative to total generated electricity.

Just last year, Indonesia announced its goal of adding 4,000 to 5,000 MW of new geothermal energy capacity by 2015, and its eventual goal of 10,000 MW of geothermal capacity by 2025.

Table 10
Renewable Portfolio Standards or Targets for Case Study Countries and Indonesia

USA	California's RPS is 33% by 2020; others are set at a lower level, except Hawaii at 40% by 2030. No federal standard; however, 29 states and District of Colombia have set mandatory standards and another 7 states have voluntary standards
China	Renewable energy as a percentage of total primary energy of 15% by 2020 Specific Targets: Wind: 100 GW on-grid by 2015, 5 GW offshore and 30 GW offshore by 2020; solar: 15 GW by 2015; hydro: 284 GW by 2015; biofuels: 5 million metric tons of ethanol used 2011-15.
India	Renewable energy as a percentage of total primary energy of 15% by 2020 Specific Targets: Wind: 9 GW by 2012; solar: 20 GW on-grid by 2022, 2 GW off-grid by 2020, 20 million solar home systems by 2022, and 14 GW solar thermal collector areas or 10 million square meters by 2022.
Brazil	Non Specific Targets: Wind: 11.5 GW by 2020; small hydro: 6.4 GW by 2020; and biomass: 9.1 GW by 2020.
Thailand	Renewable energy, overall, as a proportion of total primary energy and the proportion of renewable energy-generated electricity relative to total generation are, respectively, 20% and 14% by 2022. Specific Targets: Wind: 1.2 GW by 2022; solar: 2 GW by 2022; hydro: 1.6 GW by 2022; biomass: 3.6 GW by 2022; biogas: 600 MW by 2022; municipal solid waste: 160 MW by 2022; and new alternative energy: 3 MW by 2022.
Bangladesh	Renewable energy generated electricity: 10% by 2020 Specific Targets: Solar: 500 MW by 2015; solar off-grid: 2.5 million solar home systems for residential units by 2015; biomass: 2 MW by 2014; biogas: 4 MW by 2014; bio-digesters: 150,000 installed by 2016.
Indonesia	Renewable energy, overall, and renewable energy-generated electricity targets: 25% by 2025 (Vision 25/25) and 15% by 2020, respectively Specific Targets: Wind, solar, hydro: 1.5% combined by 2025; geothermal: 6.3% in primary energy and 12.6 GW electricity by 2025; and biofuel: 10.2% in primary energy by 2025.

Source: REN21: *Renewable Energy Policy Network for the 21st Century*

It is important to note, however, that an RPS measure, by itself, does not ensure the financing required to implement a successful renewable energy program. It has to be

accompanied by 1) adequate FITs for smaller projects, 2) the availability of project grants, and 3) appropriate fiscal incentives. With the exception of China, Indonesia's RPSs not only compare favorably with, but are also more ambitious than, any of the other case study countries. In Indonesia's case, the targets are attainable if the right measures and incentives are put into place. An obstacle, however, is that Indonesia's corporate incentives may not be as robust as those of the case study countries.

3.2.4 Cap and Trade Programs or Emissions Trading Scheme

The Kyoto Protocol under the United Nations Framework Convention on Climate Change is a 1997 international treaty that came into force in 2005. Under the treaty, most developed nations agreed to legally binding targets for their emissions of the six major greenhouse gases. Emission quotas (known as "assigned amounts") were agreed by each participating Annex I country, with the intention of reducing overall emissions by 5.2% from their 1990 levels by the end of 2012. The United States is the only industrialized nation under Annex I that has not ratified the treaty, and is thus not bound by it.

The Protocol defines several mechanisms ("flexible mechanisms") that are designed to allow Annex I countries to meet their emission reduction commitments (caps) with reduced economic impact. Under Article 3.3 of the Protocol, Annex I parties may use GHG removals, from afforestation and reforestation (forest sinks) and deforestation (sources) since 1990, to meet their emission reduction commitments. They may also use international emissions trading. Under the treaty, nations that emit less than their quota will be able to sell assigned amount units to nations that exceed their quotas. It is also possible for Annex I countries to sponsor carbon projects that reduce GHG emissions in other countries. These projects generate tradable carbon credits that Annex I countries can use in meeting their caps. The project-based Kyoto mechanisms are the Clean Development Mechanism (CDM) and Joint Implementation (JI). The CDM covers projects taking place in non-Annex I countries, while JI covers projects in Annex I countries. CDM projects are supposed to contribute to sustainable development in developing countries, and also generate "real" and "additional" emission savings.

Emissions trading (the cap and trade scheme) is a "market-based" mechanism, where a limit on access to a resource (the cap) is defined and then allocated among users in the form of permits. Compliance is established by comparing actual emissions with permits surrendered including any permits traded within the cap. In emissions trading where GHG are regulated, one emissions permit or allowance is considered equivalent to one metric ton of carbon dioxide (CO₂) emissions. Other names for emissions permits are carbon credits, assigned amount units, and Certified Emission Reduction units (CER). These permits or units can be sold privately or in the international market at the prevailing market price. These trade and settle internationally and hence allow the transfer of allowances between countries.

Each international transfer is validated by the United Nations Framework Convention on Climate Change (UNFCCC) and within the EU is also validated by the European Commission.

Trading exchanges have been established to provide a spot market in permits, as well as futures and options markets to help discover a market price and maintain liquidity. Carbon prices are normally quoted in Euros per metric ton of carbon dioxide or its equivalent (CO₂e). Other greenhouse gases can also be traded, but are quoted as standard multiples of carbon dioxide with respect to their global warming potential. These features reduce the quota's financial impact on business, while ensuring that the quotas are met at a national and international level.

The cap and trade schemes in operations are as follows:

- The European Union's Emission Trading Scheme (EU ETS) is the first cap-and-trade program for reducing heat-trapping emissions, and is designed to help European nations meet their commitments to the Kyoto Protocol. This program includes 27 countries and all large industrial facilities, including those that generate electricity, refine petroleum, and produce iron, steel, cement, glass, and paper.
- The Regional Greenhouse Gas Initiative is a cap-and-trade program, the first US program formed in 2003 that covers a single sector – electricity generation – in nine northeastern and Mid-Atlantic States. The program aims to achieve a 10% reduction in emissions from power plants by 2018.
- Western Climate Initiative (WCI) was started in February 2007 by the governors of five western US states to evaluate and implement ways to reduce their GHG emissions and achieve related co-benefits. These states and future participants in the initiative also committed to set an overall regional goal to reduce emissions (set in August 2007 as 15% below 2005 emission levels by 2020), participate in a cross-border GHG registry to consistently measure and track emissions, and adopt clean tailpipe standards for passenger vehicles. By July 2008, the initiative had expanded to include two more states and four Canadian provinces. But by the end of 2011, however, only California and the Canadian provinces remained in WCI.
- WCI launched its Compliance Instrument Tracking System Service on 1 January 2013 for its cap-and-trade program, which uses a market-based mechanism to lower GHG emissions. California's program is second in size only to the EU ETS based on the amount of emissions covered. In addition to driving emission cuts in the ninth-largest economy in the world, California's program will provide critical experience in how an economy-wide cap-and-trade system can function in the United States. California's emissions trading system will reduce GHG emissions from regulated entities by more than 16% between 2013 and 2020. It is a central component of the state's broader strategy to reduce total GHG emissions to 1990 levels by 2020.
- The Swiss ETS started in 2008 with a four-year voluntary phase as an alternative option to the CO₂ levy on fossil fuels. Revised regulations entered in force on 1 January 2013. The scheme subsequently became mandatory for large, energy-intensive industries. Medium-sized industries may choose to voluntarily opt-in. In

the 2013-2020 mandatory phase, participants in the ETS are exempt from the CO₂ levy.

- The Tokyo Metropolitan Government Greenhouse Gas Emissions Trading Scheme launched on 1 April 2010, requires installations under the cap to reduce CO₂ emissions for the first phase (FY 2010 to FY 2014) by an average of 6% compared to the base year emission. Installations are defined as energy consumers equal to 1,500 kiloliters or larger per year (crude oil equivalent). It is expected that the second phase (FY 2015-FY 2019) will be capped at an average 17% reduction below base year emissions.
- The New Zealand ETS is a partial-coverage all-free allocation uncapped highly internationally linked emissions trading scheme. It was first legislated in the Climate Change Response (Emissions Trading) Amendment Act 2008 in September 2008 under the Fifth Labor Government of New Zealand and then amended in November 2009 and November 2012 by the Fifth National Government of New Zealand.
- The Carbon Pricing Mechanism in Australia, commonly referred to as a carbon tax, was introduced by the Gillard Government on 1 July 2012. It requires businesses emitting over 25,000 metric tons of CO₂ emissions annually to purchase emissions permits. The scheme directly affects approximately 300 “liable entities” representing the highest emitters in Australia. Australia plans to start a national cap-and-trade system with linkages to the EU ETS in July 2015.
- China has embarked on one of the largest endeavors in climate economics ever, to establish a national carbon emission trading system by 2015. As a first step, carbon-trading pilots have been initiated in seven provinces and cities. The success or failure of those experiments will to a large extent determine the future of climate policies in China. While the attempts to develop a domestic carbon trading scheme are sincere and ambitious, there are considerable difficulties. Many of the challenges are not particular to China, but are common to any emission trading system. But there are also more profound worries about how to operate a market-based instrument given the current shortcomings of the Chinese market system in general.
- Brazil enacted its National Climate Change Policy in December 2009 (Law No. 12.187/2009). This law sets out Brazil’s policy on climate change and aims to promote the development of a Brazilian market for emissions reduction. In this law, the country adopted a national voluntary commitment to reduce GHG emissions by 36.1% to 38.9% compared to business as usual projections for 2020, as communicated to the UNFCCC. As part of its activities under the World Bank’s Partnership for Market Readiness, the Brazilian Government is looking into market instruments that can be implemented to meet their voluntary GHG reduction commitment under the Policy and reduce overall mitigation costs. The options assessed include a domestic ETS, sectoral crediting approaches, and a carbon tax.

- South Korea will have an operational ETS by 2015, and South Africa will have a carbon tax in place in the same year. Sixteen developing countries, including Indonesia, China, India, Brazil, Thailand, Vietnam, Costa Rica, Jordan and Chile, are working with the World Bank's Partnership for Market Readiness to develop some form of ETS. The World Bank's initiative to help countries develop a carbon market mechanism now supports about 30 governments in Asia, Africa, and South America.
- The World Bank Partnership for Market Readiness will provide support to Indonesia for the development of its own ETS, since the country has already requested it. Developing countries sell the carbon credits to developed countries that signed mandatory GHG emission reductions as offsets.

Currently there are a number of global exchanges trading in UNFCCC-related carbon credits: the Chicago Climate Exchange, European Climate Exchange (both exchanges were acquired by the InterContinental Exchange, which is also acquiring NYSE Euronext), NASDAQ OMX Commodities Europe, PowerNext, Commodity Exchange Bratislava, and the European Energy Exchange. On 6 December 2007, the New Zealand Emission Unit Register was established. At least one private electronic market has been established: the CantorCO₂e in 2008. On 23 August 2010, Tokyo's carbon credits started trading on the Japan Climate Exchange. On 14 November 2012, California's Air Resources Board conducted its first quarterly auction through WCI's Compliance Instrument Tracking System Service. In December 2012, BVRio Green Stock Exchange started to trade Rio de Janeiro carbon credits. On 18 June 2013, Shenzhen Carbon Exchange started trading carbon credits.

Many companies now engage in emissions abatement, offsetting, and sequestration programs to generate credits that can be sold on one of the exchanges.

3.2.5 Feed-In Tariffs

As a step-up from the US PURPA Regulation [1978], a FIT is a policy mechanism designed to accelerate investment in renewable energy technologies. It achieves this by offering long-term contracts to renewable energy producers, typically based on the cost of generation of each technology. It is typically offered to smaller projects in selected target technologies while the larger projects are competitively bid.

A FIT is like an open call for unsolicited proposals, as it sets a fixed price for electricity generated from renewable energy sources. There are many variations on FIT design. The levels of support provided under FITs also vary widely and are affected by technology cost, resource availability and installation size and type (e.g., ground versus rooftop-mounted solar PV systems). Utilities are obliged to buy electricity at the government-set tariff from any independent power provider or developer capable of achieving financial close in their service area. Various measures are taken in some case study countries to offer higher FITs for very small projects, or for projects located in certain underdeveloped areas, and in some

cases, to enable a re-set of the FIT at some future date in an effort to motivate the developer to incorporate technological improvements in its plant.

The case study countries also vary in their use of FITs. For example, the US, China and India do not use FITs for hydroelectric power as that technology is deemed to be competitive with fossil fuels. Brazil and Bangladesh do not use FITs at all currently. None of the case study countries has published FITs for geothermal, except Indonesia, because these are relatively larger projects that are normally competitively bid. Bangladesh is in the process of addressing this shortcoming with support from USAID. Brazil appears to rely on very large hydro, solar or wind projects, all of which are competitively bid.

Table 11 provides a comparison of FITs for the case study countries and Indonesia for six technologies. It is noted that Indonesia does not yet have a FIT for solar or wind.

Indonesia is about to launch a tender for 150 MW of new solar PV energy. The tender is expected to be issued in July 2013, leading to the installation of at least 150 MW of new PV plants in the country. The tender will last for two months. Indonesia's Ministry of Energy and Mineral Resources is also planning a new FIT for solar. The new FIT will reportedly be up to \$0.25 per kWh, and will be applicable for 20 years.

Table 11
FITs Offered by Case Study Countries and Indonesia per kWh

	USA	China	India	Thailand*	Indonesia
Solar	\$0.06-0.54	\$0.154	\$0.342-.397	\$0.265	None
Wind	\$0.066-0.18	\$0.075-.90	\$0.339-.469	\$0.116-.149	None
Hydro	None	None	None	\$0.26-50	\$0.065-.10
Biomass	\$0.073	\$0.75	\$0.019-.505	\$0.010-.017	\$0.085-.140
Biogas	\$0.073-.15	None	\$0.019-.461	\$0.010-.017	None
Geothermal	None	None	None	None	\$0.105-.195

* Thailand offers a bonus for offsetting diesel in three southern provinces of \$0.030-.50 depending on the technology.

3.2.6 Corporate Tax Incentives

Apart from FITs, most of the case study countries provide tax exemptions, incentives or rebates (“grants” or “buy-downs”) to renewable energy projects. Although the US FITs are lower than those of the other case study countries, its fiscal incentives appear to be more generous. A very successful example is the US Production Tax Credit, which is a fiscal benefit in addition to the Investment Tax Credit and accelerated depreciation, extending a tax credit of \$0.019/kWh generated during the fiscal year to renewable generation. The basic idea behind the Production Tax Credit is to motivate investors to 1) optimize the technology during implementation to improve the reliability of generation and 2) ensure the incorporation of the latest generation enhancement technologies.

All of the case study countries provide some other combination of fiscal benefits, such as 1) sales tax exemptions for renewable energy equipment, 2) accelerated depreciation, and 3) investment tax credits. Moreover, in some cases the sale of bulk power from the generator to the distributor does not attract VAT. These measures are summarized in Table 12 and compared with Indonesia.

Table 12
Corporate Income Tax (CIT)-Related Incentives Offered
by Case Study Countries and Indonesia

	USA	China	India	Thailand	Indonesia
CIT Rate	35%	25%	32.45%	20%	25%
Tax Holiday	No	No	10- years, subject to 20% minimum alternative tax	8-years	None
CIT Reduction	No	Yes, 15%, all renewable energy technologies	No	Yes, 20%, all renewable energy technologies	Yes, for foreigners, 5% each year up to 6 years
PTC	Yes, all renewable energy technologies \$01.1-02.2/kWh	No	No	No	No
ITC	Geothermal, micro-turbine 10% solar, small wind and fuel cell 30%	10%	No	No	No
MACRS*	Yes, 5-year recovery	No	Yes, 80% solar and 15% wind	No	Yes
VAT Exemption	USA does not have VAT system	Waste residuals biodiesel oil recycled goods, MSW and 50% wind	No	Yes	Yes geothermal
Duties on imported equipment	No	Yes	Yes	Yes	Yes
R&D activity	Normal deductions against income	150% against CIT	200% against CIT	200% against CIT	Normal deductions

*MACRS is the acronym for the Modified Accelerated Cost Recovery System, US tax depreciation system.

3.2.7 Finance Infrastructure

Most renewable energy systems differ from conventional energy in terms of their cost structure. While conventional energy systems based on fossil fuels have moderate or low up-front capital costs, they are characterized by high operating costs due to their fuel consumption. In contrast, renewables generally have no fuel costs (sunlight, wind, geothermal heat, etc.), but require a fairly high initial investment.

Being risk-adverse, financial communities in most countries have been reluctant to invest in emerging renewable energy technologies. A combination of public and private financing mechanisms are important in overcoming market barriers to finance, especially when they are combined with monetary incentives (Table 13), along with good planning and coordination between national and provincial governments, public awareness campaigns, workforce training, and the creation of rules and standards for the installation and interconnection of renewable energy systems. Small-scale renewable energy projects, particularly, require considerable support.

In Indonesia, the issue of market acceptance may be exacerbated because small-scale renewable projects are not required to be tendered, which creates special problems. One of these is that unsophisticated investors with limited capital and experience are attempting to develop the smaller utility-interactive projects. In some instances, as has happened in other countries, their objective appears to be to lock up a site with all the relevant permits and, subsequently, sell off the rights to develop the site to some third party, thereby capturing some profit. The problem with this is that they end up locking the site, in some cases, indefinitely, to the overall detriment of the national policy.

An enormous number of unsolicited proposals, over 1,000 by some counts, are being promoted by developers who, in many cases, lack the capacity and financial wherewithal to develop their projects in a manner that facilitates the achievement of financial close. Moreover, and as mentioned previously, these proposals are reviewed at the local level by individuals who may not have the requisite capacity to make sound decisions.

It is believed that cluster tendering is a possible solution to this problem, i.e., tendering several promising and well-selected projects at once where bidders can bid for one, several, or all of the projects simultaneously. ICED and the Indonesia Infrastructure Guarantee Fund (IIGF) have discussed the possibility of collaborating on such a structure as a means of attracting higher-quality developers. During tendering, this would enable an in-depth examination and due diligence review of each bidder's qualifications, including financial capacity, teaming arrangements and level of experience.

Table 13 outlines the domestic public and private financing mechanisms available in the case study countries.

Table 13
Domestic Public and Private Financing Mechanisms in the Case Study Countries and Indonesia

	USA	China	India	Brazil	Thailand	Bangladesh	Indonesia
Grants	At the federal level, the DOE and USDA provide grants, as do some states	Yes, Stimulus for solar PV and wind energy in rural areas	Yes, IREDA	Yes	Yes, REDP and ENCON Fund	Yes, IDCOL, Green Fund	MCC Prosperity Fund
Loans	Local green bonds	Yes, China Development Bank and China Industrial Bank	Yes, IREDA, NCEF and Power Finance	Yes, BNDES, NORDESTE	Yes, REDP and ENCON Fund	Yes, IDCOL, Green Fund	Yes, IIF, MCC Prosperity Fund
Equity	Yes	Yes	Yes	Yes, through Electro-bras	Yes, and ENCON Fund	Yes, IDCOL, Green Fund	No
Cap and Trade Mechanisms	Yes, in California and the Northeast	Pending 2013	Yes	Pending 2013	Pending 2014	Under consideration	No
Private Venture Capital	Yes	Yes, CVCA	Yes, IVCA	Yes, ABVCAP	Yes, TVCA and ESCO Funds	Yes, first VC started in 2012	Yes, AMVI
Guarantees	Yes, through DOE and USDA	No	Pending, partial risk guarantee fund	BNDES for MSMEs	ESCO	No	Yes, IIGF

The table identifies those case study countries whose public and private sectors extend financial mechanisms to renewable energy project developers, including viability gap funding (grants), debt from public sector entities, and/or equity. In the case of the USA, which has long-standing experience with renewable energy, developers have access to loans, equity and private venture capital through private banks and other private financial institutions. In the emerging market countries, much more dependence is placed on public sector institutions. Nevertheless, in all the case study countries as well as in Indonesia, there appears to be access to private venture capital and equity funds.

In addition, as mentioned earlier, Bangladesh Bank has put into place a fund that acts as a rediscount facility available to commercial banks, so that debt providers (whether private or public) can rediscount their loans to eligible renewable energy projects at a fixed rate, with matching maturities and locked-in margins. This has the added benefit of reducing the risk profile of renewable energy projects for the lenders as liquidity and rate risks are eliminated.

Indonesia, whose focus on renewable energy has had a later start, has organized the IIGF and IIF under a holding company (a guarantee fund and a debt fund). But these funds can only be used to support tendered projects, and hence cannot deal with unsolicited proposals. The arrangements, as they stand, appear to address the larger, publicly solicited projects.

3.2.8 Other Measures

The above summary of financing available for renewable energy projects does not include the research and development efforts governments finance in order to assist with the commercialization of these technologies. For example, the US Government, through Sandia National Laboratories, a wholly-owned subsidiary of Lockheed Martin Corporation, and the National Renewable Energy Laboratory (NREL), owned and managed by the Alliance for Sustainable Energy, had budgets of \$2.4 billion and \$389 million in 2011. NREL was allocated over \$3.2 billion between 2002 and 2011. China and the US spent approximately \$1.9 billion and \$1.8 billion in 2012, while Brazil and India spent \$110 million and \$86 million in corporate and government funded R&D in 2012 according to Bloomberg New Energy Finance.

Moreover, Incentives that are provided to renewable energy are sometimes financed through so-called system benefits charges or public benefit funds, which are collected from electricity consumers through a small surcharge applied per kWh of consumption. These charges are collected into renewable energy and energy efficiency funds, which are then used to support these technologies. Thirty US states and the District of Columbia currently have some form of a public benefit fund (PBF) or system benefit charge program. Many state lawmakers began to adopt PBF policies in the late 1990s following the restructuring of the electrical industry. Policymakers see PBFs as a useful mechanism for funding energy efficiency, renewable energy, and low-income assistance programs and projects.

In addition, India has enacted a Carbon Tax of 50 INR/MT on coal produced and imported in 2010. On 1 April 2007, Boulder, Colorado implemented the US's first tax on carbon emissions from electricity, at a level of approximately \$7 per ton of carbon. These taxes are similar to, but usually much higher than, the system benefits charges. There are similar emission-related taxes, such as CO₂ or SO₂ taxes. These taxes are meant to correct a market failure that does not incorporate the external costs of fossil energy sources in the heat and electricity sectors. Such taxes, mostly used in Europe, make it easier for energy from renewables to compete in the marketplace.

Net metering is another policy that aids the development of the renewable energy industry. Similar to feed-in tariffs, it lets on-site producers feed excess electricity into the public grid at the retail price, and participants only pay the utility for whatever electricity they consume above their own production. Net metering is mostly used to support smaller, distributed energy systems that are installed on or near buildings. Thirty-nine US states have net metering rules, as well most of the case study countries.

In the United States, the home mortgage company Fannie Mae offers its customers an increased mortgage if they can show that their energy bills are reduced through renewable energy and energy efficiency measures. This leverages extra financing for those who could otherwise not afford to implement these measures in their homes. Likewise, many green investment funds and major energy producers have started investing in large-scale renewable energy projects. Also, the flexible carbon trading mechanisms under the Kyoto Protocol have leveraged new capital for renewable energy projects in developing countries that have ratified the Protocol, including the case study countries, and both governments and corporations provide capital to buy carbon credits from such projects.

4 Lessons Learned that May Apply to Indonesia

This report summarized the lessons learned from the USA, China, Brazil, India, Thailand and Bangladesh to assist Indonesia in stimulating commercially proven small- and medium-scale utility-interactive renewable energy generating projects. The emphasis was on five renewable generation target technologies: hydroelectric, biomass, solar, wind, and biogas.

This section first focuses on administrative, fiscal and financial initiatives that the GOI could take to stimulate small-scale, utility-interactive, renewable generation. It is followed by a description of measures Bank Indonesia could take to improve lending to this target market.

4.1 GOI Initiatives

According to REN21's *Renewables 2012: Global Status Report*, the GOI has set an aggressive RPS requiring that it and PLN produce or procure renewable energy for up to 25% and 15% of the country's primary energy and electricity production by 2025. Within these targets, there appear to be certain set-asides for each of the target technologies in addition to that set for geothermal technology (a 6.3% share of primary energy by 2025). The term "set-aside" or "carve-out" refers to a provision within an RPS that requires PLN to use specific renewable resources (geothermal, wind, solar, hydro, biofuel and biomass energy) to account for a certain percentage of primary energy by 2025, as noted in Table 10. It is not clear from this table whether this is a fixed set-aside or merely an estimate of what is possible. Nonetheless, the overall objective was relatively aggressive compared to those of other case study countries. Aggressive objectives require bold policies and measures in order to enable the objectives. However, it is currently not clear that the GOI will be able to move quickly enough to institute the measures necessary.

Subsidy Reductions

Subsidies on fossil fuels and electricity will need to be reduced fairly rapidly. As most policy makers have recognized, it does not make much sense to be paying subsidies for fossil fuels while, at the same time, paying preferential tariffs and/or extending corporate incentives to a significant percentage of those utility-interactive stations that generate electricity from renewable sources. The experience of Brazil in completely eliminating its subsidies to the fuel and electricity sectors in a relatively short period of time appears to be a case study that should be read by policy makers.

Cluster Tendering

Since most small utility-interactive projects will be implemented in outlying areas, policy coordination at the national, provincial, district and municipal levels needs to be

strengthened. As mentioned, developers face a variety of obstacles in obtaining the necessary permits to advance their project undertakings. At the same time, local officials lack the capacity to properly assess the unsolicited proposals that are being made. One way of attempting to resolve this problem is through a “selective reliance” PLN could place on provincial, district and municipal governments for “cluster tendering.” In this case, several projects would be prepared, approved and tendered at once, with tenderers given the option of bidding on all, some or just one of the projects being considered. This is likely to be a more fruitful alternative than encouraging unsolicited proposals from anyone who feels like submitting one. The recently proposed 150 MW solar PV tender, in coordination with a new FIT for solar PV, seems to be a cluster tender candidate for the wide displacement of solar PV in Indonesia.

While unsolicited proposals tend to exacerbate issues related to policy coordination or capacity at the local level, cluster tendering ameliorates these issues. It enables local governments to coordinate their efforts with each other and PLN during the planning and project preparation stages. For example, preliminary environmental clearance and/or other social safeguard issues can be dealt with early on with assistance from the MOE; management performance specifications and standards can be drafted and incorporated in the project agreement that is annexed to the tendering documents, including specifics with regard to reporting and corporate responsibility standards, and the timetables and conditions related to the issuance of permits and licenses can be divulged in the same documents.

The bidding criteria could be based on a parameter such as the tariff (with the proviso that bids cannot exceed the FIT) and/or the quality, experience and financial wherewithal of the bidders. By using competitive tension for the rights to the projects, the participation of the IIGF (through its guarantee mechanism), IIF and Pusat Investasi Pemerintah (PIP) (through their debt instruments) may be allowed or encouraged, if needed. Cluster tendering will lead to better-quality developers, will spread the risks of site selection as well as implementation, and may be key to “professionalizing” this segment of the market. In any case, it is a good way for local officials to “learn by doing.” As mentioned above, a test case of cluster tendering is emerging now that ICED and IIGF have promised to collaborate in putting one together.

FIT and Corporate Incentive Reviews

A sound review of the FITs and corporate incentives is needed to ensure adequacy. While the FITs adopted by PLN for geothermal, small-hydro and biomass appear adequate, a comparison of Indonesia’s corporate incentives with those of the case study countries (see Section 3.2.5) would suggest a need for reassessment.

Assistance to Small Projects

Indonesia does not offer PTC, ITC or grants to small projects, when they may be needed for the financial sustainability of the project. Some countries offer a variety of ITC and/or grant

programs to encourage the use and development of renewables and energy efficiency. Most programs offer support for a broad range of technologies, while a few focus on promoting a single technology, such as photovoltaic systems. Most PTC, ITC and/or grant programs are designed to pay down the cost of eligible systems or equipment, making unviable projects reasonably viable at the posted feed-in tariff. This is especially important in Indonesia, where developers are supposed to cover their own transmission costs to the grid. Projects that otherwise could be located in ideal sites have to be moved elsewhere because they are too far from the grid, making the project too expensive to implement.

To deal with this problem, government could move to create a public benefit fund. Most PBFs were developed by US states during the electric utility restructuring era, in the late 1990s, to ensure continued support for renewable energy, energy efficiency and low-income energy programs. These funds are commonly supported through a very small surcharge on electricity consumption (e.g., in the case of the USA \$0.002/kWh). This charge is sometimes referred to as a “system benefits charge.” PBFs commonly support rebate programs, grant and loan programs for renewable generation, research and development, and energy education programs. Such a fund would only benefit small utility-interactive projects.

Alternatively, if a PBF is not possible, such a fund could be created from the national budget savings that result from the gradual withdrawal of subsidies to fossil fuels or some other mechanism such as a carbon tax, as India and the US state of Colorado have recently implemented.

Emission Trading Scheme

Reportedly, Indonesia has requested assistance in developing an emission trading scheme from the World Bank’s Partnership for Readiness. Assistance may have commenced on 12 November 2010 with an expression of interest delivered by the country to the World Bank Group. An ETS, sometimes referred to as a “cap and trade,” is a market-based approach used to control pollution by providing economic incentives for achieving reductions in the emissions of pollutants. A central authority such as the MOE sets a limit or cap on the amount of a pollutant that may be emitted. The cap is allocated or sold to firms in the form of emissions permits, which represent the right to emit or discharge a specific volume of the specified pollutant. Firms are required to meet the cap or pay penalties for failing to do so. Alternatively, firms can purchase carbon emission reduction (CER) certificates equivalent to their emissions above the cap. Firms that need to increase their volume of emissions must buy CERs from those who emission levels are lower than those set by cap. The transfer of CERs is referred to as a trade. In effect, the buyer is paying a charge for polluting, while the seller is being rewarded for having reduced emissions. Thus, in theory, those who can reduce emissions most cheaply will do so, achieving the pollution reduction at the lowest cost to society. Renewable generation plants emit small levels or, in some cases, no emissions, and hence are entitled to CER certificates, for up to an amount which equals the cap allowed for the typical “reference” power station. The developer can sell the CERs in the open market to polluters who cannot meet the cap and hence use the proceeds as “equity”

or “special dividend” resulting from the start-up operations of the renewable power station. This becomes an important incentive which the GOI is not funding, but rather enabling through the newly created cap and trade market.

Renewable Energy Laboratory

The GOI, as far as the authors of this report are aware, has not yet considered creating a renewable energy laboratory, such as the USA’s National Renewable Energy Laboratory, which is exclusively dedicated to the research, development, commercialization and deployment of renewable energy. Most of the case study counties have an agency that is responsible for this kind of research and development. In Bangladesh, for example, the University of Dhaka has assumed this role. Alternatively, or simultaneously, a university or government ministry could create the Indonesian equivalent of the US database that describes each state’s practices, incentives and lessons learned in implementing renewable energy and energy efficiency. The latter is housed at North Carolina State University and paid for by the Department of Energy. Since Indonesia’s provinces and districts have considerable autonomy, the measures and incentives provided by each local government would be of great interest to other local governments with less experience.

Ban on Deforestation

As a closing note, according to the UN Framework Convention on Climate Change, Indonesia’s land use change and forestry emissions (largely from deforestation) as a percentage of GHG emissions in 2005 were 50.8%, which implies deforestation is a primary source of GHG emissions. In 2010, Norway embarked on a \$1 billion program to save Indonesia’s forests, which was renewed in May 2013. However, forests are still being cleared by fire on Sumatra, as recently evidenced by the haze in Malaysia and Singapore. The GOI needs to embark on a ban, or strict enforcement, of deforestation, possibly including jail time for infractions of the ban. New renewable energy additions bodes well for Indonesia’s renewable energy targets of the RPS of primary energy and electricity produced, but barely puts a dent in GHG emissions if deforestation is allowed to continue.

5.2 Bank Indonesia Initiatives

On the basis of the analysis conducted, it appears that Indonesia is implementing what is regarded as a generally favorable framework for attracting investment into renewables. In particular, the emerging IRR of Law 32 [2009], Environmental Protection and Management, if issued in the manner described in Section 2.2, will be of great benefit in creating awareness among financial institutions of the importance of assessing environmental issues and social concerns in prudential lending. That, in turn, is likely to be a key driver within the banking sector in understanding current and evolving renewable energy technologies for generating electricity, processing biofuels for transportation, and incorporating energy-efficient practices in corporate operations. All evidence points to the fact that China and

Bangladesh have enjoyed a tremendous interest in renewables since they passed green banking acts in 2010 and 2011.

In addition to the issuance of its aforementioned IRR, to further ensure prudent lending practices in Indonesia, Bank Indonesia should consider the following measures:

Debt Fund

Bank Indonesia could create a debt fund, as was done in Bangladesh, in order to enable commercial banks to borrow long-term, fixed-rate funds so they can offer renewable energy developers financing on the basis of locked-in margins and matched maturities. This is particularly important to developers implementing small-scale (less than 10 MW) utility-interactive power plants that rely on renewable feedstocks. A fund of this nature would have the advantage to the lender (as well as the developer) of reducing liquidity and interest volatility risks.

Capacity Building

Increasing the frequency of workshops, through ICED, would improve skills within the banking community on the extension of credit based on cash flow techniques as well as the identification, assessment and mitigation of risk with particular emphasis on “project structuring,” i.e., improving the risk profile of projects through the use of the international and domestic mechanisms and instruments described in Section 3.2.6. There does not appear to be a widespread inclination, or perhaps awareness, to use available instruments to mitigate risk and improve credit quality.

Information

There is an opportunity to improve access to information on renewable energy by creating a first-class library and online database that are readily accessible to banks, developers and other parties in BI offices throughout the country, or in suitable alternative locations.

Staffing

BI should ensure that a dedicated team, such as exists currently within the ICED Project, continues to assist lenders and developers once the USAID-sponsored project ends in September 2014.

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Annex A: Case Studies

A.1 United States of America

Socio-Economic Survey

According to the IMF, the United States is the largest and most technologically powerful economy in the world, with a real per capita GDP of \$49,922 and a real GDP of \$15,685 billion in 2012 for a population of 314 million. In this market-oriented economy, private individuals and business firms make most of the decisions, and the federal and state governments buy needed goods and services, predominantly in the private marketplace.

US business firms enjoy greater flexibility than their counterparts in Western Europe and Japan in decisions to expand capital plant, to lay off surplus workers, and to develop new products. At the same time, they face higher barriers to enter their rivals' home markets than foreign firms face entering US markets. US firms are at or near the forefront in technological advances, especially in computers and in medical, aerospace, and military equipment; their advantage, however, has narrowed since the end of World War II. The onrush of technology largely explains the gradual development of a "two-tier labor market" in which those at the bottom lack the education and professional/technical skills of those at the top. Since 1975, practically all the gains in household income have gone to the top 20% of households. Since 1996, dividends and capital gains have grown faster than wages or any other category of after-tax income.

Imported oil accounts for nearly 55% of US energy consumption. Crude oil prices doubled between 2001 and 2006, the year home prices peaked; higher gasoline prices ate into consumers' budgets and many individuals fell behind in their mortgage payments. Oil prices climbed another 50% between 2006 and 2008, and bank foreclosures more than doubled in the same period. Besides dampening the housing market, soaring oil prices caused a drop in the value of the dollar and a deterioration in the US merchandise trade deficit, which peaked at \$840 billion in 2008. The sub-prime mortgage crisis, falling home prices, investment bank failures, tight credit, and the global economic downturn pushed the United States into a recession by mid-2008. GDP contracted until the third quarter of 2009, making this the deepest and longest downturn since the Great Depression. To help stabilize financial markets, in October 2008 the US Congress established a \$700 billion Troubled Asset Relief Program (TARP).

Long-term problems include stagnation of wages for lower-income families, inadequate investment in deteriorating infrastructure, rapidly rising medical and pension costs of an aging population, energy shortages, and sizable current account and budget deficits, including significant budget shortages for state governments.

In 2010, the US produced and consumed 4.12 trillion kWh and 3.89 trillion kWh of electricity of which 75.5%, 9.9%, 7.7% and 4.8% are generated from fossil fuels, nuclear, hydro and other renewable energy. The US is a major importer of electricity from Canada at 45.23 billion kWh.

Legislative Initiatives Affecting Renewable Energy Development outside the United States

Negotiations on the Kyoto Protocol to the United Nations Framework Convention on Climate Change (UNFCCC) were completed on 11 December 1997, committing the industrialized nations to specified, legally binding reductions in emissions of six “greenhouse gases.” (The six GHG covered by the Protocol are carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulfur hexafluoride (SF₆). The most prominent of these, and the most pervasive in human economic activity is CO₂, produced when wood or fossil fuels such as oil, coal, and gas are burned.) The Protocol entered into force on 16 February 2005, and its emissions reduction targets are binding on the 35 industrialized countries that have ratified it. The United States disengaged from the Protocol in 2001 and has not ratified it. As structured in the negotiations completed in 1997, this treaty would commit the United States, if it were to ratify the Protocol, to a target of reducing GHGs by 93%, or 7% below 1990 levels during a commitment period from 2008 to 2012.

On 25 June 2013, President Obama mandated the Environmental Protection Agency to work with the states and develop a national renewable portfolio standard of reducing GHG emissions by 17% below the 2005 base year by 2020. This coincides with the second commitment period (2013-2020) of the Kyoto Protocol, where the EU is mandated to meet GHG emissions 20% below the 1990 base year by 2020. Further, he ordered the Interior Department to initiate enough private, renewable energy capacity on public lands to power more than 6 million homes by 2025; the Department of Defense, the largest energy consumer in the USA, will install 3 GW of renewable power on its bases.

Legislative Initiatives Affecting Renewable Energy Development in the United States

The US Congress has in the past taken actions that have set the stage for renewable energy development, including:

- ***Clean Air Act [1963] and Subsequent Amendments.*** Congress passed the first Clean Air Act (CAA) in 1963, creating a research and regulatory program in the U.S. Public Health Service. The Act authorized the development of emission standards for stationary sources, but not mobile sources. The 1967 Air Quality Act mandated the enforcement of interstate air pollution standards and authorized ambient monitoring studies and stationary source inspections. In the Clean Air Act Extension of 1970, Congress greatly expanded the federal mandate by requiring comprehensive federal and state regulations for both industrial and mobile sources.
- ***Clean Water Act and Preceding Legislation [1972].*** The Clean Water Act (CWA) of

1972 is the primary federal law in the US governing water pollution. It established the goals of eliminating releases of high amounts of toxic substances into water, eliminating additional water pollution by 1985, and ensuring that surface waters would meet standards necessary for human sports and recreation by 1983.

- ***Energy Policy and Conservation Act [1975] (EPCA).*** The EPCA responded to the 1973 oil crisis by creating a comprehensive approach to federal energy policy. Its primary goals are to increase energy production and supply, reduce energy demand, provide energy efficiency, and give the executive branch additional powers to respond to disruptions in energy supply. Most notably, the EPCA established the Strategic Petroleum Reserve, the Energy Conservation Program for Consumer Products, and Corporate Average Fuel Economy regulations.
- ***Public Utility Regulatory Policies Act [1978].*** This Act is a part of the National Energy Act. It is meant to promote greater use of domestic renewable energy. The law forced regulated electric utilities to buy power from other more efficient producers, if that cost was less than the utility's own "avoided cost" rate to the consumer; the avoided cost rate was the additional costs that the electric utility would incur if it generated the required power itself, or if available, could purchase its demand requirements from another source.
- ***Renewable Energy and Energy Efficiency Technologies Competitiveness Act [1989].*** The Act sets specific goals for the US on wind, photovoltaics, and solar thermal energy programs. For wind energy, the specific goals include, "improving design methodologies and developing more reliable and efficient wind turbines to increase the cost competitiveness of wind energy." Research efforts emphasized: "(i) activities that address near-term technical problems and assist private sector exploitation of market opportunities of the wind energy industry; (ii) developing technologies such as advanced airfoils and variable speed generators to increase wind turbine output and reduce maintenance costs by decreasing structural stress and fatigue; (iii) increasing the basic knowledge of aerodynamics, structural dynamics, fatigue, and electrical systems interactions as applied to wind energy technology; and (iv) improving the compatibility of electricity produced from wind farms with conventional utility needs." The legislation continues by specifying goals for the Wind Energy Research Program, "(i) reduce average wind energy costs to 3 to 5 cents per kilowatt hour by 1995; (ii) reduce capital costs of new wind energy systems to \$500 to \$750 per kilowatt of installed capacity by 1995; (iii) reduce operation and maintenance costs for wind energy systems to less than one cent per kilowatt hour by 1995; and (iv) increase capacity factors for new wind energy systems to 25 to 35% by 1995."
- ***Solar, Wind, Waste, and Geothermal Power Production Incentives Act [1990].*** This act is an amendment to the Public Utility Regulatory Policies Act (PURPA) of 1978 and the Federal Power Act. The main purpose is to remove the size limitations placed on renewable energy facilities, such as solar and wind, to receive PURPA

benefits. This will encourage the development of renewable energy production. The act specifies the qualification, application, and construction deadlines and requirements for the facilities. The act was amended in 1991, because of a technicality, to include renewable energy facilities of all sizes.

- ***Energy Policy Act [1992]***. EPACT replaced the National Energy Conservation Act (NECPA) and focused on the following issues: water conservation, Federal Energy Efficiency Fund, utility incentive programs, financial incentive program, demonstration of new technology, the General Services Administration's Federal Building Fund, energy savings performance contracts, energy audit teams, energy-efficient product procurement, U.S. Postal Service and Congressional building regulations, and fleet management. Title XII of this Act directly discusses renewable energy and amends the Renewable Energy and Efficiency Technology Competitiveness Act of 1989 by, "(1) implement a five-year program to further the commercialization of renewable energy and energy efficiency technologies by soliciting proposals for demonstration and commercial application projects; and (2) establish an Advisory Committee on Demonstration and Commercial Application of Renewable Energy and Energy Efficiency Technologies." It further instructs the Secretary to, "(1) prepare and submit to the Congress a three-year national renewable energy and energy efficiency management plan with specified contents; (2) establish a renewable energy export technology training program for individuals from developing countries; (3) make Renewable Energy Advancement Awards in recognition of developments that advance the practical application of certain renewable energy technologies; and (4) study and report to the Congress on whether certain conventional taxation and ratemaking procedures result in economic barriers to, or incentives for, renewable energy power plants compared to conventional power plants."
- ***Energy Policy Act [2005]***. The act was an attempt to combat growing energy problems. It changed US energy policy by providing tax incentives and loan guarantees for energy production of various types.
- ***Energy Independence and Security Act of [2007]***. The stated purpose of the Act is "to move the United States toward greater energy independence and security, to increase the production of clean renewable fuels, to protect consumers, to increase the efficiency of products, buildings, and vehicles, to promote research on and deploy greenhouse gas capture and storage options, and to improve the energy performance of the Federal Government, and for other purposes."
- ***American Reinvestment and Recovery Act [2009]***. The Act was a stimulus package in response to the late-2000s recession. The primary objective for ARRA was to save and create jobs almost immediately. Secondary objectives were to provide temporary relief programs for those most affected by the recession and invest in infrastructure, education, health, and renewable energy in extending tax incentives.

- **Clean Energy Standard Act [2012].** This act created a national, market-oriented standard for electric energy generation that stimulates clean energy innovation and promotes a diverse set of low- and zero-carbon generation solutions in the U.S. at the lowest incremental cost to consumers. Clean energy is defined to mean:
 - Electricity generated at a facility placed in service after 1991 using renewable energy, qualified renewable biomass, natural gas, hydropower, nuclear power, or qualified waste-to-energy
 - Electricity generated at a facility placed in service after enactment that uses qualified combined heat and power (CHP), generates electricity with a carbon-intensity lower than 0.82 metric tons per megawatt-hour (the equivalent of new supercritical coal), or as a result of qualified efficiency improvements or capacity additions at existing nuclear or hydropower facilities
 - Electricity generated at a facility that captures and stores its carbon dioxide.

Federal Investments in Renewable Energy

The United States Department of Energy (DOE) announced on 29 September 2008 that it would invest \$17.6 million, subject to annual appropriations, in six company-led, early-stage photovoltaic (PV) projects under the Solar America Initiative's "PV Incubator" funding opportunity. The "PV Incubator" project was originally designed to fund prototype PV components and systems with the goal of moving them through the commercialization process by 2010. The 2008 award was the second funding opportunity released under this initiative. Up to \$35.4 million was scheduled to be invested in these projects, with a cost share from industry of about 20%.

The Sun Shot Initiative was announced by the DOE in February 2011, with the objective of making solar energy cost-competitive with other forms of electricity by the end of the decade. The goal was to make the price of solar electricity reach about \$0.06 per kilowatt-hour by that timeframe. At that price, it will be cost-competitive with other non-renewable forms of electricity. This in turn will enable solar-generated power to grow from less than .05% of electricity supply, to roughly 14% by 2030 and 27% by 2050. The program name was derived from President Kennedy's "Moon Shot," the name he gave in 1962 to the stated objective of reaching the moon within the decade.

On 7 December 2012, the DOE announced a \$29 million investment in four projects that will help advance affordable, reliable clean energy for U.S. families and businesses. The \$29 million would be separated into two investments:

- \$21 million investment over five years to design plug-and-play photovoltaic systems that can be purchased, installed, and operational in one day

- \$8 million investment in two projects to help utilities and grid operators better forecast when, where, and how much solar power will be produced at U.S. solar energy plants.

Actually, the US Government has a long history of supporting R&D, not only in solar but also in hydroelectric and geothermal power generation.

For example, the Geothermal Technologies Office (GTO) researches, develops, and validates innovative and cost-competitive technologies and tools to locate, access, and develop geothermal resources in the United States. GTO works in partnership with industry, academia, and DOE's national laboratories on research and development activities focused on four areas:

- Enhanced geothermal systems
- Hydrothermal and resource confirmation
- Low-temperature resources
- Systems analysis.

Geothermal energy, a virtually untapped energy resource derived from the earth's heat, is more vital today than ever. It supplies clean, renewable power around the clock, emits little or no greenhouse gases, and takes a very small environmental footprint to develop. By developing, demonstrating, and deploying innovative technologies, GTO's efforts are helping stimulate the growth of the geothermal industry within the renewable energy sector and encouraging the quick adoption of technologies by the public and private sectors.

According to its website, the GTO's vision is to provide the nation with an abundant, clean, and renewable baseload energy source by:

1. Accelerating near-term hydrothermal growth by:
 - Lowering the risks and costs of development and exploration
 - Lowering the levelized cost of electricity (LCOE) to 6 cents/kWh by 2020.
2. Accelerating the development of 30 GWe of undiscovered hydrothermal resources by:
 - Securing supply with enhanced geothermal systems
 - Demonstrating 5 MW reservoir creation by 2020
 - Lowering the LCOE to 6 cents/kWh by 2030.

Federal Tax Incentives

The US Government, in the *Energy Policy Act of 1992*, provided Renewable Energy Production Tax Credits (PTC), Investment Tax Credits (ITC) and Treasury Cash Grants (TCG)

for eligible renewable energy projects. PTC is a per kWh credit for electricity generated from renewable sources, ITC is a percentage of capital expenditures and TCG is alternative to ITC. Eligible sources include wind, solar, biomass, geothermal, landfill gas, municipal solid waste and qualified hydroelectric and wave/tidal power facilities. The incentives are shown in Table A-1.

Table A-1
US Renewable Energy Tax Incentives, 2010

Tax Credit	Incentive	Eligibility	Expiration
PTC	\$22/ MWh for first 10 years of operation	Wind, closed-loop biomass, geothermal	Wind projects placed in service on or before 12/31/2012 (temporary extension). Other eligible technologies placed in service on or before 12/31/2013.
	\$11/ MWh for first 10 years of operation	Open-loop biomass, qualified hydro, landfill gas, municipal solid waste (MSW), marine and hydrokinetic power	Placed in service on or before 12/31/2013.
ITC	30% of qualified capital expenditures	Solar, small wind, and fuel cells (subject to maximum)	Placed in service on or before 12/31/2016, but solar reverts to 10% in 2016.
	10% of qualified capital expenditures	Geothermal, CHP, and micro-turbines subject to maximum	Placed in service on or before 12/13/2016, geothermal has no expiration.
TCG	30% of qualified capital expenditures for qualified solar, fuel cell (subject to cap), small wind, biomass, geothermal, landfill gas, waste, hydro, and marine and hydrokinetic facilities 10% of expenditures for geothermal heat pumps, microturbines (subject to cap) and CHP	ITC-eligible projects can qualify for cash grant. 30% credit for solar, fuel cells, small wind; 10% credit for geothermal micro-turbines and CHP	Construction must begin by 12/31/2011; submit an application before 10/01/2012 and placed in service before the PTC or ITC credit expires by 2014 (2013 for wind) or 2017.

Table A-1
US Renewable Energy Tax Incentives, 2010

Tax Credit	Incentive	Eligibility	Expiration
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Note: values stated for PTC are 2010 and are adjusted for inflation each year.

Source: World Resources Institute.

Also, there is Modified Accelerated Capital Recovery System, which is a system of rules and schedules for accelerated depreciation. A 5-year depreciation schedule is allowed for all ITC-eligible technologies as well as large wind projects. For some biomass property, the schedule is 7 years. The ITC maximum incentive for fuel cells and micro-turbines is \$3,000/kW and \$200/kW.

The PTC is nonrefundable, whereas the taxpayer's tax liability offsets the tax credit only (a dollar-for-dollar offset per year, whereas the refundable portion is in excess of tax liability or refund). The 2013 Obama Budget proposes to make the PTC permanent and refundable, although the magnitude has to be clarified. In the case of ITC, credits in excess of the current year tax may be carried forward to offset future taxes, with limitations. Therefore, younger companies without net taxable earnings have an extended period of time to claim the credit going forward. The original PTC expired on 31 December 2008. However, there have been extensions to 1 January 2013, but a new extension died and went to the House & Ways Committee.

Despite coming under fire for controversial green energy loans during his first term, President Obama's proposed 2014 budget would permanently extend tax credits for renewable energy production, energy efficiency, and advanced energy manufacturing. "To provide a strong, consistent incentive to encourage investments in renewable energy technologies and to help meet our goal to double generation from wind, solar, and geothermal sources by 2020," according to the White House, "the Budget would make permanent the tax credit for the production of renewable electricity." The budget dedicates \$23 billion for incentives for renewable energy production and energy efficiency programs over the next decade. An additional \$2.5 billion in tax credits would be given to companies that invest in advanced energy manufacturing projects, such as facilities for green energy manufacturing, bringing the total amount of clean manufacturing tax credits to \$4.8 billion when combined with credits from the stimulus package.

US States and Renewable and Alternative Energy Standards

The US Federal Government does not have a national renewable portfolio standard (RPS). Instead, twenty seven (27) states and the District of Columbia adopted their own RPS policies, 4 states adopted alternative energy standards, 7 states adopted either renewable or alternative standards, and 12 states have done nothing.

The basic features of each state-level RPS are similar. Electricity suppliers are required to purchase specified percentages of renewable energy, with each state adopting a distinct

definition of what constitutes renewable energy. Most states allow for flexibility by using tradable renewable energy credits (RECs). Some states' RPSs use tiered classification systems that provide increased credit levels for more expensive technologies. New Jersey and Pennsylvania, for example, place solar in a separate classification and mandate a strict procurement percentage from this more expensive resource. Many of these programs are in the early stages of development and conclusive data on their impact are not available.

Experience has demonstrated that a feed-in tariff is both the least expensive and the most effective means of developing renewable power. Investors need certainty, which they receive from a feed-in tariff. In 2010, the Federal Energy Regulatory Commission (FERC) ruled that states were able to implement above-market feed-in tariffs for specific technologies.

Other states have initiated regulations that stimulate specific technologies. For example, states that have passed RPS or renewable electricity standard (RES) laws have relied on the use of solar renewable energy certificates (SRECs) to meet state requirements. This is done by adding a specific solar carve-out to the state renewable portfolio standard (RPS). The first SREC program was implemented in 2005 by the state of New Jersey and has since expanded to several other states, including Maryland, Delaware, Pennsylvania, Ohio, Massachusetts, North Carolina and Pennsylvania.

A SREC program is an alternative to the feed-in tariff model popular in Europe. The key difference between the two models is the market-based mechanism that drives the value of the SRECs, and therefore the value of the subsidy for solar. In a feed-in tariff model, the government, or in the case of the United States, the state, sets the value for the electricity produced by a solar facility. If the level is higher, more solar power is built and the program is more costly. If the feed-in tariff is set lower, less solar power is built and the program is ineffective. The problem with SRECs is a lack of certainty for investors. A feed-in tariff provides a known return on investment, while an SREC program provides a possible return on investment.

State Cap and Trade Frameworks

Over a dozen states have adopted cap and trade policies restricting GHG emissions. Under the Regional Greenhouse Gas Initiative (RGGI), ten northeastern states had agreed to cap CO₂ emissions from power plants for six years, starting in January 2009. RGGI members include Maine, New Hampshire, Vermont, Connecticut, New York, Delaware, Massachusetts, Maryland and Rhode Island. Governor Chris Christie removed New Jersey in 2011. Several states and Canadian provinces act as observers. By 2020, participating states will be required to cut utility CO₂ emissions by 10% below 1990 levels. For comparison, the EU aims to reduce emissions to 20% below 1990 levels by 2020.

The cap on allowed emissions from regulated power plants was 165 million tons in 2013, but actual 2012 emissions were only 91 million tons. Emissions were lower than previously anticipated due to low natural gas prices, prompting a conversion to the lower-emitting

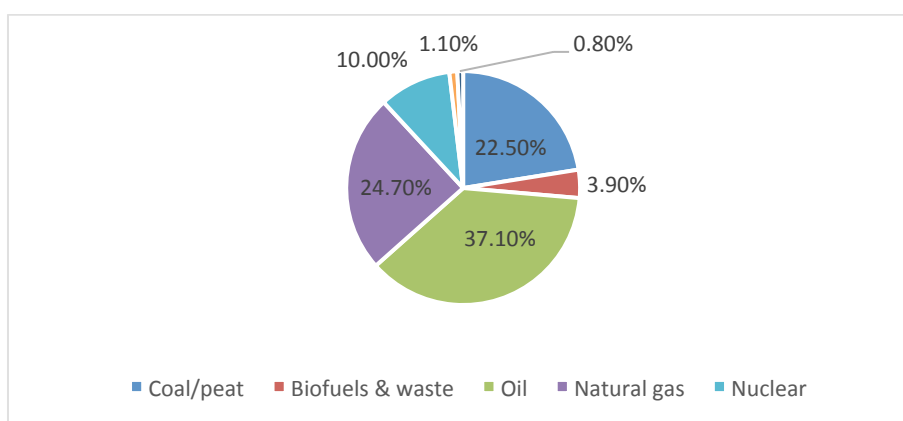
fuel, and to a lesser degree energy conservation and the recession. In February 2013, a major change in the RGGI cap was announced, lowering it to 91 million tons for 2014 with 2.5% annual reductions until 2020.

More recently, the Western Climate Initiative (WCI) was formed. Launched in February 2007, this program is a collaboration among the Governors of Arizona, California, New Mexico, Oregon and Washington to meet the regional challenges raised by climate change. It calls for a regional target for reducing GHG emissions through a cap and trade program by 2020 of 15% below 2005 levels. By July 2008, the initiative expanded to Montana and Utah and the Canadian provinces of British Columbia, Manitoba, Ontario and Quebec. On 18 November 2011, Arizona, Montana, New Mexico, Oregon, Utah and Washington left WCI. As of December 2011, the remaining WCI members are California and the Canadian provinces British Columbia, Manitoba, Ontario, and Quebec.

Specific Energy Technologies

According to the International Energy Agency, 2,163 million metric tons oil equivalent (MTOE) of the United States's total primary energy supply in 2009 was generated from fossil fuels (84.3%), nuclear (10%), and renewable energy (5.8%), as shown in Figure A-1:

Figure A-1
Total Primary Energy Supply by Source, 2009



Source: International Energy Agency Key World Energy Statistics, 2009

- **Wind:** At the end of 2012, the United States ranked second in the world in nameplate wind capacity, with 60,007 MW or 21.6% of the world total, while China ranked first. At the end of 2012, there were over 200,000 wind turbines operating, with a total nameplate capacity of 282,482 MW.

The US wind industry generates tens of thousands of jobs and billions of dollars of economic activity. Wind projects provide local taxes or payments in lieu of taxes, and strengthen the economy of rural communities by providing income to farmers with wind turbines on their land. Wind energy in many jurisdictions receives financial or other support to encourage its development. Wind energy benefits from subsidies in many jurisdictions, either to increase its attractiveness, or to compensate for subsidies received by other forms of production with significant negative externalities.

Wind energy has grown exponentially in the last decade, with an average increase of 29.7% per year. At this rate of growth of 29.7%, the U.S. would obtain 20% of its energy from wind by 2020.

- **Solar:** In 2012, 3,313 new megawatts (MW) of PV had been installed, which amounts to a 76% growth over 2011's total installed base of 4,383 MW, which itself was 73% over 2010's installed base of 2,528 MW. Growth projections indicate that upwards of 5,140 MW of new PV power will come online in 2013. Current trends indicate that a large number of PV power plants will be built in the south and southwest areas, where there is ample land in the sunny deserts of California, Nevada, Arizona, and even other states such as Louisiana. The United States has considerable solar power activity and many utility-scale solar power plants. The largest solar power installation in the world is the Solar Energy Generating Systems facility in California, which has a total capacity of 354 MW. The large growth in solar power throughout the US has been augmented by a number of legislative

and financial incentives taken in key states. A few examples of financial incentives are set forth below:

- The California Solar Initiative has set a goal to create 3,000 MW in the state of new, solar-produced, electricity by 2016. To assist in meeting that goal, the San Francisco Board of Supervisors in July 2008 passed financial incentives of up to \$6,000 of the purchase price of a homeowner system and up to \$10,000 for businesses.
- In 2008, the cities of Berkeley and Palm Springs, California initiated a revolutionary pilot program where homeowners were able to add the cost of solar panels to their property tax assessment and pay for them out of their electricity cost savings. Under this innovative arrangement, the two cities lend money to a homeowner for a solar system, to be repaid via an additional tax assessment on the property for 20 years. This allows the installation of the solar system at “relatively little up-front cost to the property owner.” The Property Assessed Clean Energy loan (or PACE, as it is referred to), is available in 28 states. Freddie Mac and Fannie Mae, the two largest mortgage guarantors, have objected to the repayment of solar loans being senior to mortgage loans, and some states have relegated PACE loans to junior loans. A house bill (HB) 2599 was introduced to prevent interference with the PACE program by other lenders, but it has not passed. The principal feature of the program is that the balance of the loan is transferred to the new owners in the event the property is sold, and the loan is paid for entirely through electric bill savings. Unlike a mortgage loan, no funds are transferred when the property is sold; only the repayment obligation is transferred. PACE programs are currently operating in California, Colorado, Florida, Maine, Michigan, Missouri, New York, and Wisconsin, and are on hold in many others, pending resolution of the Freddie Mac, Fannie Mae objection.
- New Hampshire passed a program for a \$3,750 residential rebate for a solar system up to 50% of its cost, for systems sized at less than 5 kWp (\$6,000 from 1 July 2008 until 2010).
- Similarly, the Louisiana legislature approved a 50% tax credit up to \$12,500 for the installation of a wind or solar system, regardless of size.
- In March 2013, Lancaster, California became the first U.S. city to mandate the inclusion of solar panels on new homes, requiring that “every new housing development must have at least 1 kilowatt installed per house.”

Other developments have also accelerated the development of solar renewable power.

In 2006, for example, companies began to be organized that offered a free solar panel installation to homeowners in return for a 25-year contract, or power

purchase agreement (PPA), to purchase electricity at a fixed price, normally below current electric rates. These investors then turned around and sold the electricity they purchased to third parties at a fixed price, pocketing the margin between the purchase and sale price. By 2009, over 90% of commercial PVs installed in the United States used a PPA, mostly by utilities under the RPS or by third party power traders. Approximately 90% of the PVs installed in the United States were in those states that not only set RPS standards but also specifically address the requirement for long-term PPAs.

- **Geothermal:** Geothermal energy in the United States continues to be an area of considerable activity. In 2012, the United States led the world in geothermal electricity production, with 3,386 MW of installed capacity. The largest group of geothermal power plants in the world is located at The Geysers, a geothermal field in California. The United States generates an average of 15 billion kilowatt hours of geothermal power per year, comparable to burning some 25 million barrels of oil or 6 million short tons of coal per year.

Geothermal power plants are largely concentrated in the western states. They are the fourth-largest source of renewable electricity in the United States, after hydroelectricity, biomass, and wind power. A geothermal resource assessment shows that nine western states together have the potential to provide over 20% of national electricity needs.

As of August 2008, 103 new projects are underway in 13 US states. When developed, these projects could potentially supply up to 3,979 MW of power, meeting the needs of about 4 million homes. At this rate of development, US geothermal production could exceed 15,000 MW by 2025. The most significant catalyst behind new industry activity is the Energy Policy Act [2005]. This Act made new geothermal plants eligible for the full federal production tax credit, which previously was available only to wind power projects and certain kinds of biomass. It also authorized and directed increased funding for research by the DOE for enhanced geothermal systems (EGS), and enabled the Bureau of Land Management to address its backlog of geothermal leases and permits.

The United States pioneered the first EGS effort, then referred to as hot dry rock (HDR), at Fenton Hill, New Mexico, with a project run by the government-owned Los Alamos Laboratory. It was the first attempt anywhere to make a deep, full-scale HDR reservoir, and efforts there spanned 1974 through 1992, in two phases. When it was ultimately terminated, EGS research languished until 2002. In 2002-2010, DOE supported several EGS projects in California, Nevada and Oregon, funded an important 2006 study by the Massachusetts Institute of Technology (MIT, discussed below), and supported AltaRock Energy, a private company in EGS technology development.

A 2006 report by MIT, funded by DOE, documents the most comprehensive analysis to date on the potential and technical status of EGS. The 18-member panel reached several significant conclusions:

- *Resource size:* The report calculated the US total EGS resources from 3-10 km of depth to be over 13,000 zettajoules (ZJ), of which over 200 ZJ would be extractable, with the potential to increase this to over 2,000 ZJ with technology improvements. This would be sufficient to provide all the world's current energy needs for several millennia. The report found that total geothermal resources, including hydrothermal and geo-pressured resources, to equal 14,000 ZJ – or roughly 140,000 times the total U.S. annual primary energy use in 2005.
- *Development potential:* With a modest R&D investment of \$1 billion over 15 years (or the cost of one 400 MW coal power plant), the report estimated that 100 GWe (gigawatts of electricity) or more could be installed by 2050 in the United States. The report further found that “recoverable” resources (accessible with today's technology) were between 1.2–12.2 TW for the conservative and moderate recovery scenarios, respectively.
- *Cost of electricity generated:* The report found that EGS could be capable of producing electricity for as low as 3.9 cents/kWh. EGS costs were found to be sensitive to four main factors: 1) temperature of the resource, 2) fluid flow through the system measured in liters/second, 3) drilling costs, and 4) power conversion efficiency.

The DOE's Geothermal Technologies Office (GTO) is committed to developing and deploying a portfolio of innovative technologies for clean, domestic power generation.

The U.S. Geological Survey (USGS) estimates:

- 9,057 MW of potential from domestic, conventional, identified geothermal systems.
- 30,033 MW of potential from conventional, undiscovered geothermal resources.
- 517,800 MW of potential from unconventional (high temperature, low permeability) enhanced geothermal systems resources.
- **Hydropower:** The first hydroelectric plant was completed on Niagara Falls, NY in 1881. By 1920 as 40% of the power produced in the United States was hydroelectric, the Federal Power Act was enacted into law. The Act created the Federal Power Commission to regulate hydroelectric power plants on federal land and water. As the power plants became larger, their associated dams developed additional purposes to include flood control, irrigation and navigation. Federal funding became necessary for large-scale development and federally owned corporations, such as the Tennessee Valley Authority (1933) and the Bonneville Power

Administration (1937) were created. Additionally, the Bureau of Reclamation, which began a series of western U.S. irrigation projects in the early 20th century, was now constructing large hydroelectric projects such as the 1928 Hoover Dam.

The U.S. Army Corps of Engineers was also involved in hydroelectric development, completing the Bonneville Dam in 1937 and recognized by the Flood Control Act [1936] as the premier federal flood control agency. Hoover Dam's initial 1,345 MW power plant was the world's largest hydroelectric power plant in 1936; it was eclipsed by the 7,600 MW Grand Coulee Dam in 1942, which is being upgraded to 10,080 MWs. Today, the US is the world's fourth-largest producer of hydroelectricity at 250 TWh (behind China, Canada and Brazil) and 79.511 GW of installed capacity, or 5.74% of its total electric capacity.

- **Biomass:** A variety of biomass feedstocks are currently used to generate electricity and produce heat and liquid transportation fuels. According to the EIA, biomass contributes nearly 3.9 quadrillion British thermal units (Btu) (quads) and accounts for more than 4% of total U.S. primary energy consumption.

Although biomass ranks well below petroleum, natural gas, and coal and is about one-half of nuclear, it surpasses other renewable sources. In 2009, the share of biomass in total U.S. energy consumption exceeded 4% for the first time. In the last 30 years, the share of biomass in total primary energy consumption has averaged less than 3.5% (EIA, 2010). However, there has been a gradual increase in biomass consumption that started in the early 2000s. This increase is due to ethanol production.

The electric power sector represents a small percentage of total biomass consumption (8%) and uses a variety of biomass feedstocks: fuel wood, municipal solid waste (MSW) biomass, MSW landfill gas, and bio-solids (or sewage sludge). In 2009, nearly 60% of biomass-derived electric power consumption was from MSW sources. Transportation accounts for 31% of the total US consumption of biomass, with ethanol used in gasoline blending accounting for almost 90% of the total. Biodiesel accounts for 8%, and the remainder is E85 (85% ethanol fuel) and other biomass liquids. The industrial sector accounts for 44% of total biomass energy consumption. Most of this amount, nearly 90%, is wood and wood waste. MSW, landfill gas, and biosolids account for the remainder.

A.2 People's Republic of China

Socioeconomic Survey

With approximately 1.35 billion people in 2013, China is the most populous country in the world, representing approximately 19.1% of the global population of 7 billion people. According to the IMF, its 2012 real GDP was \$8,227 billion and real GDP per capita was \$6,076.

China has a low population growth rate, which was 0.46% in 2012. This low growth is a result of the One Child Policy, which has been in effect since 1979. With this growth rate, which is expected to be stable in the next decade, the population is expected to reach approximately 1.4 billion in 2020. As a result of the current urbanization process (47% in 2010 and 2.3% annually 2010-15), up to 0.4 billion rural people are expected to move to urban centers in the future. The labor force of the country was 1 billion in 2012, or 74% of the total population. As a result of these demographic trends, China is now one of the most rapidly aging countries in the world at a median age of 36 years, in comparison to India (26.5), US (37), Japan (45) and Germany (45),

The global economy has been deeply affected by the unprecedented growth rate of the Chinese energy sector with its efforts to keep pace with the huge demands required for both industrial production and private consumption during the last decade. The electricity production of China is dominated by coal, which represents four-fifths, or 79%, of its electricity. According to the World Coal Association, China was the world's largest coal producer at 3,471 MT in 2011, of which 81.5% was steam coal, 14.5% coking coal, and 4% brown coal. China is the world's largest coal importer at 190 MT in 2011, of which 77% is steam coal. China holds the third-largest coal reserves (114.5 billion metric tons or 13.9% of the world's total reserves after the US and Russia).

China has also become the world's largest emitter of carbon dioxide at 8.32 billion metric tons in 2010. In 2012, China's installed generating capacity was 1.15 terawatts (69.5% fossil fuels, 21.8% hydro, 7.6% other renewable and 1.1% nuclear), while it produced and consumed 4.94 quadrillion watt hours and 4.69 quadrillion watt hours of electricity. China has the most rapidly expanding economy in the world, and being the most populous country as well, electricity demand growth has been particularly high. Electricity consumption in China has been growing by 12.5% annually in the last decade. Similar high growth is expected to continue, with total consumption reaching approximately 6,400 billion kWh by 2020. In the recent past, China has had the highest growth rate of electricity consumption per capita of the Brazil, Russia, India, and China (BRIC) countries; specific demand rose by about 12% annually between 2000 and 2008. Consumption in China was only 990 kWh per capita in 2000, but had climbed to 3,476 kWh per capita by 2011. This remarkable growth was mainly triggered by state-controlled infrastructure projects and by the continuous expansion of export-focused industries. China is expected to exhibit similar growth in the forecast period, resulting in electricity consumption of approximately 4,500 kWh per capita in 2020.

China has enormous coal reserves, while its oil and natural gas reserves are limited. Coal will likely retain its dominant role in electricity generation through to 2020, although due to environmental considerations and the spreading use of other energy sources, its share in the generation mix could slip to around 70%. In response to the country's energy saving and emission reduction targets, nuclear and hydro plants are expected to offer unprecedented opportunities.

As a result of the efforts made by the government over decades, China has a unique and relatively efficient regulatory and institutional framework in place. The approach of the government in the electricity sector is conservative, as energy is considered to be a strategic national commodity. Therefore, the current strong state control over the power sector is expected to remain in place, leaving only limited room for private investors to enter the market.

A huge proportion of China's population has not yet achieved high living standards, and tariffs are therefore a sensitive issue; subsidies remain large in some cases. The main priorities for the regulator in the future are to:

- Support the implementation of energy saving technologies which directly influence the prices of energy and electricity.
- Continue to boost the application of energy-saving technologies in the residential and commercial sectors.
- Support the construction of new power plants using highly efficient supercritical and ultra-supercritical technologies to replace obsolete, small-scale generation facilities.
- Transform the grid to provide more space for the development of the power generation sector.
- Set standard rules for market players that better serve their interests to substantially ease the difficulties of foreign enterprises seeking to enter the Chinese market.
- Draw up policies and continue the macro-control of the electricity industry.
- Focus on electricity pricing, as this influences the decision-making of all electrical enterprises and their investment priorities.
- Reduce subsidies to international standards in order to make prices more cost reflective.

Legislative Initiatives Affecting Renewable Energy Development in China

- ***The 11th Five-Year Plan (2006–10)*** set an energy-savings target of 20%, and the

country has adopted administrative, legal, and economic measures to achieve this goal. During the first three years of the plan, China's energy intensity— its energy consumption per unit of GDP—fell by just over 10%, saving 290 million tons of coal equivalent and reducing the country's greenhouse gas emissions by 750 million metric tons of carbon dioxide-equivalent.

- ***Renewable Energy Law, 1 January 2006 and Amended in 2009 and 2011.*** This law was enacted to promote the development and utilization of renewable energy, improve the energy structure, diversify energy supplies, safeguard energy security, protect the environment, and realize the sustainable development of the economy and society. Renewable energy in this law refers to non-fossil energy: wind, solar, water, biomass, geothermal, and ocean. The Government encourages and supports various types of grid-connected renewable power generation and power purchase agreements from renewable energy providers, non-grid rural development and solar energy in workplaces and residence,s as well as the development of biomass (natural plants and organic waste), energy crops (herbage and wood used for energy) and biological liquid fuels (methanol, ethanol, bio-diesel and other liquid fuels derived from biomass resources). The government budget establishes renewable energy development funds to support the following:
 - Scientific and technological research, standard establishments, and pilot projects for the development and utilization of renewable energy
 - Construction of renewable energy projects for domestic use in rural and pasturing areas
 - Construction of independent renewable power systems in remote areas and islands
 - Surveys, assessments of renewable energy resources, and the construction of relevant information systems
 - Localized production of equipment for the development and utilization of renewable energy.

Financial institutions may offer preferential loan with financial interest subsidies to renewable energy development and utilization projects that are listed in the national renewable energy industrial development guidance catalogue and conform to the conditions for granting loans. The government grants tax benefits to projects listed in the renewable energy industrial development guidance catalogue, and specific methods are to be prepared by the State Council.

In 2009, the 2006 Renewable Energy Law underwent several amendments. First, the new regulation legally binds electricity grid companies to buy all of the renewable electricity generation and guarantees priority power dispatching to power produced

from renewable sources. Grid companies are simultaneously expected to improve transmitting technologies and enhance grid capacity to further facilitate the integration of electricity from renewable sources. In the case of non-compliance with the imposed electricity purchasing mandate, the regulation initiates a penalty system. Responsive companies are required to pay a fine that equals double the economic loss experienced by renewable electricity producers. Second, the amendment gives the State Council's energy and finance departments, in collaboration with the state power regulatory agency, the responsibility for designing annual renewable energy power generation targets. Third, the amendment to the 2006 Renewable Energy Law initiates the Special Fund for Renewable Energy that will finance research and development and support mini- and off-grid renewable electricity generation projects in rural and remote areas. In addition, the Special Fund will act as a central mechanism allocating government funding and redistributing the Renewable Energy Premium, a subsidy set up in 2006 to balance the extra cost of integrating renewable-sourced power.

In 2011, The 2006 Law then included the Renewable Energy Surcharge, a fixed tariff added to the price of each kWh of electricity sold through the grid. The revenue generated by this premium is then to be split among power distributors and utilities to balance the higher price they have to pay for electricity from renewable energy sources. In 2006, the renewable energy premium amounted to Chinese Yuan (CNY) 0.001/kWh, equivalent of \$0.00015. Successive revisions increased it to CNY 0.004/kWh in 2009 to keep pace with the dramatic increase in electricity generation from renewable sources. In December 2011, the subsidy for on-grid price of electricity generated from renewable energy sources increased to CNY 0.008/kWh, equivalent of \$0.0012.

- ***Medium and Long-Term Development Plan for Renewable Energy, September 2007.*** The 2007 Plan calls for the investment of 2 trillion Yuan (~\$263 billion) before 2020 on renewable energy development in China. Including hydroelectric power, China's renewable energy output equals approximately 8% of all energy consumed in the country. The Plan calls for that percentage to rise to 10% by 2010 and 15% by 2020. By 2020, the Plan calls for the development of a total of 300,000 MW of hydropower, 30,000 MW of wind power, 30,000 MW of biomass, 1,800 MW of solar power, 300 million square meters coverage of solar hot water heaters, 44 billion square meters of methane gas per annum, and 50 million tons per year of bio-fuels. China will not affect food security in developing bio-fuels, but its emphasis will be on developing non-grain plants to be converted into bio-fuels.
- ***The Twelfth Five-Year Plan for Renewable Energy (2011-2015).*** China's total consumption of renewable energy will reach 478 million metric tons of coal equivalent (MTCE) by the end of 2015, which accounts for 9.5% of the total energy consumption. Also, development indicators for different types of renewable energy are outlined, as follows:

- The installed capacity of hydropower will reach 260 GW
- Grid-connected wind power will reach 100 GW
- Pumped storage hydropower capacity will reach 30 GW
- Solar PV power will exceed 21 GW
- Biomass will reach 13 GW
- Geothermal, tidal power capacity will reach 110-120 MW
- Ocean power will reach 50 MW
- Nuclear will reach 40 GW.

Beijing's 12th Five-Year Plan on Renewable Energy outlines targets for renewable energy, and how they fit into Beijing's economic development. By 2015, the city's renewable total energy consumption will reach 5.5 Mtce, accounting for 6% of the city's total energy consumption.

Central Government Investment in Renewable Energy Development

According to the Bloomberg New Energy Finance Unit's *Global Trends in Renewable Energy Investments 2013*, global new investment in renewable energy in 2012 decreased to \$244.4 billion or 12% from 2011. China was responsible for over 27% of the total new investment volume, spending \$66.6 billion on renewable energy in 2012, a 22% increase over the 2011 level. In addition, \$33 billion was invested in large hydro (>50 MW) in 2012 (22 GW), of which China accounted for 63.6% or 14 GW.

- *Brightness Program [1996]*. This umbrella program includes the Township Electrification Program (TEP) (implemented 2002-2003), and the Village Electrification Program. The TEP included 20 MW of solar PV and wind, and 200 MW of small hydro to provide electricity for more than 1,000 townships (the official statistic is 1 million people total). To date, the central government has invested \$240 million to provide hardware. Thus, it is designed to allow the development of rural communities as well as reducing poverty. The total investment in equipment and services needed to achieve the long-term goal of reaching 23 million people is about CNY10 billion (about \$1.2 billion). The governments of Holland and Germany are giving further financial and technical support to the program.
- In 2007, the Ministry of Finance created the Special Fund for the Industrialization of Wind Power Equipment. The institution allocates funding to wind project assessments and evaluations, related technology research and development, and the construction of pilot demonstration projects. The fund also supports the production of new wind turbine equipment. Subsidies of CNY 600/kWh (\$87.41/kWh) are provided to domestic companies for the first 50 new turbines (with a minimum capacity of 1.5 MW) they produce. The 2009 Building Integrated Solar PV Program provides upfront subsidies for grid-connected rooftop and building integrated solar PV (BIPV) systems. Systems of a minimum peak capacity of 50 kW benefit from a capital premium determined annually by the government. The level of subsidies for 2009 to 2012 declined as follows: 2009: CNY 15/W, 2010: 13/W (\$ 1.95/W

equivalent) for grid-connected rooftop systems and to CNY 17/W (\$2.55/W equivalent) for BIPV, 2011: CNY 9/W, and 2012: CNY 7/W.

- *China Development Bank (CDB)*. To achieve the target of energy conservation and emission reduction under the 12th Five-Year Plan, the Bank increased support for the comprehensive upgrading of coal-fired power plants. CDB provided financing for a number of projects such as the coal-to-gas project at Tianjin Chentangzhuang Thermal Power Plant, the Jilin Fengman Hydropower Station, the Guanyinyan Hydropower Station on the Jinsha River of Yunnan Province, the acquisition of Hong Kong Qingshan Power Generation Company by China Southern Power Grid, and Shanxi Pinglu 200 MW Wind Power Generation. In 2012, new lending to the electric power sector amounted to CNY 88.5 billion, with an outstanding receivable balance of CNY 721.5 billion at year-end.

The Role of Tax and other Incentives

The Chinese government's support for renewables in China includes reduced corporate income taxes, significant reductions in VAT, other tax incentives, FITs, R&D incentives, and subsidies for energy conservation technologies improvement.

Corporate Income Tax (CIT)

- A reduced CIT rate of 15% is given for qualified advanced and new technology enterprises. Applicable fields include solar energy, wind energy, biomaterial energy, and geothermal energy.
- The Clean Development Mechanism (CDM) Fund is exempted from CIT on the following income: the portion of carbon emissions reductions (CERs) proceeds that are shared by the government or grants from international financial organizations or interest income derived from capital deposits or national bonds or donations from domestic and foreign entities or individuals.
- Enterprises operating CDM projects are allowed to deduct, before CIT, the CER proceeds that are shared by the government.
- Three years' CIT exemption is followed by another three years' 50% reduction of CIT rate for income derived from specified CDM projects including hydrofluorocarbons (HFC), perfluorocarbons (PFC), and nitrous oxide (N₂O) projects, starting from the year in which the revenue from the transfer of greenhouse gas emission (GHG) reductions is first received. According to the new Administrative Measures Governing the Operation of CDM Projects in 2011, any project companies, except for 41 state-owned enterprises, shall apply for approval from the National Development and Reform Commission (NDRC) at the provincial level first, which would then submit preliminary review opinions to the central NDRC for further review. (According to the old measures, all CDM project companies applied directly to the central NDRC for approval.) The new measure also changes the sharing percentage in

the proceeds from the transfer of emission reductions units between the State and companies involved in N₂O and PFC projects.

- Three years' CIT exemption is followed by another three years' 50% reduction of the CIT rate for income derived from qualified environmental protection and energy or water conservation projects, starting from the year in which the first revenue is generated. Applicable fields include biomaterial energy, synergistic development and utilization of methane, and technological innovation in energy conservation and emission reductions.
- Ten percent of the amount invested in the qualified equipment is credited against CIT payable for the current year, with any unutilized investment credit eligible to be carried forward for the next five tax years if such equipment is qualified as special equipment related to environmental protection, energy, or water conservation and production safety.
- Only 90% of the revenue derived from the transaction is taken into account for CIT computation purposes – if such revenue is derived from the use of specific resources associated with the synergistic utilization of resources as raw materials in the production of goods.
- A deduction is given of 150% of qualified R&D expenses incurred for CIT computation purposes.

Value Added Tax (VAT)

- A 50% refund of VAT is paid on the sale of wind power.
- A 100% refund of VAT is paid on the sale of biodiesel oil generated by the utilization of abandoned-animal fat and vegetable oil.
- VAT paid on the sale of goods produced from recycled materials or waste residuals is refundable.
- VAT is exempt on the sale of self-produced goods including recycled water, qualified powdered rubber made out of obsolete tires, re-trodden tires and certain construction materials made from waste residuals (with a minimum percentage of 30%).
- VAT is exempt on sewage treatment, garbage disposal and sludge treatment services.
- VAT is refundable on sale of recycled resources from 1 January 2009 to 31 December 2010. The refund rates for 2009 and 2010 were 70% and 50%, respectively.

- In November 2011, the government authority expanded the scope of sales of self-produced goods/ products by using the prescribed recycled materials, waste residuals and agricultural residuals that are eligible for VAT refunds at rates ranging from 50 to 100% of the VAT payable. The rates may vary depending on the nature of recycled materials or residuals utilized.

Financial Subsidies and Tax Incentives Available to Energy Performance Contracting

- Financial subsidies will be granted by the central and provincial government. The standard rate of subsidies at the central level is CNY 240 per ton of standard coal saved. The standard rate at the provincial level is no less than CNY 60 per ton of standard coal saved. In March 2011, the NDRC and Ministry of Finance jointly announced the second batch of qualified energy service companies (ESCOs) consisting of 523 entities. These companies can apply for financial subsidies on energy conservation management contracts entered into on or after 1 January 2012. (The first batch of qualified ESCOs was announced in August 2010 covering 461 entities). These financial subsidies have been rolled out under the jurisdiction of energy performance contracting (EPC). However, such financial subsidies should be taxable with an ESCO for CIT purposes. We are not aware of any prevailing rules that grant exemptions on such subsidies.
- A qualified ESCO taking part in an EPC project will be eligible for a tax exemption in the first three years and a tax reduction by half (an effective rate of 12.5%) over the next three years, starting from the tax year in which the revenue from the project is first realized.
- An enterprise that invests in special equipment for energy conservation will obtain a credit against its tax payable that equals 10% of the investment amount in the year in which the investment is made. Where there is not sufficient tax payable to absorb the credit in the year, the excess credit may be carried forward for up to five tax years.
- A qualified ESCO taking part in an EPC project will be provisionally exempt from the business tax on revenues received in respect of the project.
- A qualified ESCO taking part in an EPC project will be provisionally exempt from the VAT on the transfer to the energy user of goods related to the project.
- When, at the end of the term of the energy management contract (EMC), the ESCO transfers to the energy user the assets that have materialized in the course of executing the EPC project. The ESCO can do so as if these assets had been fully depreciated or amortized for CIT purposes. In the same way, when the energy user receives the project assets from the ESCO, the energy user can do so as if these assets had been so depreciated or amortized.

- When the ESCO transfers the project assets to the energy user at the end of the term of the EMC, the ESCO will not have to recognize any revenue to take into account the contributions the energy user has made to the price of the assets.
- An energy user in an EPC project can deduct reasonable expenses actually incurred in accordance with the EMC as and when they are incurred for CIT purposes. There is no need to differentiate between service fees and asset prices in claiming such a deduction.

Feed-In Tariff

With the Renewable Energy Law as revised in April 2010, the State Bureau of Energy and other departments of the State Council will promulgate guidelines on the full purchase of electricity generated by new energies. According to the revised law, the price of on-grid electricity generated by renewable energies shall be determined by the competent price department of the State Council. The council will consider the difference in areas and the electricity generated by different types of renewable energy companies.

- ***Wind Energy.*** In July 2009, China set a fixed feed-in tariff for new onshore wind power plants in a move that will help struggling project operators to realize profits. The NDRC, the country's economic planning agency, announced four categories of onshore wind projects, which according to region will be able to apply for the tariffs. Areas with better wind resources will have lower feed-in tariffs, while those with lower outputs will be able to access more generous tariffs. The tariffs per kilowatt hour are set at CNY 0.51 (\$0.075), CNY 0.54, 0.58 and 0.61, corresponding to different wind resources in different regions.
- ***Solar Energy.*** In August 2011, NDRC announced the first nationwide feed-in tariff (FIT) incentive program applicable to solar power. The NDRC FIT Circular (Fa Gai Jia Ge [2011] No. 1594) provides for the following schemes to encourage the development of solar photovoltaic power generation projects in China:
 - A FIT of CNY1.15 (approximately \$0.177) per kwh for projects that 1) are approved for construction by the NDRC prior to 1 July 2011, 2) complete construction and commence production of electricity prior to 31 December 2011, and 3) have not yet been verified by the NDRC in respect of its on-grid power tariff.
 - A FIT of CNY 1.00 (approximately \$0.154) per kWh for projects that 1) are approved for construction by the NDRC after 1 July 2011, or 2) are approved for construction by the NDRC prior to 1 July 2011 but have not yet commenced production of electricity by 31 December 2011. (Exceptions are projects in this category but located in Tibet, where an FIT of CNY 1.15 (approximately \$0.177) per kWh will apply).

- The NDRC shall have the right to make adjustments to the FIT going forward, based on factors such as investment cost changes, technology development, etc.

Before the launch of the FIT, the NDRC had been implementing a “concession rights auction” regime in granting development rights for large solar projects in China. The auction winners were usually those offering the lowest grid power prices, almost invariably large state-owned enterprise power providers. Further, there has been a tendency in the past couple of years among project developers to recklessly lower their bidding prices in order to win projects for the mere sake of market expansion, regardless of the returns that could realistically be achieved. It is expected that the FIT incentive program will effectively help improve the competition situation in the industry, as it provides reasonable room for profit to developers with relatively advanced technologies. One would expect to see an increase in solar power investment in China in the coming year as NDRC’s implementation of the FIT becomes clearer. On the other hand, the new NDRC FIT Circular specifically provides that solar power projects selected via the auction process shall not enjoy a grid power price higher than the FIT.

This provision casts some uncertainty on the future of the auction regime. While the previous projects put by NDRC to auction had been very attractive in respect of the project sizes and solar resources, understandably it would not be that attractive to compete and bid for a price that can be at best equal to the FIT. It is therefore yet to be seen how the existing auction regime for large-scale solar projects will evolve.

- **Biomass Energy.** The 2006 Renewable Energy law introduced a CNY 0.25/kWh, equivalent to \$3.7 cents/kWh, feed-in premium for biomass generated electricity. This capital premium is added to province-specific prices of coal-generated power, and is applicable over a 15-year period. In 2010, the National Development and Reform Commission announced a new national feed-in tariff for biomass power of CNY 0.75/kWh, equivalent to \$0.11/kWh. Moreover, the Chinese Government published a target for installed biomass capacity of 30 GW by 2020.

Financial Funds/Allowance

Special funds are made available to facilitate the development of renewable energy relating to the following activities:

- Scientific and technical research, standardization processes and model engineering projects
- Renewable energy projects in rural and pastoral areas
- Construction of stand-alone electricity generation systems in remote areas and islands
- Renewable energy resource surveys, evaluation and construction of information systems
- Localization of manufacturing facilities used in the renewable energy sector.

The special funds may also be deployed to compensate as additions to the revenues of electricity generation from renewable energy resources, to recover the cost deficiencies suffered by renewable energy plants and indirectly borne by the grid for the purchase of electricity from renewable energy plants. Applicants may apply for such funds with the local finance bureaus and the government agencies in charge of renewable energy projects.

Financial Subsidies for Energy Conservation Technologies Improvement

During the 12th Five-Year Plan [2011-2015], the central government will continue to arrange special subsidies to support projects to improve energy conservation technologies. In order to achieve optimum energy conservation goals, the financial subsidies are closely linked to the quantity of energy conserved on a project basis. The project companies are granted financial subsidies if they fully complete the expected goals of energy conservation. For projects in the eastern regions of China, companies may be granted a one-off reward subsidy of CNY 240 per ton of standard coal based on the annual energy consumption after the completion of the projects; for projects in the central and western regions of China, one-off reward subsidies are CNY 300 per ton of standard coal.

Provincial Incentives

In addition to the national incentives, provincial and local governments provide incentives:

- ***Shandong Province.*** In 2007, the Government of Shandong Province established a CNY 2.133 billion fund to support energy conservation and emissions reductions. Drawing on this fund, the provincial government will allocate funds to subsidize the construction and adoption of solar hot water supply systems by hotels, schools and other establishments. In 2007, a total of 15 million square meters of buildings in Shandong had a solar hot water heating system. In January 2008 it was estimated that 99% of city center residents used solar power for water heating. In January 2008, Shandong Province announced the implementation of its One Million Rooftops Sunshine Plan, designed to stimulate the integration of various renewable energy sources into building construction. The Plan targets incorporating solar and geothermal power into buildings. Following this, compulsory regulations went into effect in the cities of Yantai and Jinan for the integration of solar energy in the construction and design of certain buildings. The Shandong Province Village Renewable Energy Regulations took effect on 1 January 2008, providing subsidies for specified renewable energy technologies in farming villages. The regulations require that governments at the county level and higher incorporate special funds into their yearly budgets, which will be used to subsidize renewable energy facility construction in farming villages. Projects eligible for subsidies include the production of methane gas from animal and agricultural waste, as well as garbage. The methane can be used to produce power directly. Subsidies will also be provided to villages that use solar power for heat and hot water in new or renovated public buildings,

such as schools, hospitals and old-age homes. The regulations apply to any non-fossil fuel energy source, including solar, wind, geothermal, hydro, biomass and ocean wave energy.

- **Hainan Province.** In August 2007, the Government of Hainan Province announced its Plan for the Construction of Wind Farms. The Province will construct a total of 1234.5 MW of wind power in 12 locations over several years. The Plan envisages that four to six wind farms, each having a 250-300 MW capacity, will be operating by 2010. Wind power generating capacity is to reach 400 MW by 2015 and 600 MW by 2020.
- **Guangxi Province.** On 9 January 2008, the Government of Guangxi Province launched the 1 Million Mu Bio-Fuel Forest Project. (Mu is a Chinese unit of area; 1 Mu = 1/15 hectares.) The plan calls for Guangxi to develop a 1 million Mu (approximately 66,666.7 hectares) of jatropha tree forest (also known as small tung oil tree). The forest is estimated to yield over 160 000 MT of biodiesel, worth CNY 640 million. The Guangxi Zhilian Renewable Energy Co. Ltd. and the Guangxi Forestry Science and Technology Institute entered into an agreement, with CNY 500 million to be invested in the development of the jatropha forest, and CNY 200 million in the construction of a biodiesel refinery and production lines for related products. The trees will be planted on lands not suitable for crop cultivation, yielding fruit every 30 years. Every 1 Mu of jatropha trees will yield approximately 270 kg/mu of seeds with 60% or more extractable oil content, yielding 50kg/mu or more biodiesel after refining. Besides Guangxi Province, Sichuan, Yunnan, Guizhou and Fujian Provinces have begun developing jatropha forests.

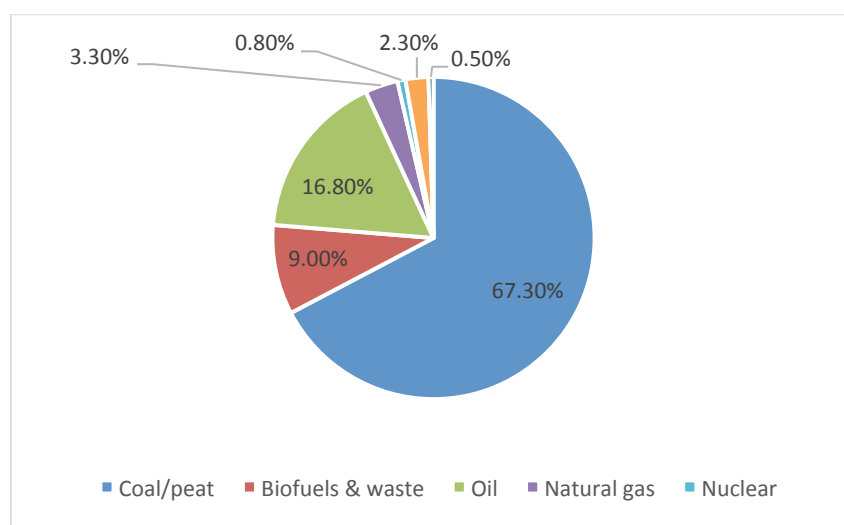
Cap and Trade Activity

China has moved forward with a pilot cap and trade system covering seven regions, including the manufacturing hub provinces of Guangdong and Hubei, as well as the cities of Beijing, Shanghai, Tianjin, Chongqing, and Shenzhen. The first of those cap-and-trade systems, in Shenzhen, began operations on 7 June 2013. By 2020, the Chinese government plans to link those regional systems into a national carbon market. Recently, the governments of China and Australia announced their intent to link the two countries' carbon markets into a regional one. In 2012, the central government chose seven manufacturing centers to begin cap-and-trade pilot programs as part of its *12th Five Year Energy Plan* to cut greenhouse gases 45% per economic unit by 2020. By 2015, when the country-wide program is fully operational, 800 million to 1 billion metric tons of emissions will be regulated in the world's biggest trading system, forecasts Bloomberg New Energy Finance.

Specific Renewable Energy Technologies

According to the International Energy Agency, in 2008, China's 2,257 million metric tons oil equivalent (MTOE) of total primary energy supply by source came from fossil fuels: 87.4%, biofuels and waste: 9.0%, hydro: 2.3%, nuclear: 0.8%, and other renewable energy: 0.5% (Figure A-2).

Figure A-2
Total Primary Energy Supply by Source, 2009



Source: International Energy Agency, *Key World energy Statistics*, 2009.

- **Hydropower.** China's installed hydro capacity in 2012 was 250 GW and constituted about 21.8% of total electric power generation capacity. In 2009, hydropower generated 615.6 TWh, which was equivalent to 16.7% of China's total electricity generation. As China's potential hydropower capacity (estimates range up to 600 GW, but currently the technically exploitable and economically feasible capacity is around 400 GW) is only about 25-30% utilized; there remains much space for further hydro development. In comparison, hydro utilization in the U.S. currently is 80% and

in Norway, Iceland, and other countries it is over 90%. Several new units are scheduled to still come on line in China and the National Development and Reform Commission in the Eleventh Five-Year Plan has set a 300 GW target for 2020. Due to China's scarcity of fossil fuels and the government's preference for energy independence, hydropower is an attractive option. Most of China's hydropower stations are located central and southwestern China, in particular in Sichuan and Yunnan provinces, where two thirds of China's untapped hydro capacity is located. The West-to-East Transmission Program (Xidian Dongsong), which is a key component of China's long-term energy strategy, plans to have a grid of UHVDC transmission lines carry bulk loads from 13 designated hydropower bases in Southwest and Western China over several thousand kilometers to the electricity-hungry coastal provinces. These hydropower bases hold 69% of China's total exploitable capacity and as of 2008, about 180 middle and large-scale dams were under construction there. Hydropower in China has been touted as a renewable and clean energy source, but this masks the fact that large dams, such as the Three Gorges Dam or the Xiluodu Dam on the lower Jinsha River, have had environmental impacts on the areas surrounding dam reservoirs. Erosion, flooding of valuable farmland, and destruction of fish breeding habitats have been typical problems.

- **Wind power.** China has the largest wind resources in the world and three-quarters of them are offshore. In 2012, China was the world's largest producer of wind power, ahead of the United States, Germany, Spain and India. At the end of 2012, wind power in China accounted for 75.6 GW of electricity generating capacity, which grew 20% over the previous year; China has identified wind power as a key growth component of the country's economy. As of 2010, China became the world's largest maker of wind turbines, surpassing Denmark, Germany, Spain, and the United States. The Chinese wind industry appears to have been unaffected by the global financial crisis, according to industry observers. China aims to have 200 GW of wind power capacity by 2020. It encourages foreign companies, especially from the USA, to visit and invest in wind power generation.
- **Solar power.** China has emerged as the world's largest manufacturer of solar PV panels in the last two years. It has become a world leader in the manufacture of solar photovoltaic technology, with its six biggest solar companies having a combined value of over \$15 billion. Around 1700 MW of solar PV were produced in China in 2007, which was nearly half of the world's production of 3,800 MW, 99% of which are exported. Following the new incentive scheme of Golden Sun announced by the government in 2009, numerous recent developments and plans announced by industry players became part of the milestones for solar industry and technology development in China. These include the new thin-film solar plant developed by Anwell Technologies in Henan province using its own proprietary solar technology, the agreement signed by LDK for a 500 MW solar project and the solar power plant project in a desert teamed up with First Solar and Ordos City. The effort to drive renewable energy use in China was further assured after the speech of the Chinese President given at the UN climate summit on 22 September 2009 in New York,

pledging that China will adopt plans targeting to use 15% of its energy from renewable sources within a decade. About 50 MW of installed solar capacity were added in 2008, more than double the 20 MW in 2007, but still a relatively small amount. In 2012, China generated 7 GW of solar power, and plans to expand the installed capacity to 20 GW by 2015.

- **Biomass and Biofuel.** China emerged as the world's third-largest producer of ethanol-based bio-fuels (after the US and Brazil) as of the end of the 10th Five Year Plan Period in 2005, and at present, ethanol accounts for 20% of total automotive fuel consumption in China. In the 11th Five Year Plan period (2006 through 2010) China planned to develop 6 megatons/year of fuel ethanol capacity, which is expected to grow to 15 megatons/year by 2020. Despite this level of production, experts say that there will be no threat to food security, though there will be an increasing number of farmers who will be “farming oil” if the price of crude oil continues to increase. Based on planned ethanol projects in some provinces in China, the output of corn would be insufficient to provide the raw material for those plants in those provinces. In the *World Economic Outlook*, the International Monetary Fund expressed concern that there would be increasing competition worldwide between bio-fuels and food consumption for agricultural products and that that competition likely would continue to result in increases crop prices. Work has begun on the ¥250 million Kaiyou Green Energy Biomass (Rice Husks) Power Generating Project located in the Suqian City Economic Development Zone in Jiangsu Province. The Kaiyou Project will generate 144 GWh/year and use 200 kilotons/year of crop waste as inputs. Bioenergy is also used at the domestic level in China, both in biomass stoves and by producing biogas from animal manure.

- **Geothermal.** Geothermal resources in China are abundant and widely distributed throughout the country. There are over 2,700 hot springs occurring at the surface, with the temperature exceeding 25°C. In 1990, the total flow rate of thermal water for direct uses amounted to over 9,500 kg/s, making China the second-largest direct user of geothermal energy in the world for non-electrical purposes. Thermal water is used directly for industrial and agricultural processes such as dyeing, drying of fruits and vegetables, paper and hide processing, air conditioning, and preheating boiler feed water. Four provinces are the most active in their use of geothermal resources for direct uses: Hebei, Tianjin, Shandong, and Tibet. Recognizing geothermal energy as an alternative and renewable energy resource, since the 1970s, China has conducted extensive explorations aiming at identifying high-temperature resources for electric generation. China has 12 major geothermal basins, amounting to an equivalent of 853 billion metric tons of standard coal equivalent, which could generate around 7 billion GWh of electricity. Despite ranking first in the world in the use of geothermal resources for all applications, as of 2010, China ranked only 18th in geothermal electric power capacity, with only 24.2 MW installed. Geothermal power is expected to provide 1.7% of China's total energy or 68.8 million TCE in 2015. By 2015, this green-energy source can replace 68.8 million tons of coal and reduce carbon dioxide emissions by 180 million tons. The only electricity-producing fields

are located in Tibet, where a 25 MW plant has been operating since 1977. According to the “2005 Chinese Geothermal Environment Bulletin” by China's Ministry of Land and Resources, the direct utilization of geothermal energy in China will reach 13.76 cubic meters per second, and geothermal energy will reach 10,779 MW, ranking it first in the world. However, such a program has not yet been started.

A.3 Republic of India

Socioeconomic Survey

India is developing into an open-market economy, yet traces of its past autarkic policies remain. Economic liberalization measures, including industrial deregulation, privatization of state-owned enterprises, and reduced controls on foreign trade and investment, began in the early 1990s and have served to accelerate the country's growth, which has averaged under 7% per year since 1997. India's diverse economy encompasses traditional village farming, modern agriculture, handicrafts, a wide range of modern industries, and a multitude of services. Slightly more than half (53%) of the work force is in agriculture, but services are the major source of economic growth, accounting for nearly two-thirds of India's output, with less than one-third of its labor force. India has capitalized on its large educated English-speaking population to become a major exporter of information technology services, business outsourcing services, and software workers. In 2010, the Indian economy rebounded robustly from the global financial crisis – in large part because of strong domestic demand – and growth exceeded 8% year-on-year in real terms.

However, India's economic growth began slowing in 2011 because of a slowdown in government spending and a decline in investment, caused by investor pessimism about the government's commitment to further economic reforms and the global situation. High international crude prices have exacerbated the government's fuel subsidy expenditures, contributing to a higher fiscal deficit and a worsening current account deficit. In late 2012, the Indian Government announced additional reforms and deficit reduction measures to reverse India's slowdown, including allowing higher levels of foreign participation in direct investment in the economy.

The outlook for India's medium-term growth is positive due to a young population and corresponding low dependency ratio, healthy savings and investment rates, and increasing integration into the global economy. India has many long-term challenges that it has yet to fully address, including poverty, corruption, violence and discrimination against women and girls, an inefficient power generation and distribution system, ineffective enforcement of intellectual property rights, decades-long civil litigation dockets, inadequate transport and agricultural infrastructure, limited non-agricultural employment opportunities, inadequate availability of quality basic and higher education, and accommodating rural-to-urban migration.

India is currently second after China in terms of population size, with about 1.22 billion inhabitants, or 17.2% of the world's total population of 7 billion people. India's population is

growing significantly compared to the other BRIC countries, with an annual growth rate of 1.7% in the last decade. This growth rate is expected to remain stable throughout the next decade, meaning India's population will match that of China soon after 2020. In spite of significant migration from rural areas to cities, 71% of the Indian population continues to live in rural areas. However, while the growth of many major cities has slowed recently, forecasts show that the urban population is going to double by 2030. Currently, India has a labor force of some 498.4 million, or around 41% of the total population. If the demographic profile of India continues to expand at a similar pace, the country will add an additional 270 million people to this pool by the end of 2025.

Although India has abundant natural resources, the country's rapidly increasing appetite for energy substantially exceeds its current production. India's installed electric generating capacity was 189.3 GW in 2009, of which 69.9% was fossil fuels. The principal fossil fuel is coal, which represents 42.3% of primary energy supply. Oil represents 23.6% of India's primary energy supply, while natural gas adds a further 7.2%. Although India's refined petroleum products production and consumption are about 3.2 million barrels (bbls)/day, which exceeds its domestic crude production of 900,000 bbls/day, it is a net importer of 3.0 million bbls/day (ranked 4th behind the US, China and Japan). Its proven crude oil reserves were 9 billion barrels in 2012. India produces 46.1 billion cubic meters of natural gas, which is below its consumption of 61.1 billion cubic meters, so it imported 12.2 billion cubic meters of natural gas in 2011 (ranked 24th in the world). Natural gas-fuelled power plants burn 35% of the total gas supply. At the same time, growing nuclear power and hydroelectricity contribute 0.7% and 1.4% of total primary energy supply. Although the renewable energy sector is growing fast, biomass and waste represented 24.5% of its total primary energy supply in 2009.

The Indian economy is among the fastest growing in the world (6.5% real GDP growth in 2012), and there is thus enormous potential for growth in the energy sector. Electricity consumption in India, at 637.6 billion kWh, was the 6th largest in the world in 2009, representing 3.3% of total world electricity consumption. India produced 880 billion kWh in 2009, making it the 6th largest producer in the world and representing 4.1% of world electricity production. India was a net importer of 10.53 billion kWh of electricity in 2009. Out of the BRIC countries, India had a 5.7% annual growth rate in electricity consumption in the last decade, the second-highest of the BRICs. Assuming the current trend is maintained to 2020, consumption in India will be 1,180 billion kWh, slightly exceeding that of Russia.

India has the lowest per capita electricity consumption of the BRIC countries, amounting to 522 kWh in 2009, despite a surge in electricity demand in the previous decade. Indeed, India had the second-highest growth rate in per capita electricity consumption among the BRIC countries from 2000-2008, the increase averaging nearly 4% annually. This increase in demand is expected to continue; however, (partly due to population growth at 1.28% in 2013) per capita electricity consumption is expected to be about 840 kWh in 2020, only 26.6% of the global average.

India's share of thermal power should barely change in the next decade, and is expected to account for 79% of total generation in 2020, although natural gas will replace oil to some extent in that figure. The growth of electricity generation in India has been hindered by domestic coal shortages and as a consequence, India's coal imports for electricity generation increased by 18% in 2010. Most significantly, nuclear is expected to account for 5.3% of the total in 2020, three times more than it does currently, and wind power will also triple its share to about 3.7%. GDP per capita in India was \$2,900 in 2008, only 33% of the world average. In spite of its high population growth rate, India's GDP per capita growth is still some 2.2 percentage points higher than the average growth rate of China and India. Overall, it is expected that India's GDP per capita will be \$6,700 in 2020, around 35% of the global average.

The privatization of the electric generation sector started in 1991–1992, although currently most units remain publicly owned, with privatized plants making up just 10–13% of total output. Private investors remain concerned about the regulatory framework, cross-subsidies and high levels of electricity theft, and this has prevented any significant private investment into the transmission and distribution segments. Observers believe that privatization would be smoother if the authorities were able to reduce network losses to 10%. The electric generation segment is expected to attract more private investment in coming years, while in the transmission and distribution segments, where state control is expected to dominate, most private investment will take the form of public-private partnership projects. The IEA estimates the Indian energy sector needs \$960 billion in investment by 2030. The generation sector requires some \$435 billion, the distribution networks \$60 billion, and the transmission grid \$165 billion.

The restructuring and privatization of the electricity sector in India began with reforms initiated in 1991. India has already shown a willingness to open its markets to international players. There is increasing private ownership in the generation sector and a planned electricity market opening in 2010 is designed to allow consumer choice between electricity supply companies. In another vital step, India is creating power exchanges to stimulate the electricity market. Legislation is currently being drafted to support this, and experts are carefully reviewing examples in the more advanced North American and European countries to find the best examples to follow. By fully implementing these activities, India should be able to successfully handle problems associated with rising energy prices, the sale of newly-built and planned electric generation units, complex issues of electricity distribution, the requirements of state subsidies, and end-user price issues.

The main priorities for the regulator in the future are to:

- Create stable policies and a competitive market environment in order to make the sector attractive to investors;
- Focus on the development of distribution and transmission networks in order to cut losses and reduce the need for excess additional generation capacity

- Continue to reduce the high level of electricity theft by strict regulation and control
- Set up a transparent, clear and stable tariff setting mechanism
- Reduce cross-subsidies to diminish mismanagement and pilferage
- Support energy saving investments to reduce the growth of demand
- Focus on energy efficiency in the residential sector.

Legislative Initiatives Affecting Renewable Energy Development in India

India's power sector has two overlapping, historic challenges. The widespread blackouts that brought much of India to a sputtering halt in 2012 were a dramatic signal of a power sector that required attention. But a challenge no less central to India's future, and arguably much more so, is that of the country's goals for renewable energy. The national government's ambitious goals for solar energy, coupled with the country's rapid progress in developing wind energy, raise many questions regarding the sources and costs of the investment that will be needed to install and operate this infrastructure. Under the Jawaharlal Nehru National Solar Mission (JNNSM), grid-connected solar PV capacity increased by 165% in 2011 to reach 427 MW. However, the ambitious targets for 4,000-10,000 MW by 2017 and 20,000 MW by 2022 may be hard to achieve under current policies and programs, and financing may be the biggest obstacle. Likewise, with 16 GW installed, India had already become the world's fifth-largest market for wind by 2011, but ambitious plans for a further expansion to 31 GW by 2017 will face similarly daunting policy and financing problems.

- ***Electricity Act 2003.*** This Act mandates that each State Electricity Regulatory Commission (SERC) establish minimum renewable power purchases, allows for the Central Electricity Regulatory Commission (CERC) to set a preferential tariff for electricity generated from renewable energy technologies, and provides open access of the transmission and distribution system to licensed renewable power generators.
- ***National Electricity Policy 2005.*** This Policy allows SERCs to establish preferential tariffs for electricity generated from renewable sources.
- ***National Tariff Policy 2006.*** This policy mandates that each SERC specify a renewable purchase obligation (RPO) with distribution companies in a time-bound manner with purchases to be made through a competitive bidding process.
- ***Rajiv Gandhi Grameen Vidyutikaran Yojana 2005.*** This policy supports the extension of electricity to all rural and below-poverty line households through a 90% subsidy of capital equipment costs for renewable and non-renewable energy systems.
- ***Eleventh Plan 2007–2012.*** This Plan establishes a target that 10% of power

generating capacity will be from renewable sources by 2012 (a goal that has already been reached) and supports the phasing out of investment-related subsidies in favor of performance-measured incentives. The Government of India targets capacity additions of 15 GW of renewable energy, which, if achieved, would bring the cumulative installed capacity to over 25 GW in 2012. The plan seeks to ensure electricity connection to all villages and poor households by 2009 and round-the-clock power by the end of the Plan, increase forest and tree cover by 5% and improve energy efficiency by 20% by the 2016-17 fiscal year.

- **National Action Plan on Climate Change (2008), (NAPCC).** The NAPCC aims to promote development goals while addressing GHG mitigation and climate change adaptation. NAPCC suggests that up to 15% of India's energy could come from renewable sources by 2020. The NAPCC includes eight focused missions, one of which is dedicated to solar energy (the others concern energy efficiency, water, sustainable habitat, and related topics). In addition, the Clean Development Mechanism (CDM) of the Kyoto Protocol is supporting the development of renewable energy projects in India. As of September 2010, there were over 500 registered CDM projects in India, and these are dominated by renewable energy projects.

During the 11th Five Year Plan period, the renewable energy sector witnessed tremendous changes in policy, with accelerated and ambitious plans to increase the contribution of solar energy in the country's energy mix. The perception that renewable energy cannot play a significant role has changed with additions in power generation capacities. The launching of the *Jawaharlal Nehru National Solar Mission* as part of the National Action Plan on Climate Change highlights the transformation of the renewable energy sector. During the 11th Plan period, there was an addition of 14,660 MW, taking the cumulative total to 24,914 MW, with wind contributing over 10,000 MW. Renewable energy also provided basic lighting services to 5,000 villages. In addition, India launched the Renewable Energy Certificate, which helps in the creation of the pan-India renewable energy market.

- **Twelfth Plan 2012-2017.** Based on the target of 12% renewable energy in the electricity mix by the end of the 12th Five Year Plan, a corresponding renewable power capacity addition of 30,000 MW has been planned. The 12th Five-Year Plan document proposes several new initiatives and progressive measures. They include, for example, the establishment of National Institute of Solar Energy, the creation of a National Bio-energy Mission, Renewable Energy Development Fund, Renewable Power Evacuation Infrastructure, and National Biomass Cook Stove Program. Parliament has released its new Renewable Energy Policy Update: *Renewable Energy in 12th Five Year Plan*. The policy update analyzes the key provisions of renewable energy in the 12th Five Year Plan document, presented at the National Development Council Meeting in December 2012. According to the Update, there is an immediate need for ensuring the government's commitment to the renewable energy targets set by NAPCC of 12% by 2017 and 15% by 2020. Although a significant amount of

stress has been laid in the 12th Plan on the implementing transmission and evacuation facilities for renewable energy, a comprehensive roadmap with time-bound milestones needs to be prepared for ensuring timely implementation.

Table A-4
Plan-Wise Growth of Renewable Power Generating Capacity
(cumulative capacity in MW)

Source	Beginning of 10th Plan	Beginning of 11th Plan	Beginning of 12th Plan
Wind	1,628	7,092	17,352
Small Hydro	1,434	1,976	3,395
Biomass Power	389	1,184	3,225
Solar	2	3	941
Total	3,453	10,255	24,914

Source: Ministry of New and Renewable Energy-Annual Report 2012-2013

Central Government Investment in Renewable Energy Development

Renewable energy in India is a sector that is still underdeveloped. India was the first country in the world to set up a ministry of non-conventional energy resources, in 1992. However, its success has been very spotty. In recent years India has been lagging behind other nations in the use of renewable energy. India's cumulative grid interactive or grid-tied renewable energy capacity (excluding large hydro) has reached 26.9 GW, of which 68.9% comes from wind, while solar PV contributed nearly 4.59% of the country's installed renewable capacity. Biofuel and waste represented about 25% of the total primary energy supply in 2009, while geothermal/solar/wind represented only 0.7%. The government provides investment support and other subsidies:

- **Foreign Direct Investment (FDI).** India India permits FDI up to 100% in the clean energy sector, subject to the provisions in the Electricity Act, 2003, so no prior approval of regulatory authorities is required.
- **Tax Holiday under the Domestic Income Tax Law.** Parties engaged in generation or generation and distribution have a 10-year tax holiday for renewable energy plants, if they begin to generate power before 31 March 2013. However, they must pay a minimum alternative tax at the rate of 20%, which can be offset in future years. It is expected a new provision will be made effective 1 April 2013.
- **Financing.** The Indian Renewable Energy Development Agency was established under the Ministry for Non-Conventional Energy Sources. This specialized financing agency promotes and finances renewable energy projects.

The Role of Feed-In Tariffs and Other Financial Incentives

In 1992, the Government of India established the Ministry of New and Renewable Energy (MNRE), the world's first ministry committed to renewable energy. MNRE is dedicated to expanding the contributions of renewable energy in all of India's end-use sectors, and undertakes policy and planning activities to that end. MNRE also supervises national-level renewable energy institutes such as the Solar Energy Centre and the Centre for Wind Energy Technology. The Indian Renewable Energy Development Agency (IREDA) provides financial support and innovative financing for renewable energy and energy efficiency projects with funds from the Indian government and multilateral lending agencies. IREDA also administers the central government's renewable energy incentive programs. Other government institutions with direct responsibilities that extend into renewable energy include several units under the Ministry of Power, the Planning Commission, and the Prime Minister's Council on Climate Change.

- **Generation-Based Incentive Scheme.** In order to attract foreign investors, the government introduced a generation-based incentive (GBI) scheme for independent power producers of wind and solar power. Under the scheme for wind power, a GBI at Rs. 0.50 per unit of electricity fed into the grid is provided for a period not less than 4 years and a maximum period of 10 years with a cap of Rs. 62 lakhs per MW. Under the Scheme for Solar Energy, the GBI supports small solar power projects connected to the distribution grid (below 33 kV) to the state utilities. Grid solar power projects in the capacity range of 100 kW to 2 MW each, connected to the high-tension grid below 33 kV are eligible under the scheme. The GBI scheme would be applicable only for those power producers who do not avail the accelerated/enhanced depreciation benefits under the Income Tax Act. The power producers who avail the benefits of the scheme will be required to furnish documentary proof to this effect. The scheme will be applicable only for those independent power producers having a minimum installed capacity of 5 MW and whose capacities are commissioned for the sale of power to the grid after the announcement of the scheme. The scheme will not be applicable to those who set up capacities for captive consumption, third-party sales, merchant plants, etc.
- **Feed-in Tariffs.** Armed with legislative powers, many State Electricity Regulatory Commissions took constructive action for the promotion of renewable energy. They: 1) issued FIT orders for technologies operating in their area of jurisdiction, 2) issued renewable purchase obligations for obligated entities, 3) determined various charges applicable for undertaking renewable energy transactions, and 4) determined/approved principles of long-term energy purchases.

Table A-4
Feed-in Tariffs by State

State/	Solar PV	Solar Thermal	Solar Rooftop	Wind	Small Hydro	Biomass	Bagasse
Andhra Pradesh	15.39	15.04	-	3.50	2.60 (1 st yr) 3.88 (10 th yr)	4.51	-
Assam	10.00	-	-	-	3.19	4.00	3.2

Table A-4
Feed-in Tariffs by State

State/	Solar PV	Solar Thermal	Solar Rooftop	Wind	Small Hydro	Biomass	Bagasse
Bihar	17.91	15.31	17.91	-	-	3.44	3.51
Chhattisgarh	15.84	13.26	15.84	-	-	4.36	-
Gujarat	15.00	11.00	12.54	3.56	3.707	4.40	4.61
Haryana	14.95	12.85	-	4.08	-	5.26	-
Himachal Pradesh	17.91	-	17.91	-	2.87	-	-
Karnataka	14.50	11.35	14.50	3.70	3.40	3.66 1 st yr 4.13 10 th yr	3.59
Kerala	17.91	15.31	17.91	3.64	2.94	-	2.76
Madhya Pradesh	15.35	11.26	15.49	4.35	5.40	3.3 1 st yr 3.79 10 th yr 5.14 20 th yr	2.80
Maharashtra	15.41	15.24	16.11	WZ-1:5.37 WZ-2:4.67 WZ-3:3.97 WZ-4:3.58	≤500 kW: 5.51 500 kW-1 MW: 5.08 1-5 MW: 4.51 525 MW: 3.87	4.98	4.98
Orissa	17.80	14.75	18.52	-	-	-	-
Punjab	17.91	15.31	-	5.07	< 5 MW: 4.26 5-25 MW: 3.65	5.05	5.05
Rajasthan	15.32	12.58	15.32	4.69	-	5.03 air cooled 4.59 water cooled	-
Tamil Nadu	15.15	13.15	14.34	5.39	3.35	4.294	4.491
Uttar Pradesh	-	-	-	-	-	5.02	4.21
Uttarakhand	-	-	-	WZ-1:5.35 WZ-2:4.35 WZ-3:3.65 WZ-4:3.20	<5 MW: 3.50 5-10 MW: 3.40 10-15 MW: 3.25 15-20 MW: 3.15 (20-25 MW) 3.00	1.995	1.995
West Bengal	16.13	-	16.13	-	-	-	2.55

Source: ABPS Infrastructure Advisory Private Limited, Mumbai, India

- **Accelerated Depreciation.** Under the domestic income tax law, renewable companies were provided with accelerated depreciation at 80%. However, recently,

the government has restricted the accelerated depreciation of 80% to windmills installed on or before 31 March 2012. Windmills installed after that date will be eligible for depreciation of 15% instead of 80% (solar remains at 80%). A company can claim either accelerated depreciation or GBI, but not both.

State Incentives

As of April 2010, 18 states had established renewable purchase obligations (RPOs) or had draft regulations under consideration with RPO requirements ranging from 1% to 15% of total electricity generation. In January 2010, CERC announced the terms of a tradable renewable energy certificate (REC) program. Under this program, 1 REC represents that 1 MWh of energy is generated by renewable energy, which is tradable. The environmental attributes can be exchanged in the form of RECs, which will be issued by a central agency set up for the administration of this program. RPOs are not yet enforced.

Cap and Trade Activity National Action Plan on Climate Change

Under the Kyoto Protocol's Clean Development Mechanism, a developed country can invest in GHG mitigation projects in a developing country. The developed country will receive a credit while the developing country gets capital and clean technology. India is the second-largest seller of carbon credits. With climate negotiations moving at a sluggish pace over the last three annual global summits, attention is shifting towards energy and climate schemes in the major economies. Under pressure after other emerging economies announced domestic schemes, India announced a target reduction of its carbon intensity: by 2020, it intends to reduce this intensity by 20% of GDP (reaching 25% of 2005). This reduction is to be achieved through India's National Action Plan on Climate Change (NAPCC) consisting of eight missions launched in 2008. Table A-5 summarizes the eight missions and their key targets.

Table A-5
The NAPCC's Eight Missions for Climate Change

Missions	Key Targets
National Solar Mission	20,000 MW by 2020 First phase (2010-12) plans for 1,000 MW
National Mission for Enhanced Energy Efficiency	Creates new institutional mechanisms to enable the development and strengthening of energy efficiency markets
National Mission on Sustainable Habitat	Promotes the introduction of sustainable transport, energy-efficient buildings, sustainable habitat, and sustainable waste management in cities
National Water Mission	20% improvement in water use efficiency
National Mission for Sustaining the Himalayan Ecosystem	Conserve biodiversity, forest cover, and other ecological values in the Himalayan Region
National Mission for a "Green India"	Seeks to afforest an additional 10 million hectare of forest lands, wastelands and community lands
National Mission for Sustainable Agriculture	Supports climate adaptation in agriculture
National Mission on Strategic Knowledge for Climate Change	Better understanding of climate science, impacts and challenges

Source: Prime Minister's Council on Climate Change

- **The Perform, Achieve and Trade (PAT)** scheme was announced as part of the National Mission for Enhanced Energy Efficiency. According to the emerging blueprint of the scheme, specific energy consumption (SEC) targets totaling 10 million metric tons of oil-equivalent (MTOE) by 2015 are to be fixed for large energy-using installations around the country and credits called Energy Saving Certificates or ESCerts issued for those exceeding the energy saving goals under the SEC targets. The credits can then be sold to installations which fail to meet their required cuts, thereby enabling the formation of a new market-based mechanism over a period of time.

- The PAT Scheme can be divided into two phases:
 - SEC targets are set for each industry (designated consumers or DCs), depending on the energy consumption level. The target will specify the reduction to be reached within the first three-year period (2012 - 2015).
 - DCs must devote this three-year period to applying efficient processes and technologies in order to meet their SEC targets.
- The market registry will provide ESCerts to DCs that are able to reduce their SEC beyond the target. DCs may sell the ESCerts through the power exchange or bank them until the next compliance cycle. Auditing is conducted by accredited verification agencies. Industries that are not able to meet the SEC target will be able to purchase ESCerts.

However, a number of the original deadlines have been postponed by the Bureau of Energy Efficiency (BEE), the nodal agency for designing and implementing the PAT scheme. With assistance from the German agency GTZ, BEE will cover 478 installations in 8 sectors – thermal power stations, cement, iron and steel, fertilizers, aluminum, chloralkali, paper and textiles. Together they account for 40% of India's primary energy consumption.

Table A-6
Perform, Achieve and Trade Scheme Target Sectors and Energy

Sectors (Number of Designated Consumers)	MTOE	Share of Energy
Power (144)	160.30	69.5%
Iron & Steel (67)	36.08	15.7
Cement (85)	14.47	6.3
Fertilizer (29)	11.95	5.2
Textile (90)	3.50	1.5
Aluminum (10)	2.42	1.0
Pulp & Paper (31)	1.38	0.6
Chlori-Alkali (22)	0.43	0.2
Total (478)	230.53	100%

Source: World Resources Institute

BEE has identified various industrial units that are very large energy consumers based on minimum annual energy consumption in various sectors. For example, in the case of iron and steel, it is 30,000 tons of oil equivalent. These units are referred to as designated consumers. The highest numbers of DCs are located in Tamil Nadu (65) followed by Rajasthan (63) and Chhattisgarh (56). Originally the eight entities belonging to the India Railways were also listed, but were later removed considering the public value of the railways. The original figure of 714 DCs was revised to 478 as of 30 March 2012.

The SEC was computed using the formula below following the principle of gate-to-gate accounting for a facility:

$$\text{SEC} = \text{Total energy input to the plant boundary} / \text{Quantity of the key product}$$

For the purpose of setting the baseline the previous five years of data were used and the arithmetic average was taken. The assessment of the absolute energy saving at the end of three years (2011-2014) will be based on the following formula:

$$\text{Energy saving} = P \text{ base year (SEC base year} - \text{SEC target year)}$$

ESCCerts will be issued by a central registry in case a facility exceeds the reductions from the baseline in the target year (2014). The ESCerts will be traded among the designated industrial facilities on two exchanges: Indian Energy Exchange (IEX) and Power Exchange India Limited (PXIL).

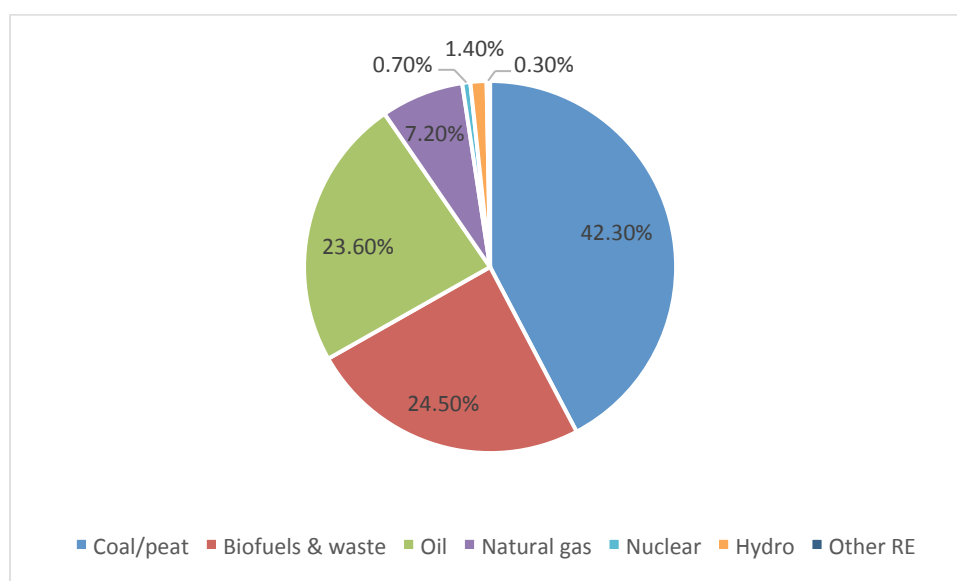
The BEE estimates that capital investments of IN R208 billion (\$4.6 billion) will be needed in the first phase from 2012-13 to 2014-15. This will deliver annual energy savings of 6.6 million MTOE by 2014-2015, worth around \$2 billion per annum. It is estimated that India's total industrial efficiency market stood at \$5.5 billion in FY 2009-10. As stated earlier, companies that exceed their energy saving targets will be able to trade the surplus. Estimates suggest the size of this energy saving (ESCCert) trading market could be around \$200 million by the end of Phase I. It is expected that trading will start once ESCerts (1 ESCert = 1 MTOE of energy) are issued to DCs that surpass their targets based on a review of their first year's performance.

Over 70% of the targeted savings are expected to come from just two sectors: thermal power and iron and steel. Analysts believe that energy-efficient solutions for these sectors will offer the maximum investment opportunity, including process control and automation technology, efficient motors, pumps, boilers, waste heat recovery, insulation material, variable frequency drives, blade fans, condensers, as well as capacitor banks.

Specific Renewable Energy Technologies

As reported by the IEA, 676 million metric tons oil equivalent (MTOE) of India's total primary energy supply by source in 2009 was generated by fossil fuels (73.1%), biofuels and waste (24.5%), hydro (1.4%), nuclear (0.7%), and other renewable energy (0.3%) (Figure A-3).

Figure 1
Total Primary Energy Supply by Source, 2009



Source: IEA, *Key World Energy Statistics, 2009*

- Wind Energy.** MNRE's official estimate of India's on-shore wind capacity potential is over 48 GW. Industry associations assert that taking into account hub heights greater than 50 m and improving conversion efficiencies from technology advancements and aggressive policy action, the potential is much greater: between 65 and 242 GW. India ranked fifth globally in total installed wind capacity in 2009, and as of March 2010, it had almost 12 GW of installed wind capacity. In addition to the sizable on-shore potential identified for India, the long coastline and prevailing wind patterns suggest substantial potential for offshore capacity, though no systematic assessment of offshore potential has yet been done.
- Solar Energy.** India is promoting three solar technologies: solar PV, concentrating solar power (CSP), and solar water heating (SWH). In June 2010, cumulative installed capacity of solar PV reached 15.2 MW, of which 12.3 MW was grid-tied (less than 0.1% of the grid-tied renewable energy capacity in India) and 2.9 MW was off-grid (0.7% of the off-grid renewable energy capacity in India). The recent JNNSM aims to dramatically increase installed PV through attractive feed-in tariffs and a clear application and administration process. Phase 1 of JNNSM targets additions of 500 MW of grid-tied and 200 MW of off-grid PV capacity by 2013. PV installations of 365 MW are approved for the state of Gujarat and 36 MW for the state of Rajasthan. By the end of Phase 3 in 2022, India plans to have 10,000 MW of grid-tied and 2,000 MW of off-grid PV. JNNSM reaches beyond installed capacity to also target the growth of India's indigenous PV industry, which has historically supplied the export market. Annual Indian module production is expected to exceed 2,500 MW by 2015, and JNNSM targets an annual production of 4,000–5,000 MW by 2022. There are currently no CSP plants in India. However, JNNSM targets the introduction and substantial additions of CSP.

Estimates across India show large areas with sufficient radiation for the development of CSP plants, and one analysis foresees the technical potential for CSP generation in India at almost 11,000 TWh per year. JNNSM envisions 500 MW of CSP by 2013 and 10,000 MW by 2022. India is on track to meet the first goal—currently there are 351 MW of approved CSP projects in Gujarat and another 30 MW in Rajasthan.

SWH technology presents an opportunity to avoid using electricity or liquefied petroleum gas for heating water in India. It has applications in the residential, commercial, and industrial sectors. India has approximately 3.5 million m² of installed SWH collector area out of an estimated potential of 40 million m². Several programs are in place to accelerate the deployment of SWH, including the provision of low-interest loans and incorporation of SWH into building codes.

- **Small Hydro.** MNRE has estimated India's small hydro potential (plants up to 25 MW) at more than 15,000 MW and is constantly revising this number upwards as new sites are identified; more than 40% of this potential has been identified in four northern, mountainous states. Capital incentives are in place to support the installation of new small hydropower plants, as well as to renovate and modernize existing plants. In addition to direct subsidies at the state and national levels, small hydro plants may also qualify for low-interest loans and income tax exemptions. CERC also offers preferential tariffs for small hydro plants, with the tariff amount and time period varying depending on project size and location. High capacity factors have been observed at small hydro plants in remote, mountainous regions of India, and small hydro has the lowest levelized costs of energy of any renewable technology in India. These factors have contributed to small hydro's importance in India's rural electrification efforts, with both federal and state programs supporting small hydro as a means of supplying electricity to villages where gaining access to the central grid is challenging.
- **Bioenergy.** Traditional biomass, such as wood and cow dung, have historically played an important part in India's energy supply, and they still supply cooking energy to almost all of India's rural population. By utilizing organic waste and agricultural output, India can incorporate modern bioenergy as a substantial part into its future energy mix. There are approximately 4 million installed household biogas plants in India with almost 4,000 additional units supplying household clusters or villages; cattle manure is the primary feedstock for these household plants. MNRE estimates that available cattle manure could support approximately 12 million household biogas plants. Larger-scale biogas facilities use industrial wastewater to generate electricity, and the 48 such plants in India have an aggregate installed capacity of 70 MW. MNRE estimates the total capacity potential for industrial biogas to be 1.3 GW. National and state programs support the expansion of biogas through technical capacity building, financial incentives, and demonstration projects, with additional deployment supported through the CDM. Solid biomass is used in India either in direct combustion or gasification to generate power or for the cogeneration of

power and heat. MNRE estimates that surplus biomass could support 25 MW of installed electricity-generating capacity and that cogeneration capabilities added to existing industries requiring process heat could add 15 GW more to the grid.

Policies in place to support biomass power generation include capital and interest subsidies and tax exemptions. CERC and several states offer a preferential tariff for electricity generated from biomass power or cogeneration plants.

Liquid biofuels, ethanol and biodiesel, are used to substitute petroleum-derived transportation fuels. Ethanol in India is largely produced by the fermentation of molasses, a byproduct of the sugar industry, and from biodiesel produced from non-edible oilseeds. India is conducting research in the area of cellulosic biomass conversion for ethanol production. Making use of one-third of the 189 million tons of surplus biomass could yield approximately 19 billion liters of gasoline equivalent each year, the equivalent of India's entire gasoline consumption. Support for liquid biofuels comes in the form of subsidized loans for ethanol production facilities, tax exemptions for biodiesel, financial incentives promoting the cultivation of *Jatropha* and other non-edible oil seeds, and research and development programs.

- ***Decentralized Energy.*** According to the International Energy Agency, in 2008, more than 400 million Indians did not have access to electricity, with electrification rates of 93.1% and 52.5% in urban and rural areas, respectively. There are several programs at the national level to promote the electrification of remote villages and below-poverty line households. Renewable technologies deployed to this end include family biogas plants, solar street lights, solar lanterns, solar PV systems, and micro-hydro plants. The Indian Government has initiated several programs, policies, and acts that focus on the development of rural energy, economy, and electrification to improve rural livelihoods with the help of renewable energy. Successful business models have been designed to address the shortcomings of global electrification efforts to increase sustainability and user satisfaction. Among these is a model for village electrification using biomass gasification where an energy service company, entrepreneur, or other actor provides system maintenance and operations, and collects feedstock.

Finding viable business solutions to supplying India's non-electrified households is perhaps the greatest single challenge and opportunity of the renewable energy market. In addition to using renewable energy to provide electricity to poor and remote citizens, renewable technologies are being used by industries that generate electricity at their facilities to supplement grid power. At the end of March 2008, the total renewable installed capacity for captive power plants greater than 1 MW was 305 MW, almost all of which was wind capacity. In addition to wind, solar thermal plants show good potential for captive power supply by integrating with the steam turbines already in place at cogeneration facilities. Industries can also deploy solar PV systems and use biodiesel in on-site generators to help power their operations.

A.4 Federal Republic of Brazil

Socioeconomic Survey

Characterized by large and well-developed agricultural, mining, manufacturing, and service sectors, Brazil's economy outweighs that of all other South American countries, and Brazil is expanding its presence in world markets. Since 2003, Brazil has steadily improved its macroeconomic stability, building up foreign reserves, and reducing its debt profile by shifting its debt burden toward real denominated and domestically held instruments. In 2008, Brazil became a net external creditor and two ratings agencies awarded investment-grade status to its debt. After strong growth in 2007 and 2008, the onset of the global financial crisis hit Brazil in 2008. Brazil experienced two quarters of recession, as global demand for its commodity-based exports dwindled and external credit dried up. However, Brazil was one of the first emerging markets to begin a recovery. In 2010, consumer and investor confidence revived and GDP growth reached 7.5%, the highest growth rate in the past 25 years. Rising inflation led the authorities to take measures to cool the economy; these actions and the deteriorating international economic situation slowed growth to 2.7% in 2011 and 1.3% in 2012. Today, unemployment is at historic lows and Brazil's traditionally high level of income inequality has declined for each of the last 14 years. Brazil's historically high interest rates have made it an attractive destination for foreign investors.

Large capital inflows over the past several years have contributed to the appreciation of Brazil's currency, hurting the competitiveness of Brazilian manufacturing and leading the government to intervene in foreign exchange markets and raise taxes on some foreign capital inflows. President Dilma Rousseff has retained the previous administration's commitment to inflation targeting by the central bank, a floating exchange rate, and fiscal restraint. In an effort to boost growth, in 2012 the administration implemented a somewhat more expansionary monetary policy that has failed to stimulate much growth.

Brazil is the most populous country in Latin America, with a total population of 201 million (in 2013). The population growth rate has been 1.43% per annum in the last decade, which is the second-highest of the BRIC countries after India. Population in Brazil is expected to grow to about 220 million by 2020. Brazil's average population density is low and the majority of inhabitants live on or near the Atlantic coast. Since the 1970s, there has been an intensive migration from rural to urban areas, resulting in 87% of the population living in urban areas by 2010. Currently, the labor force of the country is approximately 107 million, which accounts for 53% of the total population. One of the main advantages of the Brazilian economy in the global context is the high availability of labor, which is expected to remain stable in the coming years. The Brazilian GDP per capita was \$12,079 in 2012, ranked 57th in the world by the IMF, while its nominal GDP was \$2.4 trillion, ranked 6th in the world. The growth of real GDP in Brazil is predicted to be 3% in 2013 and over 4% in 2014-18. Brazil's GDP per capita is expected to grow to \$16,387 in 2018 or nearly 6% annually in the 2012-2018 period, according to the IMF.

The economy of Brazil is supported by a dynamically developing energy sector. The primary energy sources produced in the country covered 80-90% of the domestic demand in

previous years. This production is mainly supported by crude oil reserves and by robust, sugar cane-based bio ethanol output. At the same time the high rate of utilization of vast hydro reserves by large hydropower plants contributes to the country's very large share of renewables, more than 40.2% of its primary energy mix.

With CO₂ emissions from its production and consumption of fossil fuels reaching 453.9 million metric tons in 2010, Brazil is now the world's 14th emitter. Brazil possesses the largest electricity market in South America and 11th in the world, with an installed generating capacity of 106 GW (17.1% fossil fuels, 1.9% nuclear, 74.7% hydro and 6.3% other renewables) and produced 489.5 billion kWh in 2010. The electricity consumption of Brazil has grown by 3% annually in the last decade to 438.3 billion kWh in 2010. Assuming this trend continues, consumption is going to be approximately 710 billion kWh in 2020. Despite this growth, Brazil will remain the lowest electricity consumer out of the BRIC group. Brazil has achieved an average growth of 1.5% in per capita electricity consumption in the last decade. This has resulted in a per capita consumption figure of 2,200 kWh, roughly 14% below the global average. As the Brazilian economy is growing faster than the population, per capita electricity consumption is expected to be 3,200 kWh by 2020, slightly above the expected world average.

The Brazilian state still plays an important role in the electricity sector. Although privatization was initiated in the 1990s, it has slowed since the electricity crisis in 2001. Large government-controlled companies dominate the generation sector, such as Eletrobrás, CESP, CEMIG and Copel. The most notable exception is Gerasul, in the south of the country, which is now operated by France's GDF Suez. Foreign investors are cautious, partly due to the lack of a sound regulatory framework and partly due to the government's strong political control of the industry through the Ministry of Mines and Energy. The privatization of distribution companies started in the 1990s, and with the exception of some smaller state-owned companies, transmission lines have also been partly auctioned off. However, due to the precarious financial state of most state-owned electricity companies, private capital is likely to play a key role in the future development of the transmission and distribution infrastructure. The IEA estimates the total investment needs of the Brazilian electricity sector to be \$252 billion (2005 prices), divided equally between generation and transmission infrastructure. Based on this forecast, an estimated \$25 billion investment per annum is necessary for the sector.

The regulatory framework of the electricity sector in Brazil is considered to be the most developed among the BRIC countries, and Brazil is as a result a leader in privatization in the developing world. However, challenges remain, including the clarification of regulatory rules, speeding up licensing procedures, and creating better pricing mechanisms in order to attract more investment into the electricity infrastructure. The main priorities for the regulator in the future are to:

- Maintaining stable rules to reduce investment risks
- Enforcing current rules and regulations and supervising the observance of contracts

- Increasing the transparency of decision making in order to attract investors and stimulate private sector participation
- Making licensing processes easier for generation projects
- Encouraging energy efficiency programs and providing incentives for the generation of clean energy
- Creating tariff parity between the private and public sectors
- Continuing to subsidize low-income and rural populations, but removing other subsidies.

Policy and Legislative Initiatives Affecting Renewable Energy Development in Brazil

Brazil's Government Policy Background. Since 2002 Brazil has been systematically building up its National Policy on Climate Change (PNMC). This integrated plan was last updated by Congress in December 2009 and covers five key domestic areas of related policies.

- Brazil's Climate Strategy is built around a national emissions target which aims to reduce Brazil's emissions by 36-40% relative to a business-as-usual baseline by 2020. This target was signed into law in 2009.
- Agriculture and forestry together represent 80% of Brazil's emission reduction target. The key government policies in this area reflect the country's objective of zero net forest loss by 2015, with the central effort focused on reducing deforestation in the Amazon by 80% by 2020 relative to the average for the period 1996-2005 (as per the Presidency Decree of December 2010). Given the starting point of 15-20,000 square kilometers annually of deforestation across Brazil, this challenge is huge. However, by 2011, a 60% reduction had already been achieved.

Other initiatives in the policy environment include:

- **Energy Efficiency, as a Key Policy Platform.** Since 1998, the Brazilian Electricity Regulatory Agency (ANEEL) has required all utilities to invest a minimum 1% of their annual revenues into energy efficiency programs. In 2008 the electricity demand target was set to reduce electricity demand by 10% by 2030 per unit of real GDP. The other targets include reducing non-technical transmission and distribution losses by 1,000 GWh annually by 2019 and a 2,200 GWh reduction of energy use in buildings through the use of solar heating by 2015.
- **Low-Carbon Energy Generation.** The National Energy Plan aims to rapidly scale up electricity generation from renewable sources. The original policy was set out in 2002 under the *Program of Incentives for Alternative Energy Sources* (PROINFA [2002]). In the latest PROINFA update, which covers the 2010-2020 period, Brazil is targeting the addition of biomass and sugarcane bagasse capacity of 5 GW (to 9 GW), small hydro of 2.5 GW (to 6 GW) and 11 GW of onshore wind (to 11.5 GW). The first phase subsidies/incentives will be funded from the Energy Development Account.

Consumers pay into this account through an increase in their energy bills (from which low-income sectors are exempt).

- **Finance.** The Brazilian Government-owned development bank BNDES is key to providing first-round debt financing for renewable energy, grid transmission and energy efficiency programs. BNDES provided some \$3 billion, or 45%, of the total \$7 billion of climate-specific investments in Brazil in 2011. BNDES is also instrumental in managing international pledges towards avoided deforestation. It makes special financing available for these renewable projects as well (up to 70% of capital costs, excluding site acquisition and imported goods and services, at the basic national interest rates plus 2% of the basic spread and up to 1.5% of the risk spread. No interest is charged during construction. See below for a more comprehensive description of Brazil's enabling finance policy.

By early 2005, the first phase was finished and 3,300 MW were completed (1,266 MW solar, 655 MW biomass, 1,379 wind). Stage II will aim to increase the share of energy produced by renewable sources to 10% of total annual energy consumption within 20 years.

- **The National Plan on Climate Change (PNMC) Law 12.187 [2009].** In 2007, the Brazilian Government began to reformulate its response to climate change with this law. It became national policy, which also establishes a national reduction target and discusses the process to move the PNMC towards implementation. Most notably, Law 12.187 officially adopts Brazil's voluntary national greenhouse gas reduction target of between 36.1% and 38.9% of projected emissions by 2020. Its overall goal is to achieve sustainable economic and social development.

Its main points and carbon reductions by 2020 are as follows:

- Reducing deforestation of the Amazon by 80% (564 million tons of carbon by 2020)
- Reducing deforestation in the Cerrado Biome by 40% (104 million tons of carbon by 2020)
- Increasing the supply of renewable energy for a 26-33 million tons of carbon reduction by 2020
- Improving energy efficiency, for a 12-15 million tons of carbon reduction by 2020
- Recovering 15 million hectares of land degraded by cattle raising (an 83-104 million ton reduction in carbon by 2020)
- Improving the system of integrated farming, forestry and cattle raising by 4 million hectares (thus reducing 18-22 million tons of carbon by 2020)
- Expanding no-till crop planting by 8 million hectares (16-20 million tons of carbon reduced by 2020)
- Expanding biological nitrogen fixation by 5.5 million hectares (16-20 million tons of carbon reduction by 2020)
- Expanding the use of biodiesel (reducing 48-60 million tons of carbon by 2020)

- Increasing the power supplied from hydroelectric power (79-99 million tons of carbon reduced by 2020)
- Replacing charcoal from deforestation with charcoal from planted forests **in the steel industry** (reducing 8-10 million tons of carbon by 2020).

Brazil has also developed one of the largest renewable energy programs in the world. The use of ethanol as an automobile fuel, introduced in the 1970s, has evolved in the last decade with the new flex-fuel light vehicle technology. In addition, the Brazilian Government has adopted the Brazilian National Program of Production and Use of Biodiesel that establishes a mandatory addition of 5% biodiesel into petroleum diesel by 2013. This decision provides a strategic mechanism to stimulate employment and income, to reduce greenhouse gas emissions, and to decrease dependence on fossil fuels. Specific measures include:

- **National Ethanol Program National Alcohol Program (PROALCOOL) [1975].** Established on 14 November 1975 by Presidential Decree No. 76.593, the Proalcool mandate was simple to implement; alcohol plants already in operation required only simple modifications to produce ethanol. Proalcool's specific mandate was to produce 3.5 billion liters of ethanol from sugar cane by 1980.
- **National Biodiesel Production and Usage Program [2004].** The PNPB program has been focusing on organizing the biodiesel productive chain by defining financial aid, structuring the technological basis, and establishing specific regulations on the new fuel. Sectorial policies have determined the development of this productive chain. The most important action taken by the PNPB program was approving Law No. 11.097/2005, which established an addition of 2% biodiesel in the diesel oil produced in Brazil since 2008. Since then, biodiesel has been much in demand and the sector has been developing rapidly. From January 2010, the mandatory addition content of biodiesel into diesel oil in Brazil is 5%. This measure increased biodiesel from 176 million liters per year to 2.4 billion liters in 2010. The program is an initiative of President Luis Inácio Lula da Silva's government (2003-2010) to integrate a drive for energy security in the electricity generation and transportation fuel sectors with sustainable rural development. It incorporates both large-scale agribusiness and family farms across the entire span of this continent-sized nation.
- **Ten-Year Energy Plan [2011-2020].** According to the PED, electricity consumption should grow 4.6% per year through this ten-year period. To supply this rising demand, over \$100 billion will need to be invested in capacity building. The plan also calls for the doubling of wind energy, biomass to electricity, and small hydroelectric facilities during the coming decade. About 51% and 38% of the Plan's spending will be allocated to hydro and other renewable energy, while the remainder will expand transmission lines by 41,500 km and substations. The PED, and its author, the Empresa de Pesquisa Energética (EPE), also calls for a dramatic increase in oil and gas production, largely due to the exploration of **pre-salt reserves**. The PED reports that

Brazil will triple production from 2.1 million barrels a day to 6.1 million by 2020. EPE forecasts that nearly half of this production will be exported.

Central Government Investments and Subsidies of Renewable Energy Development

As of August 2011, hydroelectric power accounted for more than 71% of Brazil's domestically produced electricity and over 14.2% of its total primary energy supply in 2009. After the oil shocks of the 1970s, Brazil started focusing on developing alternative sources of energy, mainly sugarcane ethanol. Brazil is the second-largest producer of ethanol in the world and is the largest exporter of the fuel. Brazil held its first wind-only energy auction in 2009, in a move to diversify its energy portfolio.

Brazil used tax incentives beginning in the mid-1960s to initiate a reforestation program to provide for industrial wood energy and wood product needs. As a consequence of the Brazilian Forestry Code with its favorable tax incentives, the planted forest area in Brazil increased from 470,000 hectares to 6.5 million hectares by 1993. With the discontinuation of the tax incentives in 1988, plantation establishment in Brazil has slowed.

Biofuels and waste represented 32% of Brazil's total primary energy supply in 2009. The total installed solar PV capacity in Brazil is estimated to be between 12 and 15 MWp, of which 50% is for telecommunications systems and 50% for rural energy systems, which is less than 0.01% of its energy supply.

- ***Banco do Nordeste***, a regional development bank, which is Latin America's largest institution dedicated to regional development, is placing the Brazilian northeast in a prominent position as a generator and user of renewable energy. Among a wide range of financing packages offered by the institution, two directly impact the promotion of the renewable energy sector in the region: The Financing program for Preservation and Environmental Control (FNE Verde) and the Credit Line for Investment in Renewable Energy and Environmental Sustainability of the National Program for Strengthening Family Agriculture (Pronaf ECO).

The FNE Verde encourages the development of activities involving the preservation of the environmental quality of the region. Emphasis is given to projects related to energy generation from renewable sources, energy conservation with environmental gains, and the replacement of fossil fuels with other clean sources of energy. FNE Verde programs benefit from preferential financial conditions. The interest rates applied vary by the size of the borrower and by the industry, from 3.75% to 10% per annum. Moreover, the projects supported by this program have repayment terms of up to 20 years (including a grace period of 8 years). A compliance bonus of 25% on interest for enterprises located in the semi-arid region is also offered. Enterprises located outside the semi-arid region receive a 15% bonus if they pay the loan installments within the specified deadlines.

In contrast, Pronaf ECO is directed at the family farmer. Its main purpose is to

provide credit support to finance renewable energy technologies (solar, wind and biomass) for biofuel mini-mills and to replace fossil fuel technology with renewable technologies in farming equipment and machinery.

In 2009, Banco do Nordeste invested R\$628 million under the FNE Verde Program, exceeding the planned target by about 80%. An important part of the growth came via the investments made to support the deployment of wind farms. Investments supported by Banco do Nordeste in the wind energy segment are part of the Growth Acceleration Program (PAC), which is supported by the federal government. PAC aims to encourage this energy source, in conjunction with promoting the supply of electricity for local populations as part of the overall regional sustainable development plan. FNE Verde is expected to grant approximately R\$650 million in new financing solely for wind farms in the northeast, which represents more than twice the total funding provided for the segment in the past.

For the next decade, investments in wind energy are expected to continue growing. For example, the National Agency of Electrical Energy, through auctions, closed the purchase of 1,805.7 MW of wind power starting in July 2012. Thus, Brazil will quadruple its installed operating capacity from the current 602 MW to 2,408 MW. The contracts will be valid for 20 years and will require about R\$20 billion in new investments. The northeast of Brazil will host 63 of the 71 winning projects from the auction, confirming its immense potential for wind energy generation.

- **State Development Bank BNDES.** A key Brazilian Government support mechanism for the roll-out of renewable energy and transmission projects is the provision of long-term commercial term lending by BNDES, the state-owned development bank for Brazil. At the end of 2011, BNDES had a total loan book of R \$635 billion (US \$400 billion), which represented 14% growth in 2010. In 2011 BNDES made advances of \$R140 billion (US \$76 billion), with 40% of this going into large national infrastructure projects. BNDES generated a net profit for the state of \$R9 billion in 2011 and delivered an average return on equity of 15% in the last three years. BNDES' current outstanding loans total \$R6.4 billion (US \$4 billion) across 78 wind farms (including \$R228 million to EDPR).

BNDES also manages the Amazon Fund (2008) and the National Fund on Climate Change (2010). In 2011, the Climate Fund had a budget of R\$226 million, of which R\$200 million comprised loans from BNDES and R\$26 million was managed by the Environment Ministry for investment in research projects. The Climate Fund is funded by government oil revenues. The Amazon region covers 60% of Brazil's territory. The Amazon Fund will be administered by BNDES, and Norway and Germany have pledged \$610 million and \$27 million for the preservation of the Amazon Region.

The Role of Feed-In Tariffs

Brazil has no FIT policy. Its current policy for solar PV is directed to off-grid installations, where there are utility programs to provide subsidies for areas that are too far removed from the grid for connection to be feasible. A project by the International Climate Initiative (ICI) is helping Brazil set up decentralized photovoltaic systems and feed the electricity generated into the grid. By December 2012, the project reached a crucial stage: if solar electricity is produced by many small installations, for example, on the roofs of houses, there is a need for regulation on how the producer is to be reimbursed for the surplus electricity fed into the grid. Following a long advisory process involving the sharing of knowledge, the Brazilian Government has now amended legislation to introduce net metering. With this system the solar electricity generated is offset against the electricity that the owner of the installation takes from the grid. The ICI project, which is being implemented by the Deutsche GIZ and KfW, supported the consultation process with the relevant stakeholders, produced technical reports, organized study visits to Germany and advised policy-makers.

In addition, the project is assisting Eletrosul, the state energy supplier, to focus public and government attention on electricity from solar power as an alternative to fossil fuel sources and to smooth the way for a sustainable expansion of solar energy in Brazil. Thus, a large pilot photovoltaic plant with an output of over a megawatt is being installed on the roof and on the car parks of the Eletrosul building in Florianopolis. Further, high-profile solar installations are at the planning stage; for example, the football stadiums for the World Cup in Rio de Janeiro and Belo Horizonte are to be similarly equipped by 2014. The ICI project is advising Eletrosul on marketing solar electricity and is also helping other energy suppliers interested in introducing solar energy.

Provincial Incentives

In addition to the national incentives, provincial and local governments provide incentives for implementing their policies.

- **Amazonas State Climate Change Policy – Law 3135.** Limited to the Amazonas region only, this law (and complementary Law 53/2007), provides the legal framework for implementation, a financial mechanism for the payment of ecosystem services, and compensation for activities that reduce emissions from deforestation and carbon sequestration.
- **Sao Paulo Climate Change Policy- Law 14933.** The city council unanimously approved Law 14.933, which ambitiously aims to reduce Sao Paulo's citywide greenhouse gas emissions by 30% of 2005 levels by 2012 through several measures focused on transportation, renewable energy, energy efficiency, waste management, construction and land use.

Cap and Trade Activity

In 2011, land use changes, namely deforestation, accounted for 36% of Brazil's greenhouse gas emissions, while energy and agriculture accounted for 27% and 28%. Brazil has had success in reducing deforestation, so it should be able to meet its UNFCCC voluntary target of a 36.1%-38.9% reduction from 1990 levels by 2020. Brazil's first government-backed carbon trading scheme, *Bolsa Verde do Rio de Janeiro* (BVRio), a non-profit civil association created in partnership with the Department of Environment of the State of Rio de Janeiro and the Treasury Department of the Municipality of Rio de Janeiro, was launched in December 2011. Brazil had been working with the UK to develop a carbon scheme that enables economic and environmental growth. It is also considering a scheme based on carbon intensity that would monitor emissions relative to economic growth, rather than actual emissions figures.

In March 2013, BNDES and the states of Rio de Janeiro and Acre signed a technical cooperative agreement for the development of the market for environmental assets in Brazil. It will enable businesses to measure and reduce GHG emissions and to structure a network of bilateral knowledge. In Brazil, structuring a market for carbon credits is a challenge that will require the establishment of regulatory standards and metrics specific to these assets. One of the actions envisaged is that the BNDES and the governments of Acre and Rio de Janeiro constitute a working group to discuss topics such as conducting carbon inventories of companies. Another step is the definition of reduction targets and how permissions to emit carbon will be distributed among market participants. From this distribution, participating companies may trade permits among themselves or buy carbon credits from other projects. The agreement seeks to support a broad debate on environmental issues and lay the groundwork for creating the environmental assets referenced in these issues, including regulation, standards, metrics and methodologies, for example.

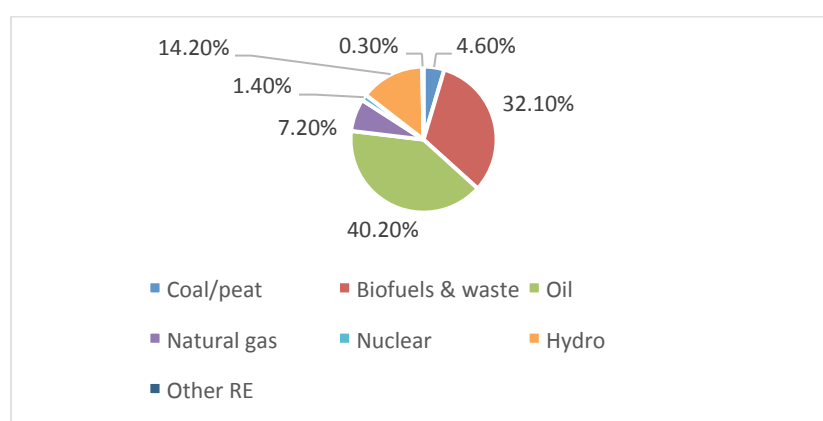
Specific Renewable Energy Technologies

According to the International Energy Agency, in 2009, Brazil's 240 million metric tons oil equivalent of total primary energy supply by source were generated by fossil fuels (53%), biofuels and waste (32.1%), hydro (14.2%), nuclear (0.7%), and other renewable energy (0.3%) (Figure A-4).

- **Wind Power.** Brazil had an installed wind power capacity of 2,508 MW at the end of 2012, up 66% year-over-year, according to the Brazilian association for wind energy, Abeeólica. There are currently 59 wind farms in operation, mostly in the northeast, making Brazil the leader in Latin America. In the last decade, the country has increased its wind power capacity 15-fold. Abeeólica estimates that, based on existing contracts, the country will have 9,000 MW of installed capacity by 2017. Wind power is now the fastest growing source of power generation in Brazil. Since 2009, when the Brazilian Government took a series of incentive measures to introduce wind power in the Brazilian energy matrix, energy auctions have already contracted about 6.7 GW of installed power. Brazil's wind power potential is

estimated at 300 GW and the energy industry expects that at least another 2 GW will be contracted per year up until 2020. While the 2009 United Nations Climate Change Conference (COP15) was taking place in Copenhagen, Brazil's national electric energy agency (ANEEL) held the country's first ever-wind-only energy auction. While focusing domestically on wind energy generation, Brazil is part of a larger international movement toward wind power as a primary source of energy. In fact, wind power has seen the highest expansion rate of all available renewable energy sources, with an average growth of 27% per year since 1990, according to the Global Wind Energy Council (GWEC).

Figure A-4
Total Primary Energy Supply by Source, 2009



Source: IEA, *Key World Energy Statistics*, 2009

- Solar Power.** In April 2012, the ANEEL board approved two new pieces of solar legislation aimed at improving the country's domestic industry. Under the first change, the Power Compensation System, ANEEL announced new regulations for net metering, meaning both businesses and residential customers can earn credits for their electricity bills, thus helping them to reduce the amount they pay for energy. The credits will be valid for 36 months. The legislation covers both micro (up to 100 kW) and macro (100 kW to one MW) photovoltaic power generation. ANEEL stated that "Customers [will be able] to gain credits on their electricity bill at both the site of power generation and remote sites by feeding excess power generation from PV systems into the distribution grid." The second change will mean Brazil's utilities are eligible for an 80% discount on taxes paid for the distribution (distribution system usage tariff) and transmission (transmission system usage tariff) of generated power from solar plants up to 30 MW. According to ANEEL, enterprises that enter into commercial operation by 31 December 2017 will have the discount of 80% applied to the tariffs during the first 10 years of operation. After the tenth year, this discount will be reduced to 50%. Meanwhile, for enterprises that enter into commercial operation after 31 December 2017, a 50% discount will be available.

The total installed photovoltaic power capacity in Brazil is negligible, other than 37 MW for telecommunications and rural energy systems. However, Brazil has

extremely favorable conditions for the production of energy through photovoltaic systems. In the northeast and midwest of the country, in particular, there is a large and constant incidence of solar radiation throughout the year. In July 2012, Sistema de Energia Renovavel (SER), a private company formed in 2011, announced plans to build 600 MW of solar capacity by 2020. In December 2012, Brazil's 21 solar power project applications totaled 1,000 MW of solar capacity. In May 2013, the government held an auction and received 13 requests for 392 MW of solar capacity, which was slightly better than its March 2013 auction of 12 requests for 300 MW.

- **Hydroelectric Power.** Hydroelectric power accounted for 71% of the total installed capacity in Brazil, at 115 GW or 82 GW in August 2011, according to ANEEL data. Brazil's technical potential for hydroelectricity generation is among the five largest in the world: it has 12% of the planet's surface fresh water as well as the conditions necessary to harness hydro potential. This potential is estimated at 260 GW, of which 40.5% is located in the Amazon Basin (in for comparison, the Paraná Basin accounts for 23%, the Tocantins Basin for 10.6%, and the São Francisco basin for 10%). However, only 63% of the potential was inventoried. The Northern Region in particular has a large untapped potential. Some of the projects currently undergoing a bidding process or already in construction in the Amazon will join the list of the ten largest in Brazil: Belo Monte (which will have an installed capacity of 11,233 MW), São Luiz do Tapajós (8,381 MW), Jirau (3,750 MW) and Santo Antônio (3,150 MW). Among the largest in operation are: Itaipu (14,000 MW, or 16.4% of the energy consumed in Brazil), Tucuruí (8,730 MW), Ilha Solteira (3,444 MW), Xingó (3,162 MW) and Paulo Afonso IV (2,462 MW).

The new hydropower plants in the Northern Region present a logistical challenge: transmission to the large cities, which are thousands of miles from the plants. This problem will be solved by the Integrated National System (SIN), a network of transmission lines and power plants that operate in an integrated manner and that cover most of the nation's territory. The Ten Year Energy Plan [2011-2020] forecasts that Brazil will have 71 new plants with 29,000 MW, of which 28 plants will be in the Amazon region with 22,900 MW.

- **Biomass and Waste.** Brazil's biomass and waste generated 32.1% of its total primary energy in 2009. According to the Brazilian Association of Industry Biomass and Renewable Energy, Brazil produced 603,393,522 tons of agricultural residue and 157,992,556 cubic meters of forestry residue, or 246,197.85 tons of oil equivalent in 2010. This generation reduces carbon emissions by 10 million tons per year as well as environmental pollution, as it uses organic garbage, agricultural remains, wood shavings or vegetal oil. Refuse cane, with its high energetic value, has been used to produce electricity.

The recent interest in converting biomass to electricity comes not only from its potential as a low-cost, indigenous supply of power, but also for its potential environmental and developmental benefits. For example, biomass may be a globally

important mitigation option to reduce the rate of CO₂ buildup by sequestering carbon and displacing fossil fuels. Renewably grown biomass contributes only a very small amount of carbon to the atmosphere. Locally, plantations can lessen soil erosion, provide a means to restore degraded lands, offset emissions and local impacts from fossil-fired power generation, and, perhaps, reduce demands on existing forests. In addition to the direct power and environmental benefits, biomass energy systems offer numerous other benefits, especially for developing countries such as Brazil. Some of these benefits include the employment of underutilized labor and the production of co- and by-products, for example, fuel wood. Nearly all of the experience with biomass for power generation is based on the use of waste and residue fuels (primarily wood/wood wastes and agricultural residues). The production of electric power from plantation-grown wood is an emerging technology with considerable promise. However, the actual commercial use of plantation-grown fuels for power generation is limited to a few isolated experiences. Wood from plantations is not an inexpensive energy feedstock, and as long as worldwide prices of coal, oil and gas are relatively low, the establishment of plantations dedicated to supplying electric power or other higher forms of energy will occur only where financial subsidies or incentives exist, or where other sources of energy are not available.

Biomass plantations are supplying energy on a commercial basis in Brazil. A combination of government policies and/or high conventional energy prices has stimulated the use of short-rotation plantations for energy. The Brazilian forest sector is one of the most developed and competitive in the world; it uses mostly eucalyptus and pine. The country holds a relevant global planting share: 6.3 million hectares, according to the Brazilian Association of Planted Forest Producers (ABRAF). Approximately one third of this area – 2.2 million hectares – corresponds to pulp and paper planted forests, including 425,000 hectares in more than 27,600 properties of independent producers – many of them participate in forest partnership programs – that supply the sector. According to an IEA study, the maximum potential for sustainable biomass production in 2050 is estimated at between 250 to 500EJ (the exajoule (EJ) is equal to 10¹⁸ joules).

- **Geothermal Energy.** Geothermal energy is virtually untapped in Brazil. According to a study published in *Pure and Applied Geophysics*, the best sites for the extraction of geothermal energy in Brazil are the younger sedimentary basins, which make the Paraná Basin a suitable place to tap geothermal. It is located in southern Brazil, where most of the population, and therefore, energy demand, are located. Measuring the geothermal gradients, it has been found that the major aquifers of this basin contain large quantities of warm water in the temperature range 40 to 90°C. The investments needed to convert this energy into electricity would appear to be too expensive at this time. However, other applications can be found for this, like residential and commercial systems utilizing hot water.

- **Smart Grid Rollout.** Investments in the smart grid in Brazil are supposed to hit \$36.6 billion by 2022, according to a report from the Northeast Group. That report says that Brazilian utilities will use smart grid investments to reduce electricity theft, improve grid reliability, and simply to build out electricity infrastructure over the coming decade. Brazilian power utilities are forecast to invest \$30 billion (\$16 billion) in the national rollout of 64 million smart meters. One of the key drivers of this policy is to dramatically reduce “non-technical losses in distribution” (power theft). Brazil’s national electricity regulator estimates losses currently run at R\$8 billion annually, and proposed to bring in legislation by the end of 2012 to require utilities to replace all meters. As in India, power theft is the key driver of this government policy in Brazil. This investment will also underwrite the development of a smart grid whereby all electricity customers will be able to optimize their energy usage via time-of-use pricing, peak demand curtailment, and the introduction of smart appliances, which are key applications of energy efficiency.

A.5 Kingdom of Thailand

Socioeconomic Survey

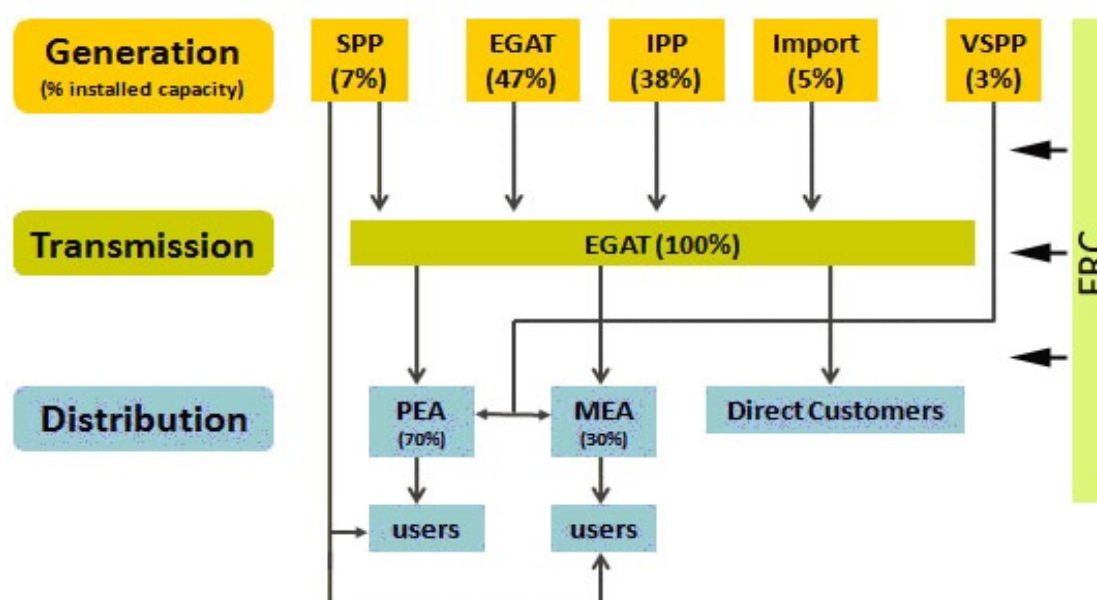
With a well-developed infrastructure, a free-enterprise economy, generally pro-investment policies, and strong export industries, Thailand achieved steady growth in 2012 due largely to industrial and agriculture exports, mostly electronics, agricultural commodities, automobiles and parts, and processed foods. Exports generated a \$366 billion GDP, according to the IMF.

Thailand is trying to maintain growth by encouraging domestic consumption and public investment to offset weak exports in 2012. Its unemployment, at less than 1% of the labor force, is one of the lowest levels in the world, which puts upward pressure on wages in some industries. Thailand also attracts nearly 2.5 million migrant workers from neighboring countries. The Thai Government is implementing a nationwide 300 baht (\$10) per day minimum wage policy and deploying new tax reforms designed to lower rates on middle-income earners.

The Thai economy has weathered internal and external economic shocks in recent years. The global economic severely cut Thailand's exports, with most sectors experiencing double-digit drops. In 2009, the economy contracted 2.3%. However, in 2010, Thailand's economy expanded 7.8%, its fastest pace since 1995, as exports rebounded. In late 2011 growth was interrupted by historic flooding in the industrial areas in Bangkok and its five surrounding provinces, crippling the manufacturing sector. Industry recovered from the second quarter of 2012 onward, with GDP growth at 5.5% in 2012. The government has approved flood mitigation projects worth \$11.7 billion, which were started in 2012, to prevent similar economic damage and an additional \$75 billion for infrastructure over the next seven years with a plan to start in 2013.

Over the past 20 years, Thailand’s electricity sector has evolved from a government monopoly to a semi-unbundled structure called the “Enhanced Single Buyer” model (Figure A-5). This model consists of the state-owned Electricity Generating Authority of Thailand (EGAT), which owns about 50% of generation assets and 100% of transmission assets. The other half of the generation assets are developed and owned by private companies, including independent power producers (IPPs), small power producers (SPPs), and very small power producers (VSPPs). IPPs and SPPs produce and sell power to the high-voltage transmission system owned by the only buyer, EGAT. VSPPs sell power through the two state-owned distribution systems, the Metropolitan Electricity Authority (MEA) and the Provincial Electricity Authority (PEA). Thailand’s policies related to energy, including electric power and renewable energy, are drafted and proposed by the Ministry of Energy (MoE). Policies related to electric power and natural gas transmission are regulated by the Energy Regulatory Commission (ERC).

Figure A-5
Thailand’s Electricity Sector
The Enhanced Buyer Model



Source: World Energy Council

In 2012, Thailand produced and consumed 173.3 billion kWh and 169.4 billion kWh, of which 89%, 10.9% and 0.2% were generated from fossil fuels, hydro and other renewable energy. Thailand is a net importer of electricity.

Legislative Initiatives Affecting Renewable Energy Development

Thai energy laws promote renewable energy in three ways. First, the 1992 National Energy Policy Act empowers the National Energy Policy Council and the Office of Energy Policy and Planning to be responsible for the making and regulating of Thailand’s energy policies and plans, including those of renewable energy. Second, the 1992 Promotion of Energy

Conservation Act, which aims at the efficient and economical production and use of energy, extends incentives to those who put efforts into energy conservation measures and disincentives to those who fail to comply; such measures can be used to promote renewable energy, especially by substituting non-renewable energy with renewable energy measures. Third, the Department of Alternative Energy Development and Efficiency, which is the key state agency in relation to the technical aspects and regulation of energy production, transmission and distribution under the 1992 Energy Development and Promotion Act, plays a significant role in renewable energy promotion. In addition, the measures related to “regulated energy,” if wisely used, can be useful in renewable energy promotion.

- ***The Renewable and Alternative Energy Development Plan, 2012-2021*** (AEDP) was approved in late 2011 and is managed by the Ministry of Energy. Thailand has to rely mainly on energy imports. Findings from 2011 data indicate that over 60% of Thailand’s primary commercial energy demand is derived from imports. Oil imports, at 80% of total domestic consumption, are trending upwards since Thailand is not capable of increasing its domestic petroleum production to meet demand. Renewable energy is being counted on as a means of significantly substituting natural gas in power generation, with solar energy, wind energy, hydroelectric power, biomass, biogas and waste/garbage being the principal targets. The AEDP targets 25% from renewable energy by 2021.

The Role of Feed-In Tariffs or Adder Program¹²

Thailand was one of the first Asian countries to implement a FIT program. Thailand’s FIT program, or “adder” program, ensures guaranteed purchases and attractive tariff rates to eligible grid-connected renewable power projects. The policy directive was initiated in 2006 and implemented in early 2007. The program is called “adder” because it adds additional payments to renewable energy generators on top of the normal prices that power producers would receive when selling electricity to the power utilities.

The MoE drafted the program and proposed it to the National Energy Policy Commission. The Energy Regulatory Commission (ERC) has legal authority to set rules and regulate the implementation of power policies, including the adder measure, in Thailand. Thailand’s adder program gives incentives for private investors to invest in renewable energy projects by guaranteeing attractive power purchasing rates. Eligible participants enter into long-term contracts with the local utility to sell renewable electricity at a pre-specified tariff for a pre-specified period of time. The program is implemented through Thailand’s three electric utilities: EGAT, MEA, and PEA. The three utilities purchase electricity from renewable electricity generators using two types of regulations: VSPP regulations for generators sized less than or equal to 10 MW and SPP regulations for generators sized greater than 10 MW and less than 90 MW. The Adder rates are distinguished by technology type, installed capacity, contracted capacity, and project location.

¹² Dr. Sopitsuda Tongsovit and Dr. Chris Greacen, “Thailand’s Renewable Energy Policy: FITs and Opportunities for International Support, University of California, 31 May 2012.

- **Eligibility.** VSPPs and SPPs that utilize solar, wind, biomass, biogas, hydro, and waste energy are eligible to participate in the program. VSPPs and SPPs may be private or public entities, but may not include the utilities themselves
- **Rate Structure.** The rate structure of the feed-in tariffs used in Thailand since 2007 is paid on top of the utility's avoided costs. This constitutes a "premium-price FIT payment." In contrast, a "fixed-price FIT payment" (Cory et al., 2009) is characterized by a FIT rate that is independent of the country's electricity prices. In June 2010 the Thai Government approved a plan to switch from a premium-price FIT payment to fixed-price FIT payment, and studies to determine the rate for each type of renewable energy are under way.
- **Rates.** Thailand's adder rates are differentiated by technology and installed capacity. In addition, "additional" adders are paid to renewable energy generators in special locations. Those special areas include the three southernmost provinces, which have experienced political unrest, and off-grid areas where the PEA typically delivers electricity from local diesel power plants. The adder rates are denominated in US dollars per kWh as shown in Table A-7.

Table A-7
Feed-in Tariff or Adder Program by Type of Renewable Energy

Type of Renewable Energy	2010	Special Adder for Diesel Replacement	Special Adder for 3 Southern Provinces	Years Supported
<i>Biomass</i>				
Installed Capacity <1 MW	0.017	0.033	0.033	7
Installed Capacity >1 MW	0.010	0.033	0.033	7
<i>Biogas</i>				
Installed Capacity <1 MW	0.017	0.033	0.033	7
Installed Capacity >1 MW	0.010	0.033	0.033	7
<i>Waste</i>				
Landfill and Digestor	0.083	0.033	0.033	7
Thermal Process	0.117	0.033	0.033	7
<i>Wind</i>				
Installed Capacity <50 kW	0.150	0.050	0.050	10
Installed Capacity >50 kW	0.117	0.050	0.050	10
<i>Small Micro Hydro</i>				
50 kW<Installed Capacity<200 kW	0.027	0.033	0.033	7
Installed Capacity <50 kW	0.050	0.033	0.033	7
<i>Solar</i>				
Installed Capacity	0.217	0.050	0.050	10

Source: World Resources Institute

The strengths of Thailand's adder program lie in rates that have been set high enough to attract private investment in diverse forms of renewable energy, and in its early framework for streamlined interconnection arrangements and standardized documents, including power purchase agreements. In addition, special rates for the three southern-most provinces help address the increased risks of investors assume when operating in those areas. The provision of special rates for remote areas with PEA-supplied diesel generation is also a good step toward advancing the concept of fossil fuel substitution and helping lower utility costs in rural areas with expensive diesel-fired generation. The rate structure is simple and easy for investors to integrate into investment plans. The program also provides secure contracts that ensure confidence to the investors that the support will not likely be revoked or cancelled once they have signed the contracts. As a result, the adder program has been able to induce impressive amount of investment in RE since its implementation in 2007.

As of December 2011, the on-grid renewable energy capacity of Thailand was 985.36 MW (589.96 MW of renewable VSPP and 395.40 MW of renewable SPP), and 7,558 MW of renewable energy capacity are in the pipeline (5,547 MW of renewable VSPP and 2,011 MW of renewable SPP). The combined renewable energy on-grid capacity and the amount in the pipeline is 8,543 MW, or around 27% of the current installed capacity in Thailand.

Central Government Investment in Renewable Energy Development

According to the Ministry of Energy's estimation, by 2021, the demand for energy in Thailand will increase from 71,728 kilo-tons of oil equivalent (ktoe) or 71.728 million metric tons of oil equivalent today, to 99,838 ktoe, or a 39.19% increase. The Thai Government is hoping to push the use of alternative and renewable energy to reach 25% of total energy consumption. This is possible because Thailand has agricultural products that can be used as energy sources such as biomass, biogas, biodiesel and ethanol. Food industries also yield a great amount of byproducts that can be made into energy from waste. Thailand's natural resources also have great potential for energy generation: the country has average sunlight energy of 18.2 MJ/m²/day, not to mention the nation's potential in wind energy. This makes Thailand attractive for alternative and renewable energy investment, and should lead to a significant decrease in GHG emissions, making Thailand a low-carbon society in the future.

The Board of Investment (BOI) in the Ministry of Industry has policies to support this type of investment by offering tax incentives and other benefits. For example, the BOI is offering tax exemptions or reductions for imported machines and materials as well as corporate tax exemptions or reductions. Incentives in other areas include permission to bring in foreign skills, permission to own land, and foreign currency cash flow. Foreign investors can also hold 100% of the shares.

Under the Investment Promotion Act B.E. 2520, the Board of Investment is authorized to grant the following incentives, guarantees and protections:

Incentives

Tax Incentives

- Exemption/reduction of import duties on machinery
- Reduction of import duties for raw or essential materials
- Exemption of juristic person's income tax and dividends
- A 50% reduction of the juristic person's income tax
- Double deductions from the costs of transportation, electricity and water supply
- Additional 25% deduction of the cost of installation or construction of facilities
- Exemption of import duty on raw or essential materials for use in production for export.

Non-Tax Incentives

- Permits for foreign nationals to enter the Kingdom for the purpose of studying investment opportunities
- Permits to bring into the Kingdom skilled workers and experts to work in investment promotion activities
- Permits to own land
- Permits to take out or remit money abroad in foreign currency.

Guarantees

- The state will not nationalize the activity of the promoted person
- The state will not undertake a new activity in competition with the promoted person's
- The state will not monopolize the sale of products similar to the promoted person's
- The state will not impose price controls on the products of the promoted person
- The state will grant permission to export at all times
- The state will not allow any government agency, government organization or state enterprise to import any kind of the product being produced by the promoted person into the Kingdom by granting import duty exemption.

Protections

- To charge extra import fees on products similar to those produced by the promoted person at a rate not exceeding 50% of the price of overseas insurance and freight charges, effective for a period of not more than one year
- If the Board is of the opinion that Section 49 is inadequate for protecting the activity of the promoted person, it will increase the measure by banning the import of products similar to the local productions
- In the case where the promoted person encounters any problem or obstacle in the course of carrying out the promoted activity, the chairman will have the power to render any appropriate assistance.

Finance

The Government of Thailand established the Energy Conservation (ENCON) Fund in 1992 to foster the expansion of energy efficiency and renewable energy projects by mobilizing and leveraging additional investments in mitigation projects. The ENCON Fund was initially available to large-scale industrial and commercial facilities and later opened up to ESCOs and small- to medium-sized enterprises. (By way of clarification, energy service companies (ESCOs) are a third party service provider that engages in a performance-based contract to implement measures to reduce energy consumption and costs in a technically and financially viable manner.) The objective of the ENCON Fund was to enable these developers to access and leverage capital, expand the ESCO market, stimulate private sector investment, and increase financial confidence in the energy efficiency and renewable energy sectors.

The Energy Efficiency Revolving Fund (EERF) was launched in 2003 to stimulate investments in large-scale industrial projects by increasing the availability of debt financing for energy efficiency and renewable energy projects while minimizing the borrowing costs to project developers. The EERF was initially funded with THB 2 billion (\$50 million) from the ENCON Fund and was managed by Department of Alternative Energy Development and Efficiency (DEDE) of the Ministry of Energy. During the pilot phase of the program, both the Global Environment Facility (GEF) and the Danish Government provided technical assistance in program design. Funding was initially available to the large energy consuming “designated” facilities; however, over time it was expanded to cover any commercial buildings, industrial facilities, and/or ESCOs.

The EERF initially provided credit lines at 0% interest to participating local banks in the range of \$2.5 million to \$10 million to finance energy efficiency projects. As the financing volume grew, banks began to co-finance projects and the interest rate was subsequently reset to 0.5% to cover administrative costs. Local banks were obligated to repay the principal and interest back to the EERF within 10 years.

Local banks were able to provide low-interest loans that covered up to 100% of project costs; loan amounts were limited to THB 50 million (\$1.4 million) per project. Loans were provided to developers of energy efficiency and renewable energy projects and ESCOs at a fixed rate between 0% and 4% (compared to the market rate of 9%), with up to a seven-year loan period. The repaid loans were then used to finance new energy efficiency and renewable energy projects, hence the “revolving” design of the Fund. The projects that required less than THB 50 million (\$1.63 million) borrowed 100% from the EERF, and those that required over THB 50 million (\$1.63 million) solicited additional debt financing from local banks.

The loan eligibility for projects was determined by local banks via loan applications and assessed by evaluating balance sheet strength and the quality of securable assets versus future cash flows and savings from the projects. As collateral for a loan, banks required mortgages over land, building(s) or equipment owned by the applicant. Since project

proponents were required to submit a feasibility study upon applying for a loan, very few applications were rejected during the program. Furthermore, the EERF was designed to remove credit risk from the government, since banks were required to repay the credit lines to the EERF in the case of default. As such, banks were also free to terminate any loan past due or in default, and restructure the loan with their own conditions to recover the losses. EERF funding covered equipment installation and upgrades consultation, civil works, piping, transportation, and tax. Land costs, land improvement costs, and building construction projects did not qualify for funding.

The banks managed the loans and reported project results to DEDE. DEDE monitored the banks' performance and measured the program's energy savings. This approach minimized government involvement in the financing process and helped to leverage bank finance. The cost to the government is the opportunity cost associated with providing 0% loans to the local banks.

By the close of the EERF program, 13 public and local banks participated in the program, which resulted in 294 projects. The total investment was THB 15,959.05 million (\$521.5 million), which consisted of THB 7,231.94 million (\$236.34) from the EERF and THB 8,727.10 million (\$285.2 million) in debt financing from local banks. The GHG emissions reductions totaled .98 million tons of CO₂e and the total financial savings were estimated to be THB 5,394 million per year (\$169 million). The majority of the projects funded involved the replacement of chillers and the installation of biogas facilities. While the ratio for financial contribution for lending under the program was initially 1:1 between DEDE and the banks, the private sector finance ratio steadily increased over time.

The ESCO Fund began in 2008 with an initial budget of THB 500 million (\$15.7 million) and was targeted at small- to medium enterprises and ESCOs. The Fund is sponsored by DEDE and is managed by two government-appointed non-profits, the Energy Conservation Foundation of Thailand (ECFT) and Energy for Environment Foundation (E for E). With a fixed government budget, both Fund managers have the ability to co-invest in energy efficiency or renewable energy projects and/or ESCOs, and also can provide assistance in securing funding from other co-investors.

Phase 1 of the project ran from October 2008 through September 2010, with a total ESCO Fund budget of THB 500 million (\$16 million), with half managed by E for E and half by the ECFT. *Phase 2* of the project ran initially from October 2010 through Sept 2012, with a total budget of THB 500 million (\$16 million), and was recently extended to end in 2013 with an expanded budget of THB 3.5 billion (\$116 million). *Phase 3* of the program is scheduled to begin in late 2013 with a budget of THB 500 million.

The ESCO Fund provides a variety of financial mechanisms project developers and/or ESCOs can take advantage of to access and leverage capital. The ESCO Fund includes six funding assistance instruments. They include: equity, venture capital, equipment leasing, partial credit guarantees, carbon credit trading, and technical assistance.

Equity Investment

The equity investment mechanism provides co-investment in energy efficiency or renewable energy projects. This mechanism increases opportunities for capital intensive-energy efficiency and renewable energy projects to expand operations. Since the ESCO Fund is government run, the main objective is to support and promote energy efficiency and renewable energy investments, and not to obtain a maximum return on projects, which is normally the case for equity financing.

- The equity investment covers 10 to 50% of the total cost of a project investment but is limited to THB 50 million (\$1.6 million) per project. The funds can be used to purchase, install and maintain new equipment;
- The internal rate of return (IRR) on projects ranges from 9-15% and is returned in the form of a dividend and/or exit price. However, E for E requires an IRR of 13%-15% on commercial projects with existing operating facilities, while new commercial projects require a discount rate (3%-4% lower than the project IRR) but not less than 9%. The project payment period can be up to 5 to 7 years, or until the project developer is prepared to buy back shares, find a strategic partner, or list its shares on the Thai stock market (SET/MAI). The exit price is set at the market rate, “put option” and “call option,” and is negotiable by the involved partners. Additionally, it is required that the ESCO fund manager has a seat on the Board of Directors of the company.
- The eligibility for equity financing from the ESCO fund is contingent on good financial standing (to be verified through the submission of balance sheets up to three years prior to submitting an application), a feasibility study with an indication of the return on the project, and an environmental assessment report.

While the equity financing mechanism has been successful in spurring new investments by helping to provide the necessary equity investment to meet bank debt/equity ratios, the cap on accessible financing of up to THB 50 million (\$1.6 million) per project remains a key barrier to investments in large scale renewable energy projects with high upfront capital costs (i.e., wind turbines, municipal solid waste power plants), since many project developers prefer not to co-invest with multiple investors. An example of how the equity fund works is provided by a review of a solar farm in Nakorn Ratchasima Province. The developer, Solar Power Company, installed a total of 6.1 MW in solar capacity, using polycrystalline PV modules. The total investment of the project was THB 723 million (\$23.5 million), of which THB 40 million (\$1.3 million) was co-invested by E for E through equity financing, and THB 434 million by a local bank through debt financing. The project’s debt-to-equity ratio was 60:40.

Venture Capital for ESCOs

ESCOs in Thailand have historically experienced constraints to accessing finance since they are small enterprises and are more technology-focused (versus capital-focused), and

therefore have low values of registered capital. The ESCO VC is a mechanism for co-investment in existing ESCOs to increase the registered capital of the company for new investments, and is intended to encourage growth in the ESCO sector.

Terms of financing:

- VC provides equity investments of up to 30% of the registered capital of the ESCO, but is limited to THB 50 million (\$1.6 million) per project. The investment payback period ends within 5 to 7 years, at which time investors can exit the project by selling shares back to the entrepreneur, finding a new strategic partner(s), or listing on Thailand's stock market (MAI/SET). The exit price is set at the market rate, "put option" and "call option," and is negotiable.
- Additionally, it is required that the investor have a seat on the Board of Directors for a period of up to 3 to 7 years.
- This mechanism remains underutilized since E for E Fund managers are reluctant to offer VC financing due to the limited amount of registered capital that ESCOs retain and therefore the large share of the company ownership that transfers to investors under this arrangement.

Equipment Leasing

Equipment leasing can be utilized for both small and large energy efficiency and renewable energy projects. This mechanism allows the ESCO fund manager to carry out an equipment purchase agreement with an ESCO and eliminates the need for project developers to invest directly in new technology equipment upgrades.

Terms of financing:

- The lessee (ESCO or project developer) will select an asset (equipment, vehicle, software)
- The lessor (ESCO fund manager) will purchase that asset up to 100% of equipment costs, up to a maximum of THB 10 million (\$325,670) per project
- The lessee will have use of that asset during the lease period of up to 5 years
- The lessee will pay a series of rentals or installments and interest (4% per year) for the use of that asset through revenue from the energy savings of the project
- The lessor will recover a large part or all of the cost of the asset plus earn interest from the rentals paid by the lessee

- The lessee has the option to acquire ownership of the asset (e.g., paying the last rental or bargain option purchase price).

In order to secure financing, ESCOs are required to be in good financial standing, use pre-approved technology and provide a guarantee savings contract. Other requirements include a feasibility study and the electricity records of the company (three years prior) upon the application submission. An example of an equipment leasing contract involves the Motorwork Factory in Samut Prakan Province, where E for E provided THB 5.56 million (\$181,073) in equipment lease financing to support the installation of a voltage regulator. The project will accrue over THB 110,000 (\$3,582) per month in energy savings.

Credit Guarantee Facility

The credit guarantee facility is modeled after Thailand's state-owned Small Business Credit Guarantee Corporation (SBCG), which is supervised and run by the Ministry of Finance. The SBCG provides partial credit guarantees to commercial banks for loans to small enterprises eligible for debt financing.

Terms of financing:

- The Credit Guarantee Facility will cooperate with financial institutions and/or credit guarantee agencies to assist project developers or ESCOs in accessing long-term loans from banks by providing credit guarantees depending on the risk of the project and limited to THB 10 million (\$334,000) per project.
- The project developers pay a fee to the guarantor (ESCO fund manager) at 1.75% per year of the guarantee amount. Currently, the design of the mechanism involves the ESCO fund manager providing funds to commercial banks to on-lend to project developers or ESCOs, and provides the partial credit guarantee to the project developer.

To date, the Credit Guarantee Facility has been underutilized, especially for large project investments, since the maximum guarantee amount is relatively small at THB 10 million (\$334,000). The mechanism could be improved by expanding the loan ceiling and also by establishing a fourth institution to take on the additional credit risk.

At the close of Phase 1 in 2010, a total of THB 330 million (\$10.8 million) had been deployed by the ESCO Fund, which resulted in a total investment of THB 3,334 million (\$109 million). During Phase 1 the majority of investments were equity investments (76%), while equipment leasing projects received (24%), and venture capital (.2%). The majority of the 33 projects were leasing projects (23), followed by equity investments (9), and ESCO venture capital (1). The types of projects supported were primarily in biomass power, followed by solar power, energy efficiency, biogas, biomass thermal energy, then solar water heater projects. By the end of Phase 2 (2012) the energy efficiency and renewable energy projects

were projected to create energy savings of 23.97ktoe/year and financial savings of THB 932.3 million (\$29 million) per year.

The key lessons that can be drawn from the ENCON Fund include the following:

- The EERF and ESCO Funds successfully expanded the availability of commercial financing for clean energy projects.
- EERF low-interest rate loans to banks and a partial credit guarantee scheme mobilized substantial private sector financing for energy efficiency and renewable energy projects and technologies. The EERF effectively reduced financial barriers for project developers and ESCOs, and cleared the path for market entry in Thailand. Moreover, these mechanisms reduced the time and management costs tied to the government by allowing local banks to on-lend public funds.
- The EERF was designed to deploy funds in a sustainable manner, which increased the number of energy efficiency and renewable energy projects overall, since loan repayments created new investments in more projects.
- The EERF required about 3-4 years to achieve a good application rate. This phase, however, was critical in boosting commercial lending confidence in the renewable energy and energy efficiency sector. The EERF and ESCO Funds were effectively used to provide commercial banks with a better and deeper understanding of project risks, costs, returns, and technology performance. The funds expanded the availability of commercial financing for clean energy projects.
- Through the ESCO Fund, the management of equity financing by a government agency allowed investors to lower their required rate of return on projects, allowed for greater control and participation of shareholders in projects, and established an approval process that mitigated the risk of project overruns and uncertain energy savings.
- ESCO Fund managers comprised technical, financial, and policy experts who mitigated project risks for the Fund and private sector investors.
- The EERF and ESCO Funds created a space for the ESCO industry to emerge, under both shared savings and guaranteed savings models. ESCO services have reduced credit risk and performance risk for energy efficiency and renewable energy projects by providing turnkey energy services with performance-based and/or lending-based contracting to energy users. Ultimately this boosted lending confidence in the sector and local banks were eventually able to lend independent of the government's involvement.
- Pilot projects were instrumental in determining the underlying economic feasibility, assessment of energy savings, and GHG reductions for specific projects.

The ENCON Fund has been critical to Thailand's AEDP 2012-2021 and Energy Efficiency Master Plan. Moving forward, it can evolve and encourage greater growth in renewable energy generation, and will be instrumental to actualizing the 25% renewable energy supply target of the AEDP 10-year plan. Currently, the political signals warrant even more investment in these sectors, and having an already experienced banking and public financing structure, the government has enabled a structure for public private partnerships. Future projects could be financed using a combination of domestic, public, multilateral, and bilateral financing to execute larger-scale projects, and to actualize the goals set forth in the AEDP 10-year plan.

Cap and Trade Activity

Thailand signed the United Nations Framework Convention on Climate Change (UNFCCC) in June 1992 and ratified the Convention in March 1995. Realizing the seriousness of climate change as a global threat, the country has been contributing to international efforts to address climate change issues as a Non-Annex I country. In February 1999, Thailand signed the Kyoto Protocol, and ratified it on 28 August 2002. As a Non-Annex I country, Thailand promotes the implementation of Clean Development Mechanisms (CDM) under the Kyoto Protocol in order to encourage clean and environmentally friendly technologies for reducing GHGs, as well as to promote the country's capability by developing sustainable business practices. The Ministry of Natural Resources and Environment (MNRE), as a result of a Cabinet Resolution, is the responsible agency for the implementation of the UNFCCC and Kyoto Protocol in Thailand. In 2007, MNRE redesigned the institutional framework for Thailand's implementation of the UNFCCC and the Kyoto Protocol.

In 2007, the Thailand Greenhouse Gas Management Organization (TGO) was created by Royal Decree. TGO is autonomous governmental organization with a specific purpose as an implementing agency on greenhouse gas emission reduction in Thailand, promoting low-carbon activities, investment and marketing of GHG emission reductions, establishing a GHG information center, reviewing CDM projects for approval, certifying certified emission reductions (CER), providing capacity development and outreach for CDM stakeholders, and performing its role as the Designated National Authority for the CDM office in Thailand.

The TGO's objectives and duties are:

- Analyzing and screening CDM projects for issuance of letters of approval and monitoring the projects
- Promoting CDM projects and the CER Market
- Serving as the national information clearinghouse for GHGs
- Managing all information on the approved CDM projects and CER values

- Enhancing the capacity building of the government and private sectors on greenhouse gas management
- Promoting public outreach regarding greenhouse gases
- Promoting and supporting all activities related to climate change mitigation.

Thailand's GHG emission by sector in carbon dioxide (CO₂) equivalent in millions for 2000 is shown in Table A-8.

Table A-8
Greenhouse Gas Emissions by Sector

Sector	CO ₂ e	% of National Total
Energy	159.39	69.6%
Industrial Processes	16.39	7.2
Agriculture	51.88	22.6
Land-Use Changes and Forests (sinks)	-7.90	-3.4
Waste	9.32	4.1
Total Emissions and Removals	229.08	100%

Source: Thailand's Second National Communication under UNFCCC

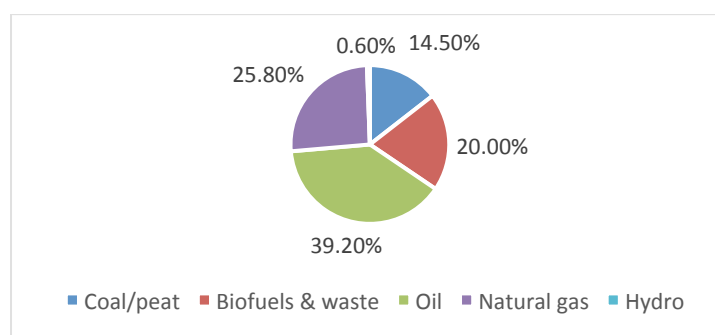
On 7 March 2013, the TGO issued letters of approval for 221 projects. The expected average annual CERS is 12,710,309 tCO₂e. Also, 125 projects have been registered. Their expected average annual CERs is 6,123,393 tCO₂e.

The World Bank's Carbon Finance Unit signed emissions reductions purchase agreements with four projects in Thailand with 4,191,615 tCO₂e total project emission reductions.

Specific Renewable Energy Technologies

According to the International Energy Agency, in 2009, Thailand's 103 million metric tons oil equivalent (MTOE) of total primary energy supply by source were generated from fossil fuels (79.5%), biofuels and waste(20.0%), and hydro (0.6%), as shown in Figure A-6.

Figure A-6
Total Primary Energy Supply by Source, 2009



Source: IEA, *Key World Energy Statistics*, 2009

The AEDP 2012-2021 targets the following renewable energy additions over a ten-year period:

Table A-9
Renewable Energy Target Additions by 2021

Sector	2021	Current
Solar	2,000 MW	75.48 MW
Wind	1,000 MW	7.28 MW
Hydro	1,608 MW	86.39 MW
Municipal Solid Waste	160 MW	1.45 MW
Biomass	3,630 MW	1,751.86 MW
Biogas	600 MW	138 MW
Geothermal	1 MW	350kW
Wave and Tidal Current	2 MW	None
Transport Fuel		
Ethanol	9ml/day	1.3ml/day
Biodiesel	5.97ml/day	1.62ml/day
Future New Fuel	25ml/day	None

Source: Department of Alternative Energy Development and Efficiency, Ministry of Energy.

The Ministry of Energy established a roadmap for promoting renewable and alternative energy development, with a target of 25% in 10 years (AEDP: 2012-2021) through the following six strategies:

1. Promoting the community to collaborate in broadening, producing and consuming renewable energy
 2. Adjusting the incentive measure on investment from the private sector
 3. Amending laws and regulations that do not benefit renewable energy development
 4. Improving infrastructure (transmission lines, power distribution lines, and activities toward the development of a smart grid system
 5. Implementing public information and capacity building programs
 6. Promoting research as a mechanism to develop the integrated renewable energy industry.
- **Solar Energy**, AEDP 2012-2021 sets a target of 2,000 MW by 2021:
 - Promote the installation of small system projects at the household and community levels, including solar PV rooftop, for 2,000 MW in 10 years. This may

include system installations on houses in general and communities, office buildings and factory roofs, housing projects or condominiums, and government buildings.

- Adjust incentive measures for investment by the private sector by adjusting the adder to the FIT.
- Amend laws and regulations that do not benefit renewable energy development.
- Improve the infrastructure system by asking the three transmission companies to expand and increase the transmission system in support of rising renewable energy and prepare for the development of a smart grid system.
- Build comprehensive knowledge of the private sector in using solar energy
- Promote research on the development of integrated renewable energy.
- **Wind Energy**, AEDP 2012-2021 sets a target of 1,200 MW by 2021:
 - Promote wind turbines in co-generation with other power generating systems in remote communities and non-electrified islands, and in agriculture, especially wind turbines for water pumping and traditional waterwheel for consumption.
 - Facilitate private sector requests for land use and the installation of wind turbines in remote areas.
 - Amend laws and regulations that do not benefit wind power, such as regulations and codes of practice for land utilization.
 - Improve the infrastructure system by establishing an extension plan for transmission and energy storage systems, for instance: pumped storage systems in area with high wind energy potential;
 - Promote establishing a network of wind energy users and producers.
 - Encourage the emergence of an industry for electricity storage devices and integrated wind power generation systems, and develop the wind turbine type appropriate for Thailand.
- **Hydropower**, AEDP 2012-2021 sets a target of 1,608 MW, excluding the 500 MW EGAT pump storage system at Lam Takong:
 - Promote hydropower generation at the village level for non-electrified households by not connecting them to the transmission system, support the construction of hydropower plant projects at the community level, and let the

local administrative organization or local people collaborate as project owners so they can learn how to self-manage and maintain their facility.

- Amend laws and regulations by solving the problems and barriers in micro-hydropower project developments that might be located in sensitive areas
- Improve infrastructure by assigning the Department of Alternative Energy Development and Efficiency (DEDE) and Electricity Generating Authority of Thailand (EGAT) to develop small hydropower systems downstream from irrigation dams and mini hydropower systems with generating capacity of 200 - 6,000 kW, and assigning EGAT to develop new pumped storage projects in two areas: Lam Takong Pumped Storage (unit 3-4) at 500 MW, and Chulabhorn Dam Pumped Storage Project with a capacity of 784 MW.
- Conduct public information campaigns on the advantages of hydropower.
- Conduct research and develop a micro-hydro run-of-river (water flow) turbine, and study and develop a low-head hydro turbine for river currents and tidal flows.
- **Energy from Waste (Municipal Solid Waste)**, AEDP 2012-2021 sets a target of 160 MW by 2021:
 - Promote and support producing energy from MSW in small and medium-size communities
 - Speed up the amendment of the Joint Venture Act B.E 2535 to allow the private sector to co-invest with local administrative organizations in producing energy from MSW of all types
 - Build up collaboration in targeted areas for the establishment of waste-to-energy systems, conduct campaigns to educate children and juveniles on the details of waste management for energy and environment at the local level
 - Study refuse-derived fuel management, research and develop the domestic production of incinerators and small waste-to-energy systems with a capacity of not over 50 tons/day, and develop the standards and appliances for producing oil from plastic waste.
- **Biomass**, AEDP 2012-2021 has targeted 3,630 MW by 2021:
 - Promote the establishment of distributed green generation (DGG) for community energy stations by the local energy enterprise groups as owners and managers, and promote the planting of fast-growing trees in any wilderness/waste land or non-utilized area, and processing for distribution as a DGG fuel.

- Adjust supportive measures, such as the adder, FIT or renewable heat Incentive, especially for DGG projects at the community level, and prepare financial support measures on improving the efficiency of old biomass power plants and change low-pressure boilers to high-pressure boilers.
- Improve infrastructure by assigning EGAT and the Provincial Electricity Authority to extend the transmission and distribution lines to support the development of biomass power plant projects, especially in areas with high potential for energy from biomass.
- Build up the collaborative process in target areas to install a biomass-derived energy system, campaign to educate children and juveniles on the detailed management of biomass for energy and environment at the local level, and establish a network of biomass energy entrepreneurs.
- Promote research by developing the production, consumption and standards for biomass pellets as future biomass fuel, develop gasification and gas engine technologies, including their extended industrial plants for domestic production, and develop biomass-to-liquid technology.
- **Biogas, AEDP 2012-2021 set this target at 600 MW by 2021:**
 - Promote and support biogas production at the household level, especially in rural communities to use in their households, and develop a biogas network for connecting to systems with excess capacity for sharing in community self-management.
 - Promote the production and consumption of compressed bio-methane gas (CBG) for transportation through price mechanisms that reflect the cost to support CBG production.
 - Study and develop laws and regulations for biogas safety standards.
 - Conduct a public relations campaign to disseminate knowledge and news and to create a good image of investment in producing and utilizing biogas safely.
 - Research and develop producing biogas from mixed wastes (co-digestion), especially the digestion of biomass resources such as water hyacinths and corn cobs to mix and ferment with animal manure, and develop the use of biogas as CBG for transportation.

- ***Geothermal Energy***, AEDP's 2012-2021 target is 1 MW by 2021:
 - There are barriers to developing geothermal sources for power generation since most domestic sources do not have a high heating value. Thus, build up the comprehensive understanding of the community on using geothermal for energy production.
 - Direct the development of geothermal energy by developing a national map of potential geothermal sources and technologies, assess the feasibility of their development using appropriate technologies, evaluate the cost-effectiveness and impacts of energy production on communities, the environment and public health, adopt technologies that utilize geothermal energy at low temperatures, and follow up on any technologies with potential and that are suitable for the local geography.
- ***Energy from Waves and Tidal Currents***, AEDP 2021-2021 targets 2 MW by 2021:
 - The main barrier in developing ocean wave and tidal current energy is the lack of data and assessments of the potential for this form of energy.
 - Identify wave and tidal energy sources and technologies that can be applied for Thailand's seas, and develop pilot projects.
- ***Hydrogen Energy and Energy Storage Systems***:
 - The main barriers to developing hydrogen energy and energy storage systems are: 1) lack of priority on planning for research and development, 2) the need to rely on overseas technologies, 3) domestic research and development are not yet broad and lack continuous budget support, and 4) no incentive measures are in place for the development and utilization of hydrogen energy in power generation.
 - To develop hydrogen energy and energy storage systems: 1) study the appropriate raw materials for hydrogen production in Thailand, 2) research and develop technologies for domestic production, storage and related devices, 3) research and develop high-efficiency, low-cost hydrogen production process, and 4) research and develop the technology for hydrogen applications in energy storage systems, including research, and demonstrate other types of potential energy storage systems, such as vanadium redox flow technology and lithium-ion battery technology.

A.6 People's Republic of Bangladesh

Socioeconomic Survey

In real terms, Bangladesh's economy grew 6.2% per year between 2007 and 2012 (it had a 2012 GDP of \$122.7 billion or \$818 per capita), despite political instability, poor infrastructure, corruption, insufficient power supplies, and the slow implementation of economic reforms. Bangladesh remains a poor, overpopulated, and inefficiently-governed nation. It is the eighth-most populous country at 163.7 million people. Although 54% of its GDP is generated through the service sector, 45% of Bangladeshis are employed in the agriculture sector. Bangladesh is one of world's largest producers of rice (4th), potato (11th), mango (9th), pineapple (16th), tropical fruit (5th), onion (16th), banana (17th), jute (2nd), and tea (11th).

Although Bangladesh is a poor developing nation, Goldman Sachs named it one of the “Next Eleven” after it coined the “BRICS” (Brazil, Russia, India, China and South Africa). The next eleven countries are Bangladesh, Egypt, Indonesia, Iran, South Korea, Mexico, Nigeria, Pakistan, Philippines, Turkey and Vietnam.

In Bangladesh, textiles and clothing is an \$18 billion industry in 2013. More than three-quarters of Bangladesh's export earnings come from the garment industry and represents 10% of GDP. The industry began attracting foreign investors in the 1980s due to cheap labor and low conversion costs. In the 2011–12 fiscal year, the industry exported \$18 billion worth of products; in contrast, in 2002, the exported amount was \$5 billion. Bangladesh has been ranked as the 2nd largest clothing exporter behind China. The industry now employs more than 4 million workers, 90% of whom are women.

Bangladesh has seen a dramatic increase in foreign direct investment, which is now about 25% of GDP. In order to enhance economic growth, the government set up several export processing zones to attract foreign investment. These are managed by the Bangladesh Export Processing Zone Authority, which generated \$32.7 billion in exports on \$2.7 billion in investments cumulatively from 1983 to 2013.

The Government of Bangladesh is planning to construct the largest deep sea port in South Asia at Sonadia Island. The BDT 500 billion project will be completed in multiple phases and enable Bangladesh to service the whole region as a maritime transport and logistics hub. India, China, Bhutan, Nepal and other neighboring countries will be able to take full advantage of the strategic location and privileges given to Bangladesh because of its least developed country status, for exporting goods that are manufactured in Bangladesh. Furthermore, a new \$7.5 billion international airport will be constructed. The airport is being modeled on Thailand's Suvarnabhumi Airport in size and capacity. To ease the chaotic traffic congestion in the capital, Dhaka, the government plans to construct more expressways, freeways, and flyovers. There is also a plan to build an overhead rapid transit called Dhaka Metro, but the progress is slow and controversial because of contracts and agreements. Recently the government launched high-speed modern trains and is also going to construct a metro rail system and high-speed electric powered inter-city bullet rail

network. More airports, bridges (such as the multi-billion Padma Bridge project) and the national highways are also being constructed to facilitate trade and regional development.

There has also been a significant growth to Bangladesh's ship building industry in the last few years. Ships and vessels are being produced by local shipbuilders. Furthermore, Bangladesh has already started taking orders and executing them perfectly for foreign companies from Germany, Denmark and other European countries that prefer the cheap market of Bangladesh over their local markets. In 2011, more than 250,000 skilled and semi-skilled workers were employed in the shipbuilding industry and the average growth of the industry is 8% annually. Bangladesh possesses the second-largest ship breaking industry in the world, which is the main source of materials for shipbuilding yards located throughout the country.

Bangladesh belongs to the group of countries with the lowest commercial energy consumption per head in the world (136 kWh). The CIA estimated that in 2009 the country produced 36 billion kWh of electricity. Of this amount, 95.8% was generated by fossil fuels and 4.0% by hydroelectric power plants. The majority of the electricity is generated by natural gas, as the country has 183.7 billion cubic meters of natural gas reserves and produces around 20 billion cubic meters annually. Sixty percent of Bangladesh's households have no electricity and in places where it is delivered, 40% of the electricity generated is not paid for. The country experiences regular electricity blackouts and shortages, and its poor reliability is often cited among the factors driving away foreign investors. The Bangladeshi government is willing to address the problem, but in general has not had enough resources to build new electric power generating plants.

Bangladesh has 15 MW of solar energy capacity through rural households and 1.9 MW of wind power in Kutubdia and Feni. The government plans to produce 5% of the country's total power generation by 2015 and 10% by 2020 from renewable energy sources such as air, waste and solar energy.

Legislative Initiatives Affecting Renewable Energy Development

The key legislative initiatives affecting renewable energy development are:

- ***Sustainable and Renewable Energy Development Authority Act, 2012.*** In June 2012, the parliament passed an act for the establishment of an institution called the Sustainable and Renewable Energy Development Authority (SREDA). It will identify, initiate, and develop national programs to promote the accelerated use of renewable energy, energy efficiency and energy conservation measures. The government is committed to facilitate both public and private sector investment in renewable energy projects and to substitute non-renewable energy with renewables for electricity production. Government policy envisions that 5% and 10% of the total electricity generation will be from renewable sources by 2015 and 2020, respectively.

- ***The Sustainable Energy for Development Program.*** This program, which is supported by the Ministry of Power, Energy, and Mineral Resources and the German Federal Ministry for Economic Cooperation and Development through Germany's GTZ Agency, is working to promote the efficient use of renewable energy through solar energy systems, biogas digesters, improved cook stoves and improved rice parboiling systems. A further focus is on energy efficiency improvement in industry and households.

Bangladesh has emerged as a model country for renewable energy use through the widespread dissemination of photovoltaic (PV) lighting systems, so-called solar home systems (SHS). SHS are gaining rapid acceptance among households and small businesses in rural areas. The Infrastructure Development Company Ltd. (IDCOL), a state-owned financing company, has facilitated the installation of about 2 million SHSs of 10-130 Wp in the country's rural villages since 2002, benefitting over 9 million Bangladeshis. The German government, through KfW and GTZ, has been actively supporting IDCOL's SHS program for several years. GTZ, with co-financing from the Dutch-German-British-Norwegian Partnership Program Energizing Development, provided funds for buy-down grants for over 180,000 such systems, as well as other support to help partner organizations develop the market for these systems. As the cash price of a SHS remains out of reach for many rural Bangladeshis, IDCOL and its partners introduced a micro-financing concept for SHS that makes their purchase affordable. Still, as many households could not afford the "normal" SHS, the SED Program, with its private sector partner organizations, developed smaller and less expensive SHSs that even poor households can afford. Light emitting diode (LED) technology was used to design 10-21 Wp solar systems that can power two to four lights and charge a mobile phone at a cost of only BDT 10,000 to 15,000. An even smaller LED solar system is now being introduced, a PicoPV solar lantern with panel sizes of less than 10 Wp that will replace the traditional kerosene lantern and provide clean quality light sufficient for reading or doing homework.

The SED Program is also working with several partner organizations to promote the industrial and commercial use of biogas technology throughout Bangladesh, since biogas digesters provide a simple, comparatively inexpensive, but highly effective way of using biomass such as chicken or cow manure to produce clean biogas for cooking and power generation. As an added benefit, the slurry resulting from biogas production provides high-quality organic fertilizer. Over 1,500 commercial biogas plants have already been installed in slaughterhouses and dairy farms; they were retrofitted to run on biogas and this approach is gaining popularity daily.

In the rice milling industry, there are over 50,000 small and medium-sized steam generating furnaces that are used to parboil 90% of the Bangladesh rice harvest right at the mills. Not only are these rice par boilers burning up almost half of Bangladesh's most plentiful source of biomass – rice husks, they are also very dangerous for operators and onlookers as the primitive boilers are not equipped with safety features. In conjunction with the Bangladesh Rice Research Institute (BRRI), the Energy Audit Cell of the Ministry of Power, Energy, and Mineral Resources, and Modern Erection Ltd., SED engineers worked to optimize the

geometry of the boiler, furnace, and chimney constituting the parboiling system for efficient combustion and heat transfer. They constructed the boiler from steel sheets and insulated it and the steam pipes with glass wool. They added a pressure gauge, a water-level gauge, a simple dead-weight safety device that opens automatically when pressure exceeds the safety limit, and a 12 meter chimney to aid in combustion and significantly reduce polluting smoke. Rice husk is fed into the furnace with the help of a blower. This new rice parboiling system, which reduces the use of rice husks by over 50%, is now being replicated across the country.

Another important achievement of SED Program is the establishment of a sustainable supply chain for improved cook stoves. The concept was originally developed by a team led by Dr. A.M. Hasan Rashid Khan at the Institute of Fuel Research and Development; it reduces the biomass required to cook a meal by 50% by adding a grate at the correct depth to focus the fire on the bottom of the utensils and small air holes in the wall of the stove just below the grate. The addition of a chimney eliminates the smoke and particles that threaten the health of women and children near the stove. SED developed the original design further, in particular, through introduction of concrete stoves that can be mass-produced and installed easily. So far, about 1,000,000 of these improved domestic stoves and over 3,000 commercial stoves have been sold under the brand name Bondhu Chula by many partner organizations and over 3,200 “sanitary shops” mobilized by SED.

SHS are affordable for many households as the ongoing SHS success story shows; improved cooking stoves cost between BDT 600-1200, depending on design, and thus are affordable for most households without the need for financing. However, access to finance is a crucial issue for the large-scale adoption of renewable energy technologies such as rice parboiling systems and large biogas plants. IDCOL is working with the commercial banking sector to facilitate access to loans for investment in such technologies. Furthermore, to reduce the interest on such loans the Green Fund set up by the Bangladesh Central Bank is a good vehicle to be utilized for such investments. The government has not only shown increasing interest in promoting and accelerating the use of renewable energy and energy efficiency improvements but also political determination to put ongoing and future efforts on a sustainable footing, as demonstrated by establishment of SREDA and development of policies and strategies for the implementation of renewable energy and energy efficiency improvement programs.

The Role of Feed-In Tariffs

In January 2013, the Bangladesh Energy Regulatory Commission (BERC) initiated a move to prepare a FIT for renewable energy to promote its use in the country. According to BERC officials, the regulatory body is working with USAID to help prepare the FIT, which will consider the socio-economic conditions and technological cost of different renewable energies such as solar, wind and biomass. The goal of the FIT is to offer cost-based compensation to renewable energy producers, providing the price certainty and long-term contracts that help finance renewable energy investments. But until a FIT is in place, the private sector and financial institutions are unlikely to come forward to invest in the sector.

Central Government Investment in Renewable Energy Development

The Infrastructure Development Company Limited (IDCOL) was established on 14 May 1997 by the Government of Bangladesh. The company now stands as the market leader in private sector energy and infrastructure financing in Bangladesh. IDCOL promotes the dissemination of solar home systems in remote rural areas through its Solar Energy Program with financial support from the World Bank, Global Environment Facility), KfW, GTZ, Asian Development Bank and Islamic Development Bank. IDCOL started the program in January 2003 and its initial target was to finance 50,000 SHSs by the end of June 2008. The target was achieved in September 2005, three years ahead of schedule and \$2.0 million below the estimated project cost. IDCOL then revised its target and decided to finance 200,000 SHSs by the end of 2009. This was also achieved by May 2009. Subsequently, IDCOL's target was to finance 1 million SHSs by the end of year 2012, a goal which was also achieved.

SHSs are sold (mostly through microcredit) by micro-financial institutions (MFIs), such as Grameen Shakti, to households and business entities in the off-grid areas of Bangladesh. IDCOL serves as a refinancing facility to the MFIs and also channels grants to reduce the SHS cost as well as support the institutional development of the MFIs. Also, it provides technical, logistic, promotional and training assistance to the MFIs. In addition to solar, IDCOL provides support in biogas and biomass power generation.

Bangladesh Bank (BB), through its Green Banking and CSR Departments, is one of the first central banks in the world with a clear vision to promote green banking to safeguard the environment from unusual weather patterns, rising greenhouse gases, and declining air quality. It has issued a number of guidelines for green banking. BB has proactively put in place a fund of BDT 2.0 billion (\$25.6 million) to refinance commercial bank lending for renewable energy generation and other environmentally beneficial projects. As of December 2012, BDT 853.54 million (\$11 million) out of a BDT 2.0 billion revolving fund has been allocated by BB to different green categories. BB is concentrating on the most effective utilization of resources (power, gas, fuel, water, paper, etc.).

It installed a solar power system on its rooftop in 2010 and implemented automation through various initiatives such as Bangladesh Automated Clearing House, Credit Information Bureau online, Enterprise Resources Planning, Enterprise Data Warehouse, National Payment Switch, e-tendering, e-recruitment, e-passes, online salary and other necessary information as part of its in-house green initiatives and activities. BB has recently taken the initiatives to convert its 30-story building into a green building with modern rain water harvesting, waste water recycling and motion sensor energy-efficient bulbs supported by window-based solar panels. All of its member banks have formulated their own green banking policies and formed green banking units on the basis of BB's guidelines.

According to BB records, commercial banks have disbursed BDT 703,633.21 million for 11,165 projects that have good environmental risk ratings and BDT 270,921.53 million in the form of green finance during 2012. Banks have been encouraged to utilize BDT 258.89

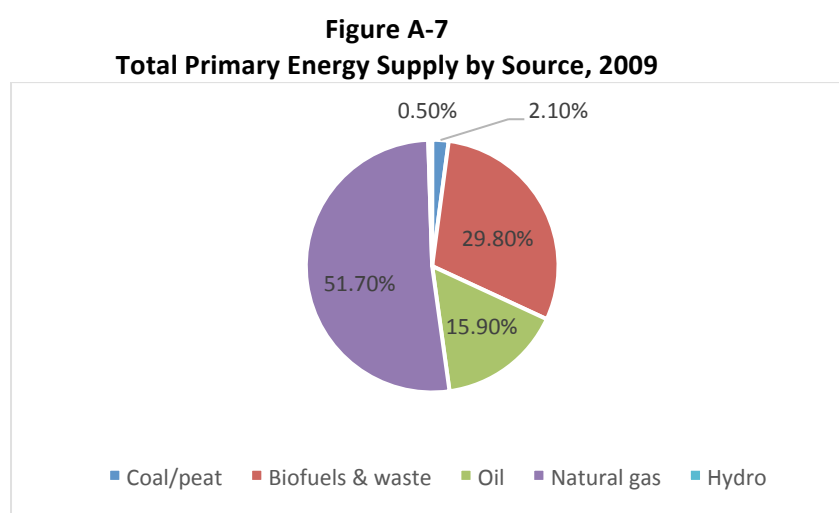
million for green events and green projects as part of their corporate social responsibility activities. They have also concentrated on green marketing, training and development through utilizing BDT 90.42 million from their funds.

Cap and Trade Activity

Bangladesh does not have any cap and trade policy. However, USAID and the USDA Forest Service have evaluated the Sundarbans, the world's largest mangrove forest. The total area of 6,017 square kilometers is 61% covered by land and 39% by water. The Sundarbans can capture 54 million tons of carbon annually, which may be worth taka 150 billion (or nearly \$2 billion) in carbon trading. Reportedly, Indonesia got \$1 billion from selling its carbon reserves to the Oslo Fund.

Specific Renewable Energy Technologies

In 2009, Bangladesh's total primary energy supply, excluding electricity, was provided by natural gas (51.7%), biomass and waste (29.8%), oil (15.9%), coal (2.1%), and hydro (0.5%), which totaled 29,599 ktoe according to IEA (see Figure A-7).



Source: IEA, *Key World Energy Statistics*, 2009

The major sources of renewable energy are:

- **Solar photovoltaic:** Solar photovoltaic (PV) systems are in use throughout the country, as mentioned previously. Assisted by development partners, Bangladesh is scaling-up solar PV systems through the Rural Electrification Board (REB), Local Government Engineering Department (LGED), Bangladesh Power Development Board (BPDB) and other agencies. The Renewable Energy Research Centre of the University of Dhaka has installed a model 1.1 kW grid-connected photovoltaic system. There is a strong potential for solar energy within the country.

- **Solar Thermal:** This technology involves harnessing solar radiation for the generation of electricity through a number of steps, finally generating mechanical energy. This technology needs to be disseminated throughout the country to supplement the power supply.
- **Wind Energy:** Wind energy has also made some inroads, but its potential is mainly limited to coastal areas and offshore islands with strong wind regimes. These coastal settings afford good opportunities for wind-powered pumping and electricity generation. At present, there are 2 MW of installed wind turbines at Feni and Kutubdia.
- **Biomass:** Bangladesh has strong potential for biomass gasification-based electricity. The more common biomass resources available in the country are rice husk, crop residue, wood, jute sticks, animal waste, municipal waste, sugarcane and bagasse. This technology can be disseminated on a larger scale for electricity generation.
- **Biogas:** Mainly derived from animal and municipal wastes, biogas may be one of the promising renewable energy resources for Bangladesh. There are now tens of thousands of household and village-level biogas plants in place throughout the country. The potential for harnessing basic biogas technology for cooking, and rural and peri-urban electrification during periods of power shortfalls, is strong.
- **Hydro:** Micro-hydro and mini-hydro have limited potential in Bangladesh, with the exception of Chittagong and the Chittagong Hill tracts. Hydropower assessments have identified some possible sites from 10 kW to 5 MW, but no appreciable capacity has yet been installed. There is one hydropower plant at Kaptai, which was established in the 1960s and has an installed capacity of 230 MW.
- **Other:** Other renewable energy sources include bio-fuels, gasohol, geothermal, river current, wave and tidal energy. The potentials of these sources are yet to be explored.

The government's targets for generation from renewable energy are shown in Table A-10.

Table A-10
Renewable Energy Target Additions and Investment by 2015 and 2010

Source	2015 (MW)	2020 (MW)
Solar	376	1480
Hydro	250	300
Biomass and Biogas	60	120
Wind	6	100
Total New Renewable Energy Generation	687	2000
Total Investment Required	\$196.5 million	\$371.5 million

Source: Dewan Mowdudur Rahman, Navid Bin Sakhawat, Riasad Amin, and Faisal Ahmed, *A Study on Renewable Energy as a Sustainable Alternative for Ensuring Energy Security in Bangladesh and Related Socio-Economic Aspects*.

Annex B: Overview of Measures, Incentives and Definitions

In 2005, China, the USA and EU-27 were the largest emitters of equivalent carbon dioxide (CO₂e): 19,490.7 million metric tons, or 44.19% of world's total GHG emissions (16.38%, 15.72%, and 12.09%, respectively), which includes six greenhouse gases, bunker fuel emissions and land-use changes. Even though they are heavy polluters, the US, China and EU-27 are at the forefront of GHG emissions reductions and provide financial incentives and government regulations and policies for renewable or clean energy.

Brazil, Indonesia and India are among the top-10 polluters at 6,769.2 million metric tons or 15.34% (6.47%, 4.62% and 3.15%, respectively), while Thailand and Bangladesh emitted 509.7 million metric tons or 1.15% (0.83% and 0.32%, respectively,) of the world's GHG emissions.

Table B-1 summarizes the administrative and fiscal incentives used to stimulate renewable energy development and reduce GHG emissions by 2020 and beyond. In the USA, some states have state-level policies, as there is no national level policy for feed-in tariffs, electricity quota obligations (renewable portfolio standards), net metering and heat obligations. China has most of the administrative and fiscal incentives of the United States, except for the incentives that are unique to individual states. In terms of the number of measures, Brazil, India, and Thailand would be next, in that order, followed by Bangladesh, which appears to have adopted the fewest measures.

Indonesia would be close to Brazil, except that it does not have feed-in tariffs for solar and wind power. We understand that Indonesia's corporate tax incentives, in the form of accelerated depreciation, are regarded by some parties as difficult to access, so we have included a question mark beside that incentive. Although Indonesia has grants, loans funds, and guarantees, access to these by small projects (less than 10 MW) is limited. For example, the grants to small-scale, utility-interactive, renewable projects are provided through MCC's Green Prosperity Fund, but they only apply to those projects located in 13 provinces. Indonesia has a loan fund (IIF) and a guarantee fund (IIGF), but these can only be employed for competitively bid projects. Hence, they are not applicable to small renewable energy projects, which according to the rules of the game, are generally sole sourced.

This report organizes incentives and policies that promote renewable energy and energy efficiency into two general categories: financial incentives, and rules, regulations & policies (roughly 20 specific types of incentives and policies).

Table B-1
Summary of Administrative and Fiscal Incentives
Used to Stimulate Renewable Energy Development

Table B-1
Summary of Administrative and Fiscal Incentives Used to Stimulate Renewable Energy Development

	Indonesia	United States	China	Brazil	India	Thailand	Bangladesh
Financial Incentive Systems							
Feed-In Tariffs	✓	✓	✓	✓	✓	✓	
Corporate Tax Incentives	?	✓	✓	✓	✓	✓	
Grants	✓	✓	✓	✓	✓	✓	✓
ESCO Financing for Energy Efficiency		✓	✓	✓	✓	✓	
RE Industry Recruitment Programs		✓	✓				
Loan Programs (or Funds)	✓	✓	✓				✓
Guarantee Fund	✓						
PACE Financing		✓					
Performance Based Incentives		✓					
Sales Tax (or VAT) Incentives		✓	✓	✓	✓	✓	
Cap and Trade Schemes		✓	✓	✓	✓	✓	✓
Research & Development		✓	✓	✓	✓	✓	
Database Library		✓	✓				
Rules, Regulations, and Policies							
Energy Standards for Public Buildings		✓	✓	✓	✓	✓	
Green Power Purchasing Standards		✓	✓	✓	✓	✓	
Interconnection Standards		✓	✓	✓	✓	✓	✓
Mandatory Utility Green Power Option		✓	✓	✓	✓	✓	
Net Metering		✓	✓	✓	✓	✓	
Public Benefit Funds		✓	✓	✓	✓	✓	✓
Renewable Portfolio Standards		✓	✓	✓	✓	✓	✓
Solar and Wind Permitting		✓	✓	✓	✓	✓	✓
MSW Safeguards		✓	✓	✓	✓	✓	✓
Biofuel Stimulus		✓	✓	✓	✓	✓	✓
Financial Incentive Systems		Why They Work to Stimulate Investment in Utility-Interactive Renewable Energy Systems					
Feed-In Tariff	FIT is a policy mechanism designed to accelerate investment in renewable energy technologies. It achieves this by offering long-term contracts to renewable energy producers, typically based on the cost of generation of each technology.						
Corporate Tax Incentives	Corporate tax incentives include tax credits, accelerated deductions and exemptions. These incentives are available to corporations that purchase and install eligible utility-interactive renewable energy, or energy efficiency systems. In a few cases, the incentive may be based on the amount of						

Table B-1
Summary of Administrative and Fiscal Incentives Used to Stimulate Renewable Energy Development

	energy produced by an eligible facility i.e., a production tax credit. Some states in the USA allow the tax credit only if a corporation has invested a minimum amount in an eligible project. Typically, there is a maximum limit on the dollar amount of the credit or deduction. Accelerated deductions and production tax credits, in particular, are shown to have greatly stimulated interest in renewable energy investment.
Grants	Some countries offer a variety of grant programs to encourage the use and development of renewables and energy efficiency. Most programs offer support for a broad range of technologies, while a few programs focus on promoting a single technology, such as photovoltaic (PV) systems. Most grant programs are designed to pay down the cost of eligible systems or equipment, making unviable projects reasonably viable at the posted feed-in tariff. Others focus on research and development, or support project commercialization. The US also offers grants for renewables and energy efficiency projects for end-users, in order to stimulate market development. Grants are usually competitive, subject to tender.
ESCO Financing Model of Implementing Energy Efficiency Projects	Energy service companies provide a broad range of comprehensive energy solutions, including the design and implementation of energy savings projects, retrofitting, energy conservation, energy infrastructure outsourcing, power generation and energy supply, and risk management, as well as innovative financing.
RE Industry Recruitment Standards	To promote project development and post-sale service to developers, the US and other countries offer financial incentives to recruit or cultivate the manufacturing, development and servicing of renewable energy systems and equipment. These incentives commonly take the form of tax credits, tax exemptions and grants. In some cases, the amount of the incentive depends on the quantity of eligible equipment that a company manufactures. Most of these incentives apply to several renewable energy technologies, but can apply to specific technologies, such as solar. These incentives are usually designed as temporary measures to support industries in their early years. They commonly include a sunset provision to encourage the industries to become self-sufficient.
Loan Programs (or Funds)	Loan programs provide financing for the purchase of renewable energy or energy efficiency systems or equipment. Low-interest or zero-interest loans for energy efficiency projects are a common demand-side management practice for electric utilities. Governments also offer long-term (sometimes subsidized) loans for a broad range of renewable energy and energy efficiency measures. These

Table B-1
Summary of Administrative and Fiscal Incentives Used to Stimulate Renewable Energy Development

	programs are commonly available to the commercial sector. Loan interest and terms vary by program; in some cases, they are determined on an individual project basis. Loan terms are generally 10 years, but can be less. Governments in India, Turkey, Brazil and China offer loans through development banks. Some US states offer loans and/or loan guarantees for renewable and energy efficiency projects.
Property Assessed Clean Energy Financing	In the USA, Property-Assessed Clean Energy (PACE) financing effectively allows property owners to borrow money to pay for renewable energy and/or energy-efficiency improvements. The amount borrowed is typically repaid over a period of years via a special assessment on the owner's property. In general, local governments (such as cities and counties) that choose to offer PACE financing must be authorized to do so by state law. The new energy systems bought by individuals can then be used to supply the utility with the electricity that is not used.
Performance-Based Incentives	Performance-based incentives, also known as production incentives, provide cash payments based on the number of kWh or Btu generated by a renewable energy system. A feed-in tariff is an example of a PBI. To ensure project quality, payments based on a system's actual performance are generally more effective than payments based on a system's rated capacity.
Sales Tax (or VAT) Incentives	Sales tax incentives typically provide an exemption from, or refund of, the state sales tax (or sales and use tax) for the purchase of a renewable energy system or energy efficiency measures. Some states have established an annual "sales tax holiday" for energy efficiency measures by annually allowing a temporary exemption, usually for one or two days from the state sales tax.
Import Tariff Relief	Certain renewable energy technologies or equipment are not available in a country, so the government provides relief from import duty on project equipment.
Cap and Trade Schemes	Cap and trade is a market-based approach used to control pollution by providing economic incentives for achieving reductions in the emissions of pollutants. A central authority sets a limit or cap on the amount of a pollutant that may be emitted. The limit or cap is allocated or sold to firms in the form of emissions permits that represent the right to emit or discharge a specific volume of the specified pollutant. Firms are required to hold a number of permits or carbon credits equivalent to their emissions. The total number of permits cannot exceed the cap, limiting total emissions to that level. Firms that need to increase their volume of emissions must buy permits from those who require fewer permits. The transfer of permits is referred to as a <i>trade</i>. In effect, the

Table B-1
Summary of Administrative and Fiscal Incentives Used to Stimulate Renewable Energy Development

	buyer is paying a charge for polluting, while the seller is being rewarded for having reduced emissions. Thus, in theory, those who can reduce emissions most cheaply will do so, achieving the pollution reduction at the lowest cost to society.
Research & Development	Many governments provide funding for R&D in renewable energy technologies in solar PV and thermal, wind, bioenergy (biomass, biogas, municipal solid waste), biofuels (ethanol, biodiesel and new fuels), tidal and geothermal. The National Renewable Energy Laboratory, located in Golden, Colorado, is the United States' primary laboratory for renewable energy and energy efficiency R&D, which was established in 1974 by the Department of Energy
Database Library	DSIRE is the most comprehensive source of information on incentives and policies that support renewables and energy efficiency in the United States, which was established in 1995 by the Department of Energy. In 2008, DSIRE started to support analysis efforts at NREL and Lawrence Livermore Laboratory.
Rules, Regulations, and Policies	
Energy Standards for Public Buildings	Many states and local governments in the US, as well as the US federal government, have chosen to lead by example by requiring new government buildings to meet strict energy standards. These include policies for green building standards, energy-reduction goals, equipment procurement requirements, and/or the use of on-site renewable energy. Many of these policies require that new government buildings (and renovated buildings, in some cases) attain a certain level of certification under the U.S. Green Building Council's Leadership in Energy and Environmental Design (LEED) program. Policies often encourage the use of on-site renewables subject to life-cycle cost analysis.
Green Power Purchasing Standards	Government entities, businesses, residents, schools, non-profits and others can play a significant role in supporting renewable energy by buying electricity from renewable resources, or by buying renewable energy credits (RECs). Many state and local governments in the USA, as well as the federal government, have committed to buying green power to account for a certain percentage of their electricity consumption. Green power purchases are typically executed through contracts with green power marketers or project developers, through utility green power programs, or through community aggregation.
Interconnection Standards	Interconnection standards specify the technical and procedural process by which a utility connects an electricity-generating facility to the grid. Such standards include the technical and contractual terms that system owners and

Table B-1
Summary of Administrative and Fiscal Incentives Used to Stimulate Renewable Energy Development

	utilities must abide by. Standards are set for interconnection to the distribution grid, while others are adopted for interconnection to the transmission level.
Mandatory Utility Green Power Option	US and European policies sometimes require electric utilities to offer their customers the option to buy electricity generated from renewable resources, commonly known as “green power.” Typically, a utility either offers green power generated using renewable resources that the utility owns (or for which it contracts), or it buys renewable energy credits (RECs) from a certified provider.
Net Metering	For electric customers who generate their own electricity, net metering allows for the flow of electricity both to and from the customer – typically through a single, bi-directional meter. When a customer’s generation exceeds the customer’s use, electricity from the customer flows back to the grid, offsetting electricity consumed by the customer at a different time during the same billing cycle. In effect, the customer uses excess generation to offset electricity that the customer otherwise would have to purchase at the utility’s full retail rate. Net metering is required by law in most U.S. states.
Public Benefit Funds	Most public benefit funds (PBFs) were developed by US states during the electric utility restructuring era, in the late 1990s, to ensure continued support for renewable energy, energy efficiency and low-income energy programs. These funds are commonly supported through a very small surcharge on electricity consumption (e.g., \$0.002/kWh). This charge is sometimes referred to as a “system benefits charge.” PBFs commonly support rebate programs, loan programs for renewable generation, research and development, and energy education programs.
Renewable Portfolio Standards	Renewable portfolio standards (RPSs) require utilities to use or procure renewable energy or renewable energy credits to account for a certain percentage of their retail electricity sales – or a certain amount of generating capacity – according to a specified schedule. (Renewable portfolio goals are similar to RPS policies, but their goals are not legally binding.) Most US states, for example, have established an RPS. The term “set-aside” or “carve-out” refers to a provision within an RPS that requires utilities to use a specific renewable resource (usually solar energy) to account for a certain percentage of their retail electricity sales (or a certain amount of generating capacity) according to a set schedule.
Solar and Wind Permitting	Permitting standards can facilitate the installation of wind and solar energy systems by clarifying the conditions and fees involved in project development. Some local

Table B-1
Summary of Administrative and Fiscal Incentives Used to Stimulate Renewable Energy Development

	governments have adopted simplified or expedited permitting standards for wind and/or solar. "Top-of-the-stack" permitting (or fast-track permitting) saves system owners and project developers time and money. Some states have capped fees that local governments may charge for a permit for a solar or wind energy system. In addition, some states have developed or supported model wind ordinances for use by local governments.
Municipal Solid Waste Safeguards	Modern landfills are well-engineered facilities that are located, designed, operated, and monitored to ensure compliance with federal regulations. Solid waste landfills must be designed to protect the environment from contaminants which may be present in the solid waste stream. Federal standards include location restrictions, composite liner requirements, leachate collection and removal systems, operating practices, groundwater monitoring requirements, closure and post-closure care requirements, corrective action provisions and financial assurance during closure.
Biofuels Stimulus	As the price of oil becomes prohibitive and countries import more oil for transport, countries create policies to stimulate biofuel replacement or mix ethanol, biodiesel or new fuels, such as algae and hydrogen-based fuels.

Negotiations on the Kyoto Protocol to the United Nations Framework Convention on Climate Change (UNFCCC) were completed on 11 December 1997, committing the industrialized nations to specified, legally binding reductions in emissions of six "greenhouse gases." The six gases covered by the Protocol are carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulfur hexafluoride (SF₆). The most prominent of these, and the most pervasive in human economic activity, is carbon dioxide, produced when wood or fossil fuels such as oil, coal, and gas are burned. The Protocol entered into force on 16 February 2005, and its emissions reduction targets are binding on the 35 industrialized countries that have ratified it; the United States disengaged from the Protocol in 2001 and has not ratified it. China, India, Brazil, Thailand and Bangladesh ratified the Kyoto Protocol in 2002; however, these developing countries, as is true with developing countries in general, did not have binding emission reduction targets.

Like Brazil, Indonesia has the potential to realize major reductions in national greenhouse gas emissions from a reduction of deforestation. Table B-2 addresses the cap and trade or emissions reductions programs or tradable RECs of the example countries.

Table B-2
Cap and Trade Framework

USA	The USA does not have a national cap and trade framework. Regional Greenhouse Gas Initiative members Maine, New Hampshire, Vermont, Connecticut, New York, Delaware, Massachusetts, Maryland and Rhode Island have agreed to reduce carbon dioxide from utilities by 10% below 1990 levels by 2020. California started a cap and trade program in 2012 and agreed to reduce GHG emissions 17% below 1990 levels by 2020.
China	China has pilot carbon trading schemes in seven provinces and cities: Beijing, Chongqing, Guangdong, Hunan, Shanghai, Shenzhen and Tianjin. Some will be launched as early as 2013. They will cover energy production and various energy-intensive industries. Officials have said a national scheme might not emerge this decade as China plans to launch additional test schemes in 2016. The government plans to cut CO ₂ emissions per unit of GDP to 40-45% below 2005 levels by 2020.
India	Trading under India's mandatory energy efficiency trading scheme is set to begin in 2014 after a three-year rollout period. It covers eight sectors responsible for 54% of India's industrial energy consumption. India has pledged a 20-25% reduction in emissions intensity from 2005 levels by 2020.
Brazil	Brazil has a voluntary emissions market. Deforestation represents 80% of Brazil's greenhouse gas emissions. <i>Bolsa Verde do Rio de Janeiro</i> , a non-profit civil association created in partnership with the Department of Environment of the State of Rio de Janeiro and the Treasury Department of the Municipality of Rio de Janeiro, was launched in December 2011.
Thailand	Thailand has a voluntary emissions market that will be launched in October 2014. Sectors have not been specified, but the government said it will discuss emissions targets with participants.
Bangladesh	Bangladesh does not have a cap and trade policy. However, USAID and the USDA Forest Service have evaluated the Sundarbans, the world's largest mangrove forest. The total area of 6,017 square kilometers is 61% covered by land and 39% by water. Sundarbans can capture 54 million tons of carbon annually, which may be worth nearly \$2 billion in carbon trading.

Annex C: Terms of Reference

The Central Bank of Indonesia (Bank Indonesia) has announced plans to develop a green banking policy that will include regulations on commercial bank activity in the financing of small-scale renewable energy projects. The renewable projects targeted include a broad range of projects, including small hydro power plants, biomass, biogas, wind power, and solar photovoltaic systems. The U.S. Agency for International Development's Indonesia Clean Energy Development (ICED) Project is currently working with Bank Indonesia on its green banking initiative.

Bank Indonesia asked ICED to conduct research on what other countries are doing to incentivize commercial bank lending to renewable energy projects. Bank Indonesia will use the examples and experience from other countries in making decisions about its own policies and programs in order to promote investment in, or access to, finance for clean energy projects. As part of its own mandate, Bank Indonesia's role is to encourage bank intermediation to support financing in those sectors with the potential for government cooperation.

The clean energy sector has limited credit available and low growth. However, the Government of Indonesia has established ambitious targets to increase the contribution of renewable energy as a means of improved energy security and reduced greenhouse gas emissions. While strategically important, renewable energy projects are not well understood by the local banking sector, and do not appear to be commercially attractive to the banking industry. Helping Bank Indonesia to support the sector is an important objective of ICED.

As lenders become involved in financing projects in the late stages of development, securing commercial lending to the sector also requires a regulatory and legal framework. In other words, the banks look to lend when the external project risks appear certain and stable. Policy instruments that governments can employ to create greater certainty include feed-in tariff policies and tax incentives such as investment tax credit or cash grants.

The Scope of Work

The Consultants shall support ICED's and USAID's efforts to the tasks below. Execution of the activities shall be under the technical direction of the Chief of Party (Mr. Bill Meade) or his designee. The Consultant(s) shall undertake the activities below and as referenced in the Consultant(s) technical approach as written in the Technical Proposal:

Task 1: Summary of Investment Incentives for Renewable Energy

Governments have employed a variety of mechanisms to increase the investment in renewable energy projects. Consultants will research banking and fiscal (tax) mechanisms that have been used by governments to stimulate investment in renewable energy projects. For each mechanism, the Consultants will provide a description of how that mechanism

affects the attractiveness or debt and/or equity financing and the commercial viability of the renewable energy project. For purposes of this study, the research should be limited to renewable energy independent power projects that are connected to utility transmission/distribution system.

Deliverable:

- Draft “Summary of Investment Incentives for Renewable Energy”

Task 2: Examples of Central Bank Initiatives for Lending to Renewable Energy Projects

There are different approaches that Central Banks have been using to promote commercial lending to renewable projects; using their own funds or other funds to either on-lend or providing direct financial incentives to commercial banks; such as subsidizing lending through direct interest rate subsidies. Alternatively, they make investments in clean energy funds, making available directly long-term lending for renewable projects. Other pricing incentives relate to reduced pricing on loans depending on the volume of lending that the sector will require.

Globally, central banks are becoming proactive in promoting renewable energy investments, and in some cases actively supporting national development banks, e.g., Turkey’s Development Bank TKSİB is directly funding several clean energy funds; similarly, the Indian Central Bank, in order to advance the Government’s National Solar Mission, eased the rules for lending to off-grid solar projects, allowing banks to extend subsidized loans. In Bangladesh, the Bangladesh Bank, the Central Bank, has launched a revolving fund of BDT 2 billion for refinancing renewable energy projects, i.e., solar energy, biogas and effluent treatment plants through commercial banks at concessionary terms and conditions.

The Consultant will prepare case studies of six (6) countries as potential models for Indonesia. The case studies, each approximately 10-15 pages, will describe the mix of banking, tax and other policies that have created a conducive investment climate for renewable energy projects. The case studies will also research trends in market development for renewable energy, e.g., investment in and installed capacity/energy production from renewable energy projects. To the extent possible, the Consultant will evaluate the effectiveness of the banking and other policy mechanisms in increasing lending to renewable energy projects. Countries to be researched will include the U.S., Thailand, Brazil, China, and India.

Task 3: Presentation to Bank of Indonesia Results of Desk Study

The Consultant will prepare a presentation reflecting the findings and conclusions of Tasks 1 and 2. ICED is preparing a Financing Renewable Energy Workshop in Jakarta, Indonesia in May 2013. The Consultant will provide a draft presentation in advance of the workshop, and will incorporate comments from Tetra Tech. The Consultant will finalize the presentation and deliver the presentation at the Financing Renewable Energy Workshop. The costs of travel and per diem should be included in the Consultant’s budget.

Deliverables:

- Draft “Examples of Central Bank Initiatives for Lending to Renewable Energy Projects”
- Draft Presentation
- Final Report incorporating comments from the ICED team
- Presentation documents in PowerPoint summarizing key findings.